

QUANTUM CRYPTOGRAPHY MARKET

GLOBAL FORECAST TO 2025

BY COMPONENT (SOLUTIONS AND SERVICES), SERVICE (CONSULTING
AND ADVISORY, DEPLOYMENT AND INTEGRATION, AND SUPPORT
AND MAINTENANCE), SECURITY TYPE, VERTICAL, AND REGION

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SUMMARY OF CHANGES

- The new research study includes the impact of COVID-19 on the overall quantum cryptography market as well as on the market segments and regions.
- The new market study includes breakup of the component segment by solution and service. The services segment includes consulting and advisory, deployment and integration, and support and maintenance. The security type segment includes network security and application security. The vertical segment is bifurcated by government and defense, BFSI, retail, healthcare, automotive, and others.
- The new study elaborates the market analysis at the regional level for the following countries: Americas, Europe, APAC, and MEA.
- The new study also comprises COVID-19 drivers for the vertical and region segments.
- The new research study consists of 25 players as compared to 21 players in the previous version of the report.
- The new study includes updated financial information/product portfolio of players. It also provides updated financial information in the context of quantum cryptography market till 2017–2019 for each company in graphical presentation.
- The new study includes updated market developments of profiled players. The current report includes the developments from 2017 to 2020 pertaining to the market.
- The new study includes quantitative data for historical years i.e. 2017-2018, base year i.e. 2019, and forecast years i.e. 2020-2025.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
ACL	Access Control List
AES	Advanced Encryption Standard
AI	Artificial Intelligence
ANR	National Research Agency
APAC	Asia Pacific
BFSI	Banking, financial services and insurance
CAGR	Compound Annual Growth Rate
CIS	Commonwealth of Independent States
CPDS	Correlated Photon Detection System
CQC2T	Centre for Quantum Computation and Communication Technologies
CQIQC	Centre for Quantum Information and Quantum Computation
CSS	Certified Security Solutions
DARPA	Defense Advanced Research Project Agency
DCMS	Digital, Culture, Media, and Sport
DoE	Department of Energy
DRC	Data Recovery Center
DST	Department of Science and Technology
EPS	Entangled Photon Source
ESG	Enterprise Strategy Group
FBI	Federal Bureau of Investigation
GSMA	Global System for Mobile Communications Association
HIPAA	Health Insurance Portability and Accountability Act of 1996
IARPA	Intelligence Advanced Research Projects Activity
ICPS	under Interdisciplinary Cyber Physical Systems
ICS	Industrial Control Systems
IDS	Intrusion Detection System
IEEE	Institute of Electrical and Electronics Engineers
IIT	Indian Institute of Technology

IPS	Intrusion Prevention System
ISSA	Information Systems Security Association
ISV	Independent Software Vendors
IT	Information Technology
LED	Light Emitting Diode
LPWAN	Low-Power Wide-Area Network
MEA	Middle East and Africa
MENA	Middle East and North Africa
MMC	Malicious Mobile Code
NATO	North Atlantic Treaty Organization
NFC	Near Frequency Communication
NICT	National Institute of Information and Communications Technology
NIST	National Institute of Standards and Technology
NRF	National Research Foundation
NSF	National Science Foundation
NTS	Never-The-Same
ORNL	Oak Ridge National Laboratory
OTA	Over-The-Air
PCI DSS	Payment Card Industry Data Security Standard
PKI	Public Key Infrastructure
POS	Point-of-Sale
PQC	Post-Quantum Cryptography
QEP	Quantum Engineering Programme
QIS	Quantum Information Sciences
QKD	Quantum Key Distribution
QRNG	Quantum Random Number Generator
QuST	Quantum Information Science and Technology
QUTEGA	Quantum technology - Fundamentals and Applications
RD&D	Research, Development, and Demonstration
RF	Radio-Frequency
RoW	Rest of World
RSA	Rivest-Shamir-Adleman

SI	System Integrator
SSP	Six-State Protocol
TTP	Trusted Third Party
UAE	United Arab Emirates
UQCC	Updating Quantum Cryptography and Communications
USD	United States Dollar
VAR	Value-Added Reseller
VPN	Virtual Private Network

1 INTRODUCTION

1.1 INTRODUCTION TO COVID-19

COVID-19 is an infectious disease caused by the most recently discovered novel coronavirus. Largely unknown before its outbreak began in Wuhan, China, in December 2019, COVID-19 moved from being a regional crisis to a global pandemic in just a few weeks. The World Health Organization (WHO) declared COVID-19 as a pandemic on March 11, 2020.

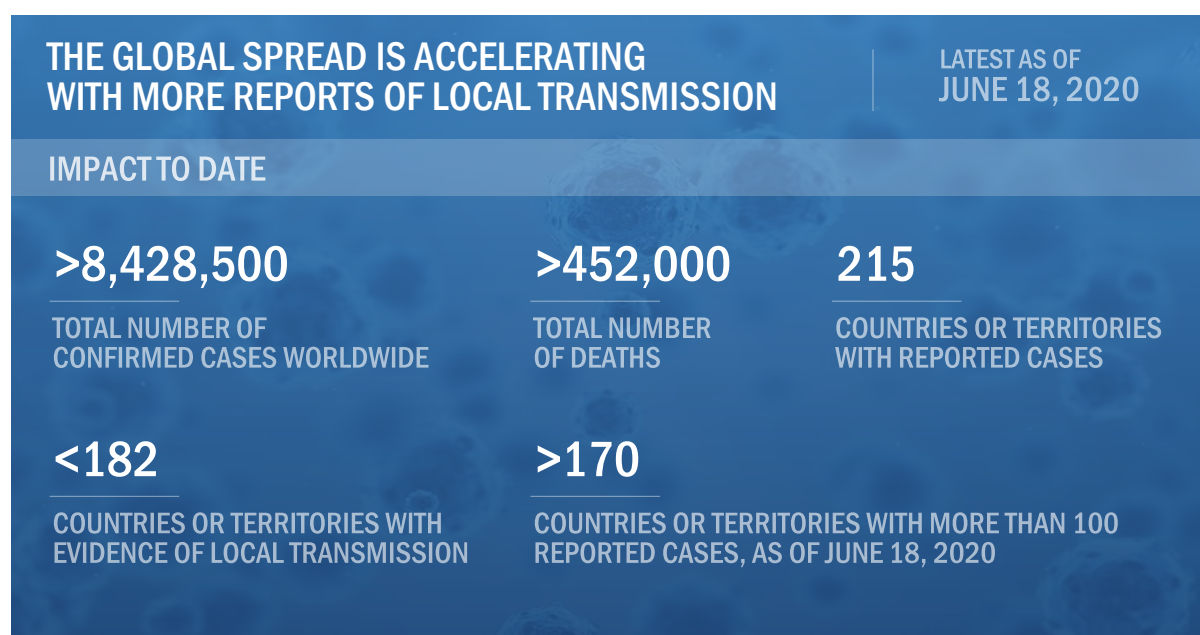
As this virus is spreading rapidly at a global level, countries across the globe are trying to prevent further contagion by adopting measures, such as social distancing, contact tracing, self-quarantining, surveillance, communication, and testing. So far, China, South Korea, Japan, and Singapore have managed to flatten the curve, while the US, Italy, Spain, Germany, France, and Iran are imposing drastic measures to slow down the spread and control fatalities.

Government agencies have announced special financial aid packages toward the development of preventive and curative drugs, the purchase of critical care medical devices, and the fast-track approval of diagnostic tests. Various organizations across the globe, such as the Bill and Melinda Gates Foundation (BMGF) and Wellcome Trust, are also coming together to accelerate and strengthen the efforts to fight the COVID-19 pandemic.

1.2 COVID-19 HEALTH ASSESSMENT

This is indeed an unprecedented crisis, which has put an enormous strain on healthcare providers across the globe. According to the WHO, within 67 days, the first 100,000 confirmed cases were reported worldwide. It took the first 11 days to reach 200,000 confirmed cases, in the next four days it reached 300,000 confirmed cases, and the count reached one million confirmed cases in another 12 days.

FIGURE 1 COVID-19: THE GLOBAL PROPAGATION

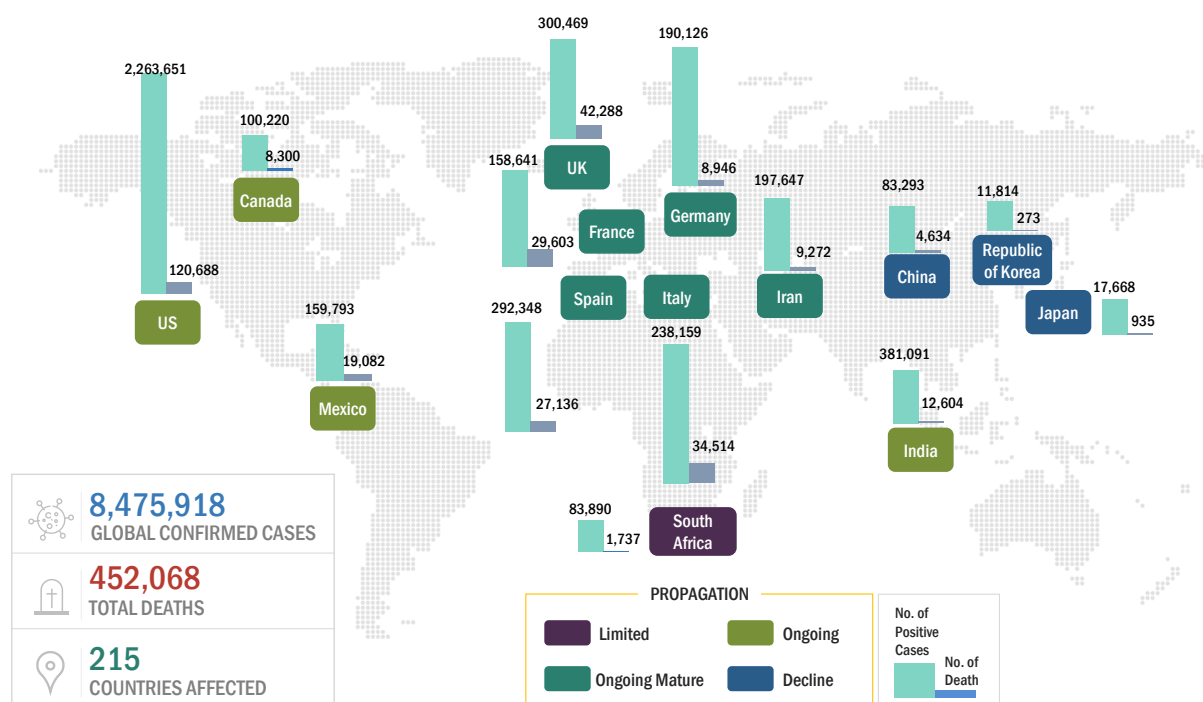


Source: WHO and ECDC

As of June 18, 2020, the virus has spread to more than 215 countries, with over 8,428,500 reported cases and close to 452,000 deaths.

The actual number of infected people is probably higher than the official total, as some mild and asymptomatic cases are not likely to be tested and counted.

FIGURE 2 COVID-19 PROPAGATION: SELECT COUNTRIES



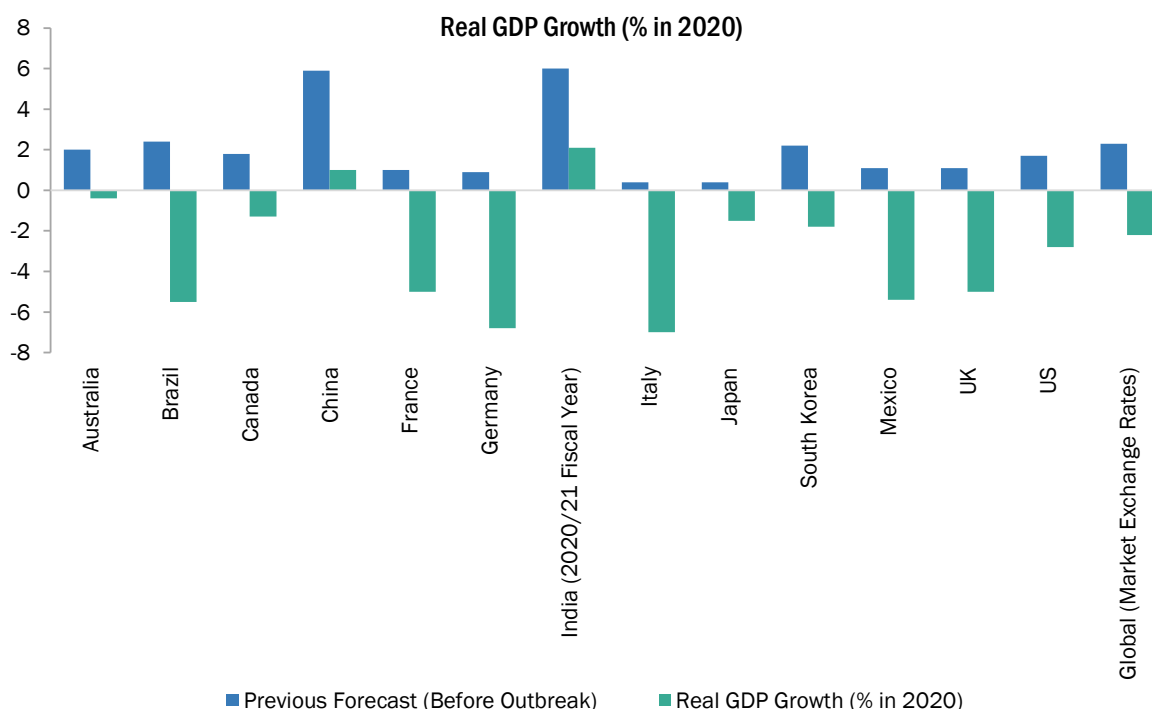
Source: WHO (World Health Organization) Coronavirus Disease 2019; data as of June 18, 2020

Disease progression patterns are similar in most countries but flattening of the curve and disease containment are largely dependent on various measures taken by each country. China and South Korea, for example, have managed to control the spread of the disease by implementing severe countermeasures, such as restricting travel; imposing city-wide quarantines, curfews, and lockdowns; building specialty hospitals; increasing medical aid; and ramping up testing. Most other countries are trying to emulate these measures to contain the spread of the disease.

1.3 COVID-19 ECONOMIC ASSESSMENT

COVID-19 has not only affected the healthcare sector but also impacted the global economy. It has had a significant economic impact on various financial as well as industrial sectors, such as energy, oil and gas, transportation and logistics, manufacturing, and aviation. It is predicted that the world economy will go into recession, due to the loss of trillions of dollars. With the increasing number of countries imposing and extending lockdowns, economic activities are declining, which would impact the global economy.

FIGURE 3 REVISED GROSS DOMESTIC PRODUCT FORECASTS FOR SELECT G20 COUNTRIES IN 2020



Source: The Economist Intelligent Unit

With the combined effects of the pandemic and the slump in global oil prices, investments are likely to contract sharply this year, especially in the energy sector. Export growth is also expected to reduce drastically. A recovery is expected in the second half of the year, provided there is no emergence of another wave of the pandemic, and countries manage to control the spread quickly and resume normalcy.

China has managed to contain the spread and is currently on track to resume normalcy. It is likely to recover in the second half of the year. However, the Eurozone, one of the hardest-hit regions, is expected to undergo a recession throughout 2020. Within the Eurozone, Germany, which is a manufacturing hub, would recover at a much slower pace than the other countries. The US economy is also likely to see a slump, with unemployment set to rise; recovery will be slow and will only be seen in the second half of the year.

1.3.1 COVID-19 ECONOMIC IMPACT—SCENARIO ASSESSMENT

Recovery from this crisis largely depends on various knowns and unknowns, which most countries will have to deal with and respond to. The speed and strength of the economic recovery will depend on the health and economic responses as well as the agility and speed of implementing various measures.

FIGURE 4 CRITERIA IMPACTING THE GLOBAL ECONOMY



In the recent past, the global economy became substantially more interconnected. The adverse consequences of various steps related to containment are evident from global supply chain disruptions, weaker demand for imported products and services, and the increase in the unemployment rate. Risk aversion has increased in the financial market, with all-time low-interest rates and sharp declines in equity and commodity prices. Consumer and business confidence have also reduced significantly.

This has magnified over time, and the spillover is seen in other countries as well. The latest forecasts by multiple sources predict a 4% decrease in the global Gross Domestic Product (GDP) in 2020 and a slow recovery, making the recession caused by this pandemic worse than the 2009 global recession. Based on these forecasts, the world economy is expected to recover by 2021; but for some countries, including the US, Germany, and Italy, recovery is predicted to take longer.

In this fast-changing environment, the full impact of the pandemic on the global economy may not entirely be known. MarketsandMarkets uses three scenario-based approaches (epidemiology and health response as well as economic response) to assess the economic impact and recovery period at the global level. Countries and regions are likely to have different impacts and recovery periods.

FIGURE 5 SCENARIOS IN TERMS OF RECOVERY OF THE GLOBAL ECONOMY

SCENARIO		
 OPTIMISTIC	 REALISTIC	 PESSIMISTIC
EPIDEMIOLOGY/CONTAGION SPREAD AND HEALTH RESPONSE		
FATALITY RATE 1.5–2%	FATALITY RATE 2–3.5%	FATALITY RATE MORE THAN 3.5%
HOSPITALIZATION RATE: LESS THAN 15%	HOSPITALIZATION RATE: 15–20%	HOSPITALIZATION RATE: MORE THAN 20%
ADDITIONAL ICU BEDS AND LESS THAN 2.5 - 5%	ADDITIONAL ICU BEDS 5%	ADDITIONAL ICU BEDS 15%
ADDITIONAL VENTILATORS 15%	ADDITIONAL VENTILATORS 15–20%	ADDITIONAL VENTILATORS 25%
ECONOMIC RESPONSE		
FISCAL STIMULUS TO THE TUNE OF USD 7.5–8 TRILLION GLOBALLY	FISCAL STIMULUS TO THE TUNE OF USD 5–7.5 TRILLION GLOBALLY	FISCAL STIMULUS TO THE TUNE OF LESS THAN USD 5 TRILLION GLOBALLY
EMERGENCY FUNDING TO SUPPORT EFFORTS TO FIGHT COVID-19 TO THE TUNE OF USD 100 BILLION GLOBALLY	EMERGENCY FUNDING TO SUPPORT EFFORTS TO FIGHT COVID-19 TO THE TUNE OF USD 75–100 BILLION GLOBALLY	EMERGENCY FUNDING TO SUPPORT EFFORTS TO FIGHT COVID-19 TO THE TUNE OF LESS THAN USD 75 BILLION GLOBALLY
MULTILATERAL AGENCIES FINANCING TO THE TUNE OF USD 100 BILLION GLOBALLY	MULTILATERAL AGENCIES FINANCING TO THE TUNE OF USD 50–75 BILLION GLOBALLY	MULTILATERAL AGENCIES FINANCING TO THE TUNE OF LESS THAN USD 50 BILLION GLOBALLY
RECOVERY SCENARIO		
▪ Milder recession in 2020 Q1 and Q2; stronger recovery in 2020 Q3 and after ▪ US: Recovery by 2020 Q3 ▪ EU5: Recovery by 2020 Q3 ▪ China: Recovery by 2020 Q2 ▪ India: Recovery by 2020 Q3	▪ Recession in 2020 Q1 and Q2; partial bounce back in 2020 Q3, then slow growth ▪ US: Recovery by 2020 Q4 ▪ EU5: Recovery by 2020 Q4 ▪ China: Recovery by 2020 Q2 ▪ India: Recovery by 2020 Q3	▪ Deeper recession in 2020 Q1, Q2, and Q3; modest rebound in 2020 Q4 ▪ US: Recovery by 2021 Q1 ▪ EU5: Recovery by 2021 Q1 ▪ China: Recovery by 2020 Q3 ▪ India: Recovery by 2020 Q4

Source: UNCTAD, Moody's, FAS, OECD, and MarketsandMarkets Analysis

With more than eight million confirmed cases of COVID-19, businesses are coping with lost revenues and disrupted supply chains as factory shutdowns and quarantine measures are spreading across the globe, restricting movement and business activities. The economic impact is mounting, and the top economies of the world are likely to enter a global recession.

To provide a perspective on various scenarios, we assume factors that would impact the economic recovery from the global recession.

In a realistic scenario, we assume that possible health and economic responses will help economies bounce back partially in 2020 Q3. In the first two quarters of 2020, countries will ramp up their testing capacities and critical healthcare services as well as increase the availability of financial aid. This will help contain the spread of the virus. The economic recession will last until the second quarter of 2020; slow recovery will be seen from the third quarter of 2020. This is likely to be a “U-shaped” economic recovery, with gradual improvement in the economy.

In an optimistic scenario, the economy is likely to recover much faster and will continue from 2020 Q3. A quick return to normalcy will be seen after the curve of infection flattens toward the end of the second quarter of 2020, due to timebound and efficient measures, both health and economic. In this scenario, most of the economies will experience a mild recession, and the growth would accelerate in 2021. This is likely to be a “V-shaped” economic recovery, with a strong and fast improvement in the economy.

In a pessimistic scenario, we assume that all measures, both health and economic, would neither help control the spread of the virus nor lead to economic recovery. Most economies would experience unprecedented economic losses, along with social and political turmoil. In this scenario, most economies will face a recession till the third quarter of 2020, and modest recovery will begin in 2020 Q4. This is likely to be an “L-shaped” economic recovery, with countries taking longer to return to their pre-crisis levels.

To assess the impact of the COVID-19 and provide market forecasts, similar scenario-based approaches have been considered in this report.

1.4 OBJECTIVES OF THE STUDY

- To define, describe, and forecast the quantum cryptography market by component (solutions and services), security type, vertical, and region
- The report includes both PRE-COVID-19 and POST-COVID-19 scenarios to analyze the quantum cryptography market. The study shows in-depth qualitative analysis along with the quantitative market estimates and forecasts for both the scenarios
- To provide detailed information about the major factors (drivers, restraints, opportunities, and industry-specific challenges) influencing the market growth
- To analyze the opportunities in the market and provide details of the competitive landscape for stakeholders and market leaders
- To forecast the market size of the segments with respect to four main regions: the Americas, Europe, Asia Pacific (APAC), and Middle East and Africa (MEA)
- To profile the key players and comprehensively analyze their market rankings and core competencies¹
- To analyze competitive developments, such as new partnerships, new contracts, and new product developments, in the quantum cryptography market

1.5 MARKET DEFINITION

Quantum cryptography is a zero-trust encryption security model that works on Heisenberg’s uncertainty principle and entangled photon states. Quantum cryptography theoretically achieves the highest level of security, thereby ensuring notification of attacks by the sender, receiver, and trusted parties. The technology is relevant for all mission-critical applications in verticals, such as government and defense; Banking, Financial Services, and Insurance (BFSI); and retail.

¹ Core competencies of the companies have been captured in terms of their key developments, key observations, and key strategies adopted by them to sustain their positions in the market

1.6 MARKET SCOPE

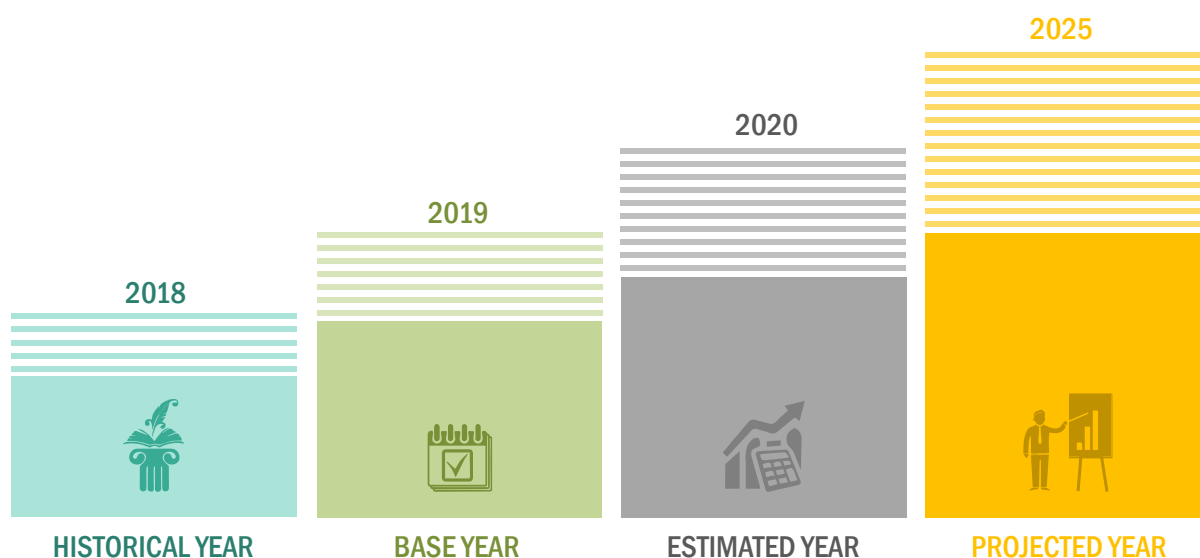
1.6.1 MARKET SEGMENTATION



*Others includes telecom, energy and utilities, logistics, and education

Source: MarketsandMarkets Analysis

1.6.2 YEARS CONSIDERED FOR THE STUDY



Note: The forecast period taken under consideration is 2018–2023

1.7 CURRENCY CONSIDERED

The base currency used in the report is the United States Dollar (USD), with the market size indicated in USD million/billion.

- For companies that reported their revenues in USD, the revenues were sourced from their annual reports.
- For companies that reported their revenues in other currencies, the average annual currency conversion rate was used for that particular year to convert the value to USD.

1.8 STAKEHOLDERS

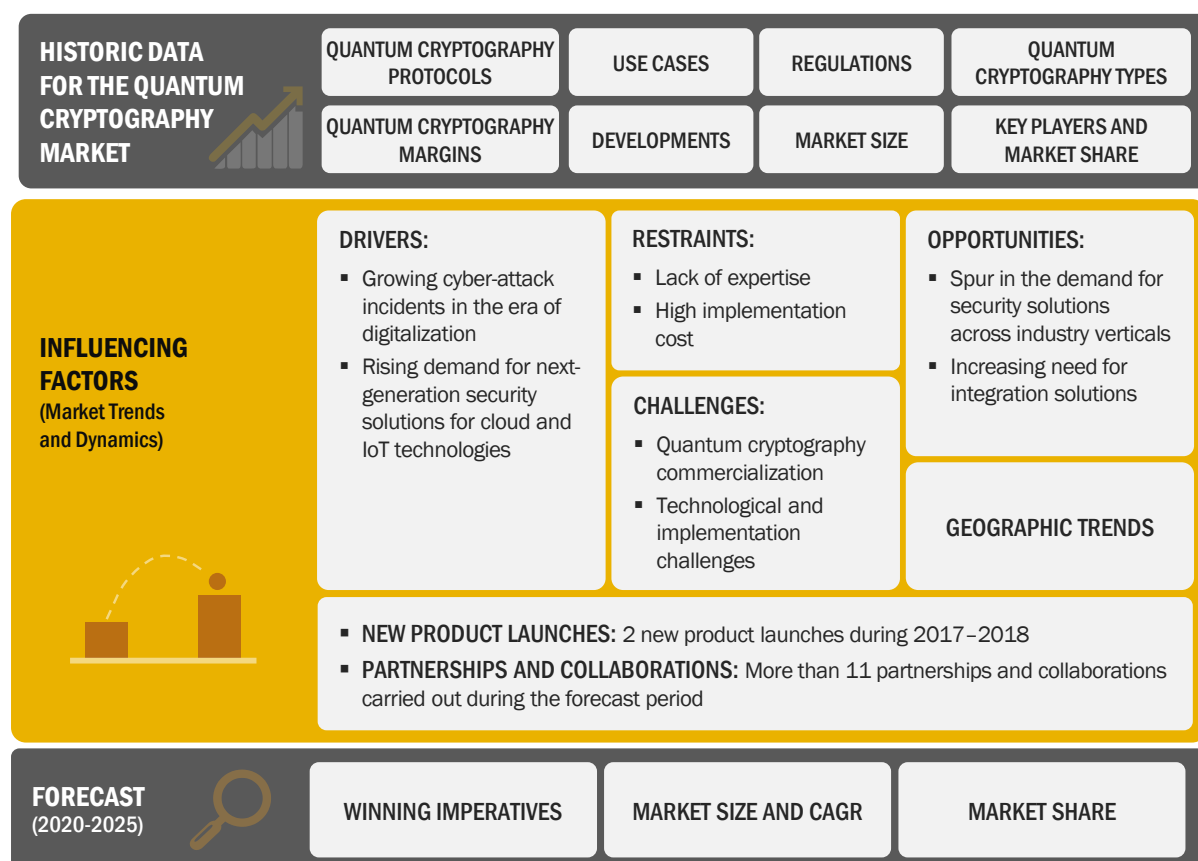
- Encryption solution providers
- Quantum cryptography vendors
- Application security service providers
- System Integrators (SIs)/network security service providers
- Consultancy/advisory firms
- Quantum computing and quantum cryptography R&D firms

2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

This research study involved the use of extensive secondary sources, directories, and databases, such as Factiva, D&B Hoovers, and Bloomberg, to identify and collect information useful for this comprehensive market research study on the quantum cryptography market. The primary sources were mainly industry experts from the core and related industries, preferred quantum cryptography solution and service providers, third-party service providers, consulting providers, end users, and other commercial enterprises. In-depth interviews were conducted with various primary respondents, including key industry participants and subject matter experts, to obtain and verify critical qualitative and quantitative information as well as assess the prospects. The following illustrative figure highlights the market research methodology applied to make this report on the global quantum cryptography market:

FIGURE 6 QUANTUM CRYPTOGRAPHY MARKET: RESEARCH DESIGN



Source: MarketsandMarkets Analysis

2.1.1 SECONDARY DATA

In the secondary research process, various secondary sources were referred to for identifying and collecting information for this study. The secondary sources included companies' annual reports, press releases, and investor presentations; and white papers, certified publications, and articles from recognized associations and government publishing sources.

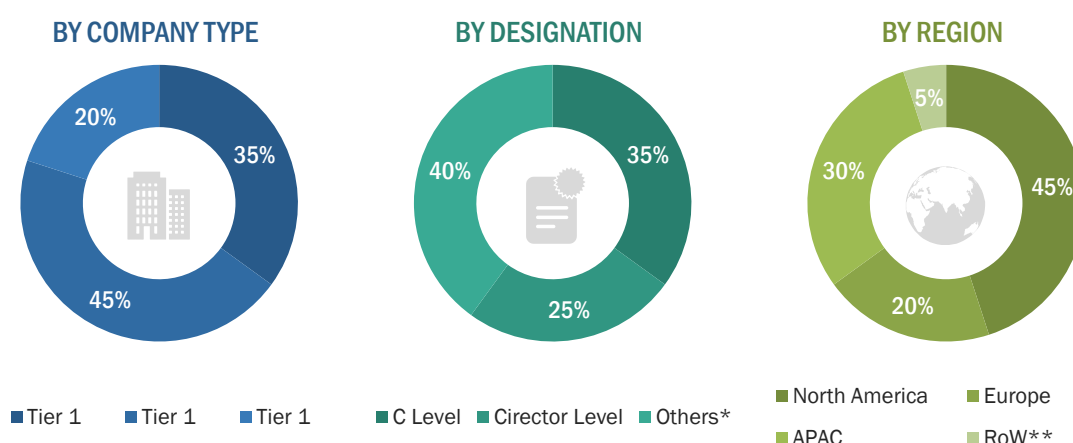
Secondary research was mainly used to obtain key information about the industry's value chain, the total pool of key players, market classification and segmentation according to industry trends to the bottom-most level, regional markets, and key developments from both market- and technology oriented perspectives.

2.1.2 PRIMARY DATA

In the primary research process, various primary sources from both supply and demand sides were interviewed to obtain qualitative and quantitative information for this report. The primary sources from the supply side included industry experts, such as Chief Executive Officers (CEOs), Vice Presidents (VPs), marketing directors, technology and innovation directors, and related key executives from various key companies and organizations providing quantum cryptography solutions and associated services. The primary sources from the demand side included end-users, such as Chief Information Officers (CIOs), consultants, service professionals, technicians and technologists, and managers at public and investor-owned utilities. After the complete market engineering (including calculations for market statistics, market breakdown, market size estimations, market forecasting, and data triangulation), extensive primary research was conducted to gather information, and verify and validate the critical numbers arrived at. Primary research was also conducted to identify the segmentation types; industry trends; key players; the competitive landscape of different market players; and key market dynamics, such as drivers, restraints, opportunities, challenges, industry trends, and key strategies.

In the complete market engineering process, both top-down and bottom-up approaches were extensively used, along with several data triangulation methods to perform market estimation and market forecasting for the overall market segments and subsegments listed in this report. Extensive qualitative and quantitative analysis was performed on the complete market engineering process to list key information/insights throughout the report.

2.1.2.1 Breakup of primary profiles



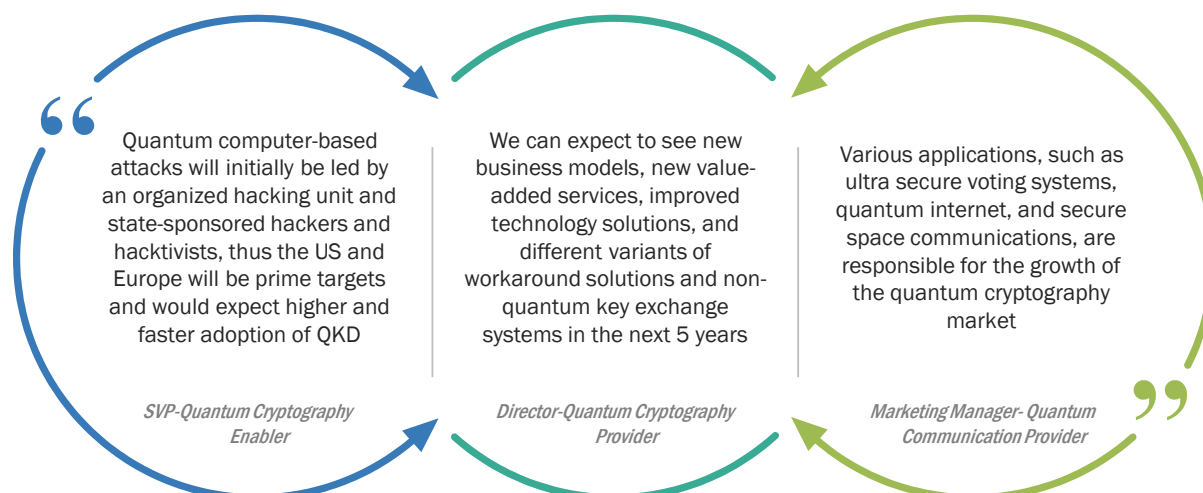
Note: Tier 1 companies' total annual revenues > USD 10 billion; Tier 2 companies' total annual revenues = USD 1 – 10 billion; and Tier 3 companies' total annual revenues = USD 500 million – 1 billion of the overall revenue

*Others include marketing managers, sales managers, and product managers

**RoW includes Latin America and MEA

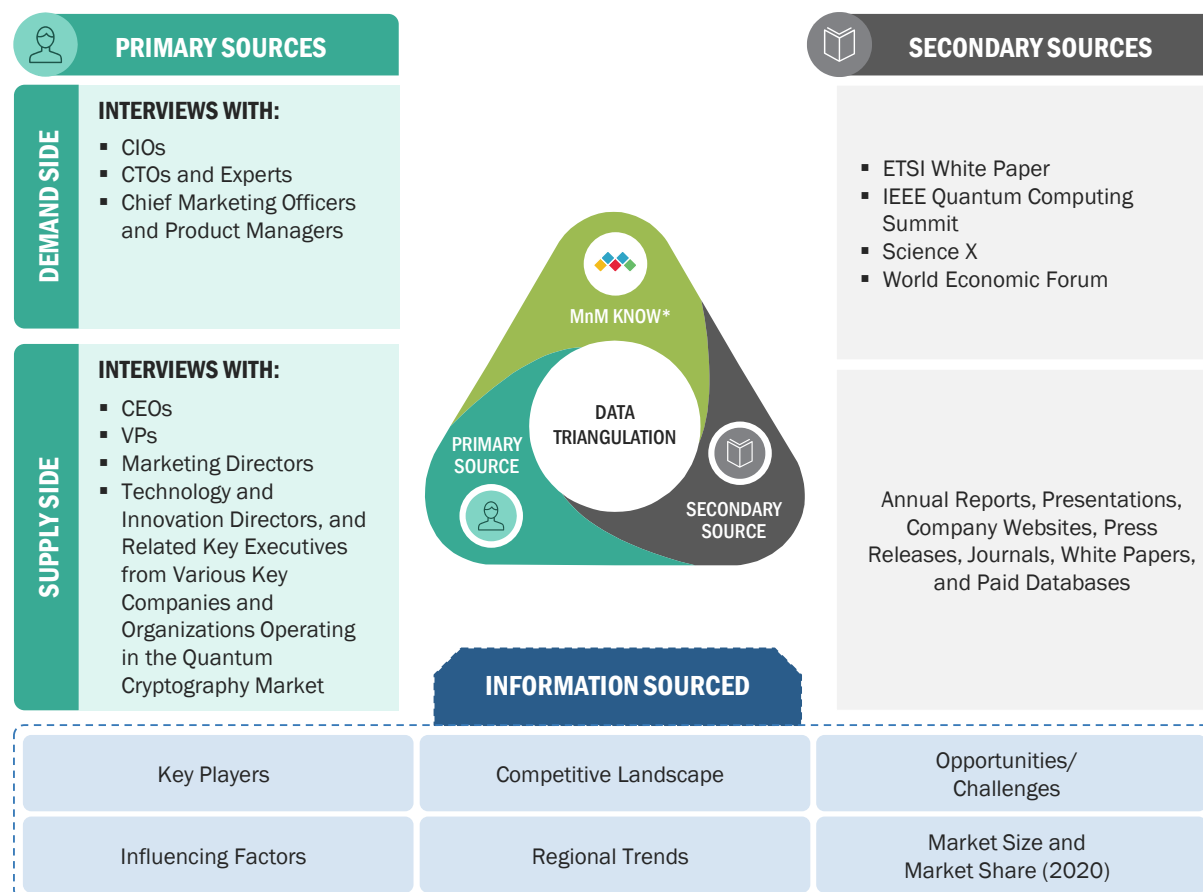
Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

2.1.2.2 Key industry insights



Source: Industry Experts

2.2 MARKET BREAKUP AND DATA TRIANGULATION



*MnM KNOW stands for MarketsandMarkets' 'Knowledge Asset Management' framework. In this context, it stands for existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform "Knowledge Store", subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources.

Source: MarketsandMarkets Analysis

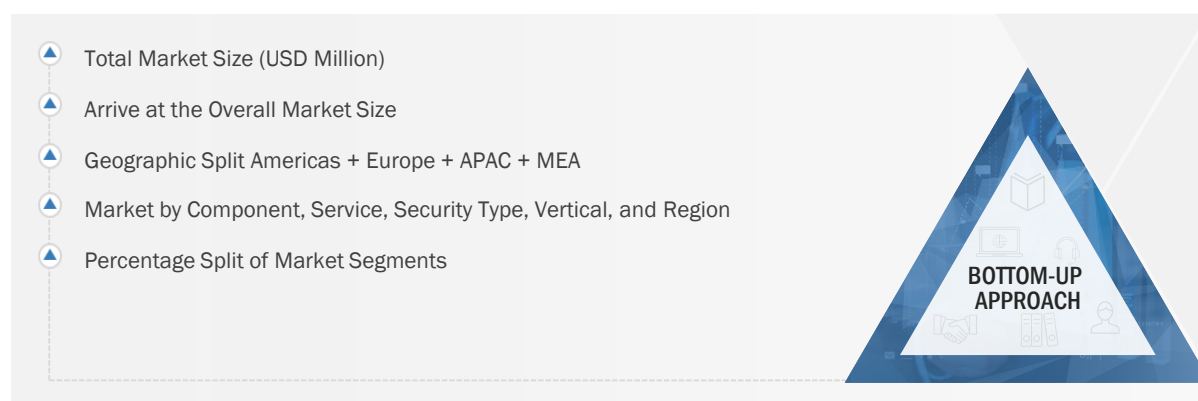
2.3 MARKET SIZE ESTIMATION

The top-down and bottom-up approaches were used to estimate and validate the market size of the global quantum cryptography market and estimate the market size of the various other dependent submarkets. The research methodology used to estimate the market size included the following details: the key players in the market were identified through secondary research, and their market shares in the respective regions were determined through primary and secondary research. This entire procedure included the study of the annual and financial reports of the top market players and extensive interviews for key insights from industry leaders, such as CEOs, VPs, directors, and marketing executives. All percentage shares, splits, and breakdowns were determined using the secondary sources and verified through the primary sources. All possible parameters that affect the market covered in this research study have been accounted for, viewed in extensive detail, verified through primary research, and analyzed to get the final quantitative and qualitative data.

The following figures are an illustrative representation of the overall market size estimation process employed for this study:

2.3.1 BOTTOM-UP APPROACH

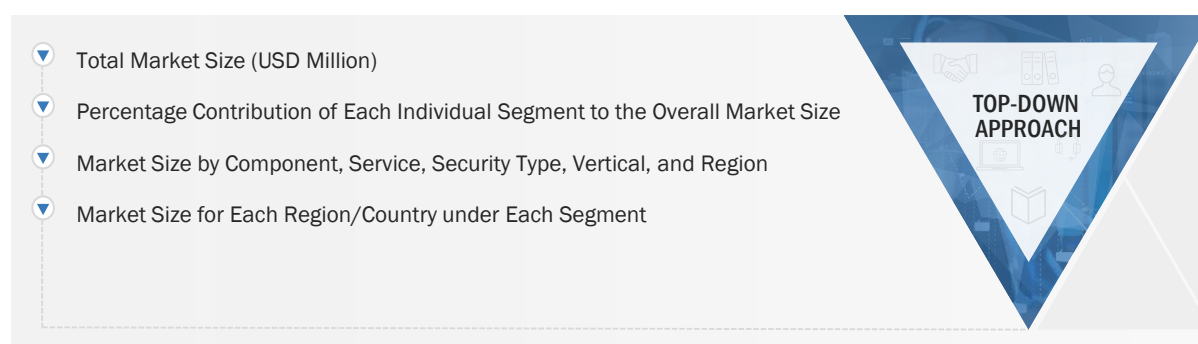
FIGURE 7 QUANTUM CRYPTOGRAPHY MARKET: BOTTOM-UP APPROACH



Source: MarketsandMarkets Analysis

2.3.2 TOP-DOWN APPROACH

FIGURE 8 QUANTUM CRYPTOGRAPHY MARKET: TOP-DOWN APPROACH



Source: MarketsandMarkets Analysis

The bottom-up procedure was employed to arrive at the overall market size of the quantum cryptography market from the revenues of the key players (companies) and their market shares. This calculation was done based on estimations and by verifying their revenues through extensive primary interviews.

Calculations based on the revenues of the key companies identified in the market led to the overall market size. This overall market size was used in the top-down procedure to estimate the size of the other individual segments [components (solutions and services), security types, verticals, and regions] via percentage splits of market segments from the secondary and primary research. The bottom-up procedure was also implemented for the data extracted from secondary research to validate the market segment revenues obtained.

Market shares were then estimated for each company to verify the revenue shares used earlier in the bottom-up procedure. With the data triangulation procedure and validation of data through primaries, the exact values of the overall parent market size and each individual market size were determined and confirmed in this study.

2.4 MARKET FORECAST

Factors that have a direct or indirect impact on the market were analyzed to estimate the overall trend of the market during the forecast period. Each factor was analyzed on a scale of 1–3, 1 being the lowest, and 3 being the highest. After analyzing each factor and its impact on the market, different forecasting techniques were used to arrive at the market size.

TABLE 1 FACTOR ANALYSIS

FACTOR	INFERENCE	IMPACT
Recent Developments	Recent developments, such as mergers and acquisition, partnerships are increasing and are more in 2018 as compared to 2017	High
Regulations	Rules and regulations laid down by various governing bodies for various industries directly impact the market	Low
IoT Adoption	IoT adoption is increasing at a high rate	High
Mobile Users	The number of mobile users is increasing at a moderate pace	Medium
Government Initiatives	There are several government initiatives taken by various governments over the last three years. Hence the impact would be high	High
Security Spending	Security spending is increasing at a high rate for the last three years. Hence the impact would be high	High
Technology Maturity	Technology Maturity of the quantum cryptography market: The technology is in the introductory phase.	High

Source: MarketsandMarkets Analysis

2.5 ASSUMPTIONS FOR THE STUDY

The following assumptions were taken into consideration to complete the overall market engineering process for the quantum cryptography market:

PARAMETER	DESCRIPTION
GLOBAL QUANTUM CRYPTOGRAPHY MARKET SIZE	Market size estimates are conservative/realistic as opposed to being optimistic/aggressive.
EXCHANGE RATE	Dollar fluctuations are not expected to be serious enough to affect the forecasts to a significant extent. Fall in the value of dollar is expected to enhance revenue growth opportunities for the US multinationals, whereas stability in the value of dollar is expected to stabilize the price of imports and exports.
QUANTITATIVE ASSUMPTION	The market size may not add up to the total, as it has been rounded off.
GLOBAL ECONOMIC CONDITION	The world economy is assumed to be stable, exhibiting a strong positive growth. This is also expected to apply to the economies of countries.
FINANCIAL STATISTICS	All revenues and financial insights were extracted from the company websites or annual reports, which are true to the best of MarketsandMarkets' knowledge.

Source: MarketsandMarkets Analysis

2.6 LIMITATIONS OF THE STUDY

- The contribution of the research companies mentioned in the company profiling section was not included in revenue calculation.
- The study does not include the market numbers of 2016, due to lack of quantum cryptography commercialization.

3 EXECUTIVE SUMMARY

In the era of the Internet of Things (IoT), cloud computing, and Artificial Intelligence (AI), data generation is increasing at a rapid pace. Encryption is one of the essential security techniques for the security of databases, networks, and applications. Quantum cryptography solutions play an important role in achieving and securing multiple entities. The traditional encryption techniques that make use of computational complexity are being used in today's scenario and have been relied on for more than the past two decades. However, these traditional techniques were not designed to prevent quantum attacks at the time of their invention, when research into quantum computation was unknown to most cryptographic experts. New cryptographic techniques are going to replace the traditional ones to provide security against quantum threats. These techniques are known as "quantum-safe" and comprise properties of light that prevent interception of messages, as well as follow standard computational procedures.

Furthermore, this system was designed to fight quantum attacks, which are evolving due to an increase in the practices of quantum computation. Quantum computers can decode conventional cryptographic information in a short timeframe, and they pose potential security threats to the organizational data, including bank account details, identity information, and other sensitive data. Hence, to move a step ahead in attaining a zero-trust security model, researchers are working toward the development of quantum cryptography solutions and services.

Many industries practice cryptographic techniques and fielded systems as a component of the network security system. These security products need to be upgraded with quantum-safe cryptographic techniques. One of the crucial factors to be considered here is the cost of transitioning into quantum-safe technologies. Usually, new technologies tend to follow a typical cycle of innovation where early adopters pay high premiums, and after a while, commoditized product offerings are available with ample competition. Quantum cryptography techniques cannot be expected to work out with the prevailing hardware systems present in most of the organizations, but in the time to come, it is expected that an ordered and cost-efficient technology transition would take place, where the new products would gradually phase-in, and old ones would be phased out. The quantum cryptography market is gaining traction due to the sharp surge in cyber-attacks over the network and endpoint communications devices.

Additionally, the outlook of organizations and government bodies toward securing the communications between businesses to obtain a smooth and secure flow of information among trusted parties is increasing. The complex architecture of communication networks and their unique characteristics make it mandatory to secure the information with the utmost level of security. The heterogeneity of endpoint devices has been the crucial aspect of network security and demands a trusted solution to ensure an unbreakable encryption solution. Quantum cryptography can provide organizations with such competent and unbreakable security.

The quantum cryptography market can be segmented into solutions and services. Quantum cryptography solutions segment includes the revenue generated from Quantum Key Distribution (QKD) platforms, QKD servers, virtual encryptors, QKD distributors, key and policy managers, software development kit (SDK) to integrate cryptography, quantum cryptography communication device, quantum security gateway, Quantum Random Number Generator (QRNG), hacker box, quantum secure communication network products, and repeaters. The services segment includes consulting and advisory, deployment and integration, and support and maintenance. Furthermore, the quantum cryptography market also includes verticals and regions. The verticals comprise government and defense, BFSI, retail, healthcare, automotive, and others (telecom, energy and utilities, logistics, and education). The government and defense vertical is estimated to hold the highest market share in 2019 and is expected to grow significantly over the next five years. BFSI and healthcare verticals are expected to grow at a rapid pace as compared to other verticals during the forecast period. Among regions, the Americas is estimated to hold the highest market share in 2019 and is projected to grow at the highest CAGR during the forecast period.

The growth of the Americas is attributed to a large number of government initiatives and heavy investments toward the research of quantum computing. Speaking about the commercialization of quantum cryptography solutions, the verticals that require highly critical data security would be the first adopters. These verticals include government, defense, and BFSI followed by high transactional verticals such as retail. Furthermore, verticals such as healthcare, telecom, energy and utilities, and logistics and transportation would be the adopters of quantum cryptography.

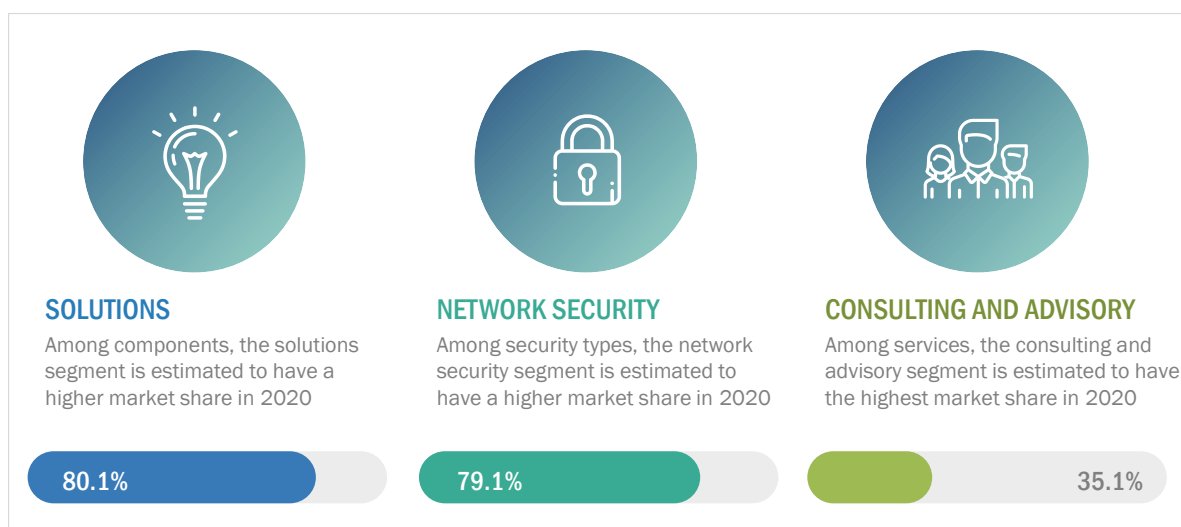
Following are the key factors driving the growth of the quantum cryptography market:

- **IoT adoption:** According to Cisco, a leading IoT provider, the number of IoT connections is expected to grow from 21.7 million in 2017 to 155.7 million by 2024.
- **Mobile subscriber penetration:** According to Global System for Mobile Communications Association (GSMA) Intelligence, the number of mobile subscribers is increasing at a significant rate.
- **Recent developments:** Partnerships, collaborations, and new product launches are increasing every year. There were 4 deals in 2016, 5 deals in 2017, and 8 deals in 2018.
- **Government initiatives:** There are several initiatives taken by various governments, such as the US National Quantum Initiative, USD 218 million investment initiative by the US Department of Energy, US quantum initiatives with Department of Energy (DoE), National Science Foundation (NSF) funding and house bill, Quantum Canada initiative, UK and Singapore collaboration initiative to invest USD 12 million for developing next-generation quantum communications networks, and China's development of quantum research supercenter.
- **Security spending:** Worldwide security spending is increasing significantly. The highest growth in spending has been witnessed in cloud security, application security, and data security.

REGION	SUBSCRIBER PENETRATION IN 2016	SUBSCRIBER PENETRATION BY 2020
Europe	84%	87%
Commonwealth of Independent States (CIS)	78%	83%
North America	81%	85%
Latin America	70%	78%
APAC	65%	76%
Middle East and North Africa (MENA)	59%	62%
Sub-Saharan Africa	44%	48%

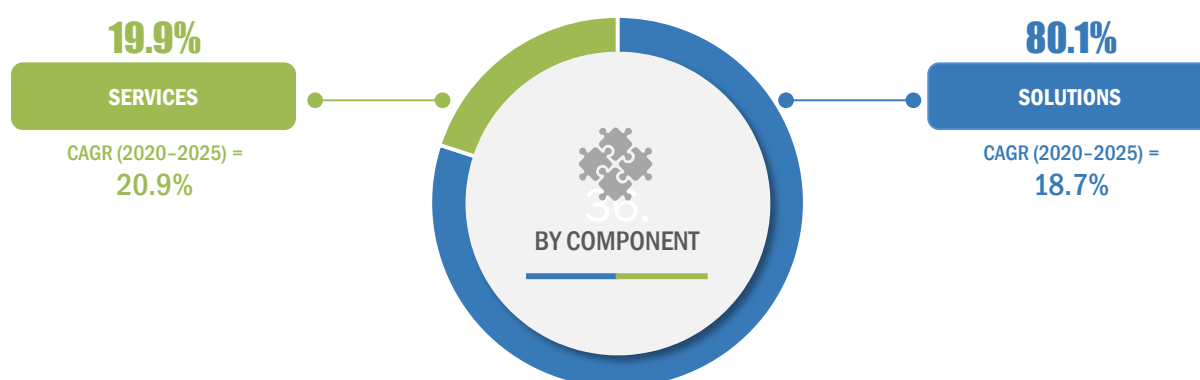
Source: GSMA Intelligence Report 2017

FIGURE 9 QUANTUM CRYPTOGRAPHY MARKET, TOP THREE SEGMENTS, 2020



Source: Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 10 SOLUTIONS SEGMENT TO HAVE A HIGHER MARKET SHARE IN 2020



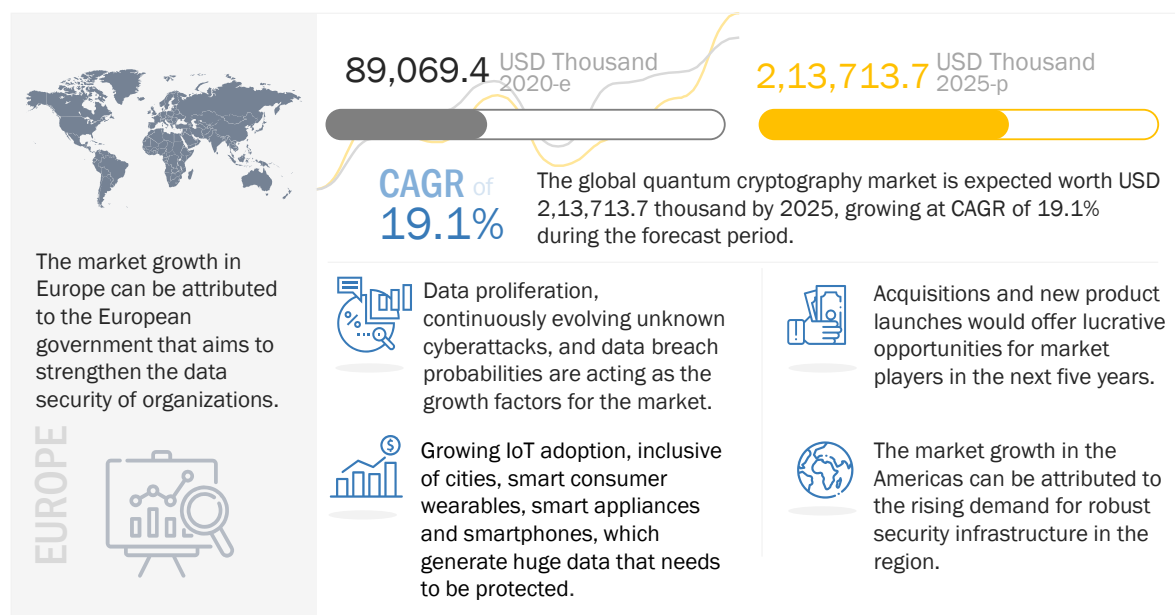
Note: The percentage figures represent the market share in 2020

Source: Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4 PREMIUM INSIGHTS

4.1 ATTRACTIVE OPPORTUNITIES IN THE QUANTUM CRYPTOGRAPHY MARKET

FIGURE 11 INCREASING DEMAND FOR CUSTOMIZED BUSINESS SUPPORT SYSTEM SOLUTIONS AND SERVICES TO DRIVE MARKET GROWTH

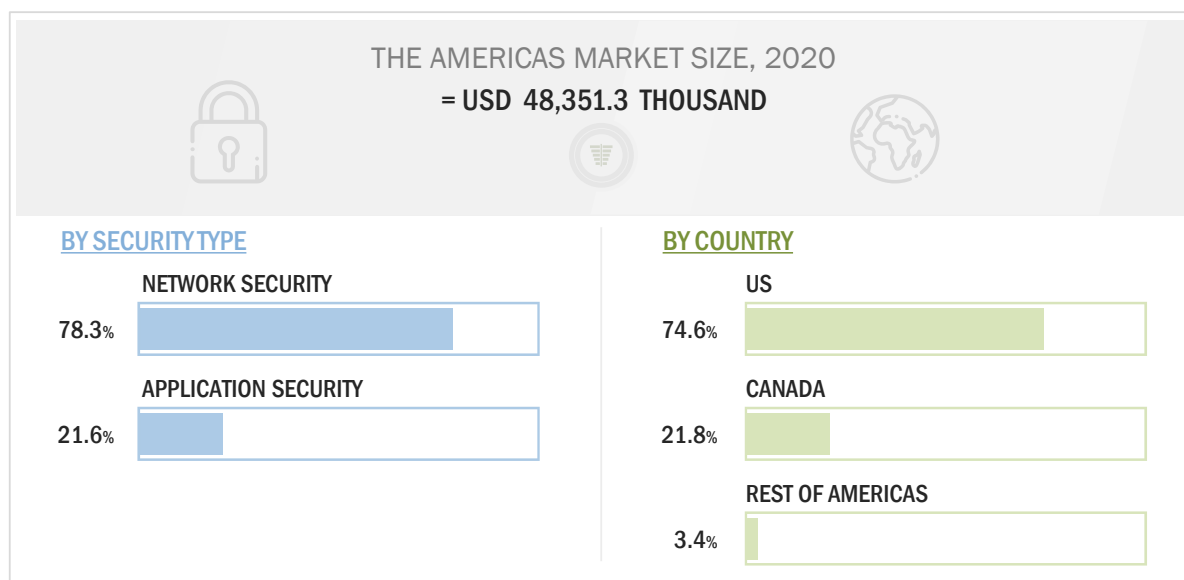


e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

4.2 QUANTUM CRYPTOGRAPHY MARKET IN THE AMERICAS, BY SECURITY TYPE AND COUNTRY

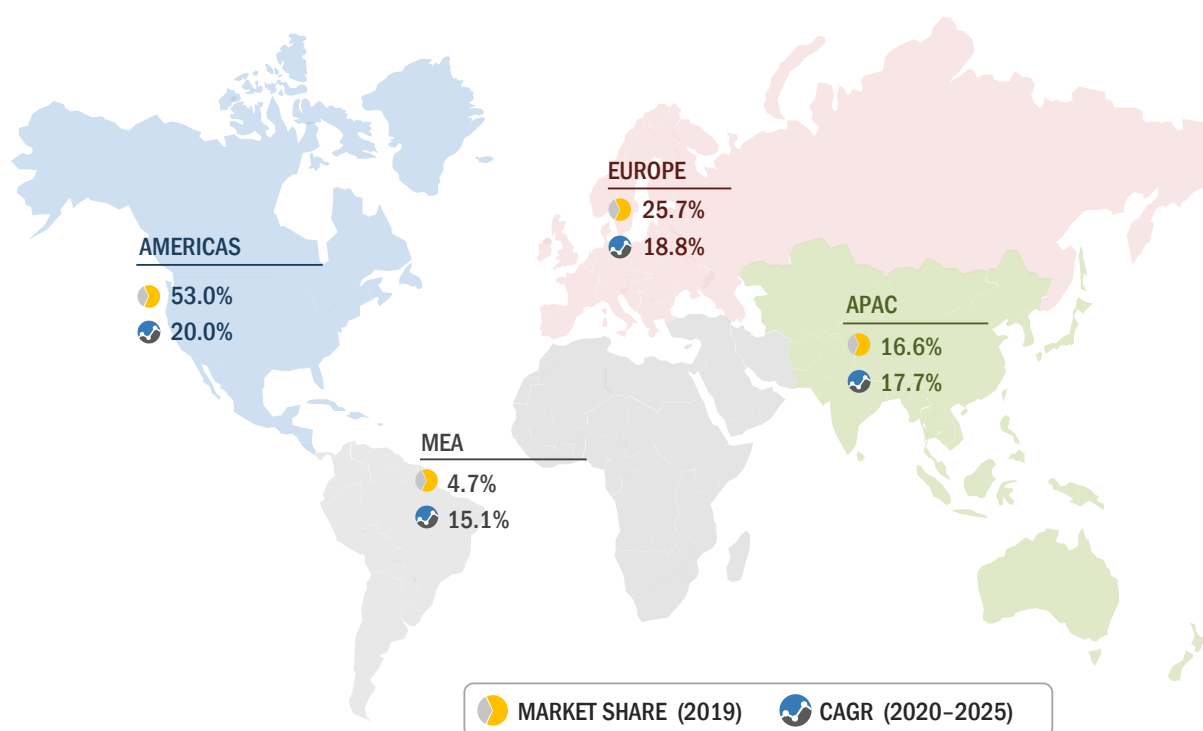
FIGURE 12 NETWORK SECURITY TYPE AND UNITED STATES TO HOLD HIGH MARKET SHARES IN 2020



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

4.3 QUANTUM CRYPTOGRAPHY MARKET: INVESTMENT SCENARIO

FIGURE 13 THE AMERICAS TO EMERGE AS THE BEST MARKET FOR INVESTMENT OVER THE NEXT FIVE YEARS



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

5 MARKET OVERVIEW AND INDUSTRY TRENDS

5.1 INTRODUCTION

The global quantum cryptography market analysis consists of components (solutions and services), security types, verticals, and regions. The quantum cryptography market is a very lucrative market for cryptography providers, as organizations across various verticals around the globe are progressively adopting quantum-safe security solutions. Furthermore, growing incidents of cyber attacks, and increasing data security and privacy concerns are expected to drive the market's growth. The market is diverse and competitive with the presence of many key players, such as QuintessenceLabs, NuCrypt, Qasky, Crypta Labs, Qubitekk, Post-Quantum, MagiQ Technologies, ISARA, ID Quantique, QuNu Labs, QuantumCTek, Quantum Xchange, Aurea Technologies, Infineon, Mitsubishi Electric, IBM, HP, NEC, Toshiba, and Microsoft.

5.2 MARKET DYNAMICS

The quantum cryptography market is in the initial phase, as the adoption of quantum cryptography solutions is at the peak only in the Americas and Europe. The major factors such as the growing incidents of cyber attacks in the era of digitalization, increasing cybersecurity funding, rising demand of next-generation security solutions for cloud and IoT technologies and evolving next-generation wireless network technologies are expected to drive the growth of the global quantum cryptography market. However, lack of expertise may limit the market growth. On the other hand, there are significant growth opportunities for quantum cryptography vendors. Factors such as spur in demand for security solutions across industry verticals and increasing need for integration solutions are expected to shape the future of the quantum cryptography market. Moreover, issues related to quantum cryptography commercialization, and technological and implementation challenges may act as challenges for the market growth.

FIGURE 14 DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES: QUANTUM CRYPTOGRAPHY MARKET



Source: Secondary Literature, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

5.2.1 DRIVERS

5.2.1.1 Growing cyber attacks in the era of digitalization

Digitalization offers various opportunities to optimize systems and operations across industry verticals. The increasing number of connected devices is giving rise to complexities in cyber risk management and also unwanted cyber incidents across the globe. As a result, cyber attacks are occurring at a greater frequency and are targeting businesses and governments. Countries, including the US, Turkey, China, Brazil, Pakistan, Algeria, India, and Europe are the main targets of cyber-attackers.

TABLE 2 GROWING NEED FOR QUANTUM CRYPTOGRAPHY WITH THE INCREASING NUMBER OF CYBER ATTACKS

YEAR	CYBERATTACKS
2011–2014	▪ India witnessed a 350% rise in cybercrime cases
2012–2015	▪ In the US, cyber incidents increased by approximately 50% against its critical infrastructure
2016	▪ According to the US Federal Bureau of Investigation (FBI) report, since 2016 more than 4,000 ransomware attacks occur daily
2017	▪ The WannaCry ransomware attack targeted more than 45,000 systems in the UK, Russia, China, Italy, Egypt, and India ▪ HACKMAGEDDON's Cyber Attacks Statistics showed that 77.4% of the attackers were motivated by cybercrimes

2018 (As of October)

- HACKMAGEDDON's Cyberattacks Statistics showed that 77.9% of the attackers were motivated by cybercrimes. Malware/ Point-of-Sale (POS), account hijacking, targeted attack, and others are the major cyber attacks recorded in 2018

Source: MarketsandMarkets Analysis, Kenes Exhibitions, HACKMAGEDDON, and World Economic Forum

Security is an important aspect. Hence the keys and digital signatures of quantum cryptography can be easily implemented in the traditional cybersecurity framework. Due to this threat landscape of cyber attacks, there is a huge adoption of quantum cryptography hardware and services.

5.2.1.2 Increasing cybersecurity funding

With the increasing number of systems, many security and privacy issues will arise, and every endpoint, gateway, sensor, and smartphone will become a potential target for hackers. There is a need for security and privacy protection in enterprises. Many governments across the world are introducing cybersecurity investment funds to reduce the risk of cyber incidents. In addition to this, according to the Cisco Utility and Energy Security report, approximately 73% of IT security professionals at utilities have faced a public security breach. To reduce the risk of cyber incidents in utilities, many governments are investing in enhancing the infrastructure and adopting emerging technologies.

TABLE 3 QUANTUM CRYPTOGRAPHY INITIATIVES

YEAR	FUND (USD)	DESCRIPTION
2017	USD 25.76 million	In the UK, the Department for Digital, Culture, Media, and Sport (DCMS) launched a cybersecurity training program to help address the problem of skills shortage across the region.
2018	USD 28 million	The US Department of Energy (DOE) announced USD 28 million support to the Research, Development, and Demonstration (RD&D) of next-generation tools and technologies to enhance the cybersecurity and resilience of the US' critical energy infrastructure.
2019	USD 15 billion	The 2019 US President's Budget includes budget authority for cybersecurity-related activities.

Source: MarketsandMarkets Analysis, CBR Government, DCMS, and US Government Publishing Office

5.2.1.3 Rising demand for next-generation security solutions for cloud and IoT technologies

Cloud services have become important due to the rise in high-capacity networks as well as the decreased cost of computers and data storage devices. Cloud storage enables users to utilize, centralize, and share resources (both hardware and software) over a network. Cloud computing is linked to cloud storage to move data to the cloud and access it from multiple devices and locations. Moreover, it helps reduce business costs and optimize the use of computing power distributed across many users and businesses. As cloud computing offers the advantage of having a centralized IT security management system across multiple applications and systems, it affects the security of the enterprise IT system. Quantum cryptography using the QKD technique secures the data transmitted over a public network between the user and the cloud.

Moreover, it provides strong encryption, quantum-safe server, endpoint encryption, and network infrastructure security. Similarly, IoT is becoming an important part of our lives, and has found a new way to address different problems of industry verticals. However, it is very much vulnerable to different security threats. The current security level is not very strong for the future applications of IoT, as the acquisition of data is increasing at a rapid pace. Researchers are working toward the development of quantum cryptography solutions to address the cybersecurity threats in IoT. For instance, in January 2018, SecureRF

Corporation, a provider of quantum-resistant security tools announced that Walnut Digital Signature Algorithm (WalnutDSA) has been accepted by the US Commerce Department's National Institute of Standards and Technology (NIST) for the evaluation of their post-quantum standardization project.

5.2.1.4 Evolving next-generation wireless network technologies

Advancements in high-speed wireless network technologies, such as 4G, 5G, and Wi-Fi, have become ubiquitous. With the increasing penetration of connected devices, deployments of enhanced wireless technologies are also increasing rapidly. Moreover, the connectivity cost has dropped significantly over the past few years, and these developments offer a significant opportunity to telecom operators to enhance and secure their network capabilities using quantum cryptography. The upcoming 5G technology is expected to contribute to the increased adoption of quantum cryptography solutions and provide an opportunity for telecom companies to increase their revenue and business opportunities by securing their network and making it more reliable for their clientele. Implementing quantum cryptography in a wireless network minimizes the environmental changes that impact quantum missions in Wi-Fi networks. Moreover, it provides unconditional security to combine with the Institute of Electrical and Electronics Engineers (IEEE) 802.11 networks and observe the behavior of the modified key distribution process under various input conditions.

5.2.2 RESTRAINTS

5.2.2.1 High implementation cost

Although the implementation of quantum cryptography solutions increases the safety of communication, these solutions are expensive and time-consuming to implement, and this results in less adoption of quantum cryptography solutions. Their implementation cost increases with increasing distance, thus, organizations have to pay huge charges for implementing and maintaining these solutions and services, and this becomes a burden for them. For instance, quantum cryptography systems cost around USD 50,000 to connect only two parties at a time. Therefore, high installation costs act as restraints for the growth of the quantum cryptography market.

5.2.2.2 Lack of expertise

The key to an effective and efficient quantum cryptography solution is to exchange the keys securely. Deploying the quantum cryptography solution can be extremely difficult and time-consuming in large advanced complex environments. There is a lack of qualified and skilled IT professionals and data scientists. According to the Information Systems Security Association (ISSA) and Enterprise Strategy Group (ESG) Survey Result 2016, the security analysis and investigations sectors suffered approximately 31% cybersecurity skills shortage. Additionally, approximately 43% of organizations are impacted by the global shortage of cybersecurity skills.

Most Small and Medium-sized Enterprises (SMEs) are not willing to pay the expected salaries to cybersecurity professionals; as a result, they hire and train their junior employees to manage cyber-related issues. The high level of complexity related to the maintenance and support of quantum cryptography solutions creates a high requirement among companies to pay high salaries to a small number of IT professionals and data scientists, who can identify and deploy the quantum cryptography server. These factors may affect market growth, as organizations are not able to find the appropriate talent. For small enterprises, it is difficult to pay high salaries. Additionally, training employees for the development of the required skills would result in additional costs for an enterprise. Therefore, it is important for an enterprise to estimate and evaluate the cost of implementation.

5.2.3 OPPORTUNITIES

5.2.3.1 Spur in demand for security solutions across industry verticals

The BFSI and government verticals are adopting security solutions to secure their transactions. Online businesses are increasing rapidly, due to which there is an increase in daily transactions. Hence, for the security of these transactions, quantum cryptography is required. Financial institutions provide the facility of online banking and online transactions to their customers, which is again expected to boost payment security, which is driven by next-generation quantum cryptography solutions. In banking transactions and applications, it becomes important to protect the client and proprietary information. Moreover, as the government firms need to transmit enormous amounts of confidential data securely at a global scale, the possible chances of cyber attacks increase. Quantum cryptography offers the best practices in the key generation for highly secure crypto operations and banking solutions. It generates keys for security applications and crypto operations, such as authentication, digital signatures, and secure access control. Furthermore, it helps protect mission-critical data for the government and provides high-performance layer encryption. Increase in demand for security solutions across verticals is expected to provide business opportunities for the quantum cryptography market.

5.2.3.2 Increasing need for integration solutions

Rising adoption of virtual and cloud storage among organizations has further added to the diversity of devices that are being used for data storage. Moreover, various operating systems have further challenged IT resources for the provision of data protection. The security of mobile transactions is obscure to most people but it is essential if they are to stay protected from malicious online attacks, frauds, and thefts. The adoption of various solutions for data protection involves fragmentation, thus generating increased demand for integrated solutions that can function across diverse platforms at low costs and with low IT infrastructure. Integration of quantum cryptography enables the robust exchange of quantum information. This would create a widespread opportunity for quantum cryptography solution providers to offer integrated solutions.

5.2.4 CHALLENGES

5.2.4.1 Quantum cryptography commercialization

Commercializing quantum cryptography is a challenge due to the lack of customer awareness, as knowledge is one of the key drivers for economic growth. For the commercialization of high-technology innovations, the enterprise commercialization team must identify, obtain, combine, and manage the required technological knowledge. Moreover, innovation could be successful if the innovation team can adhere to the employees' learning paths and create and maintain a good network. As quantum cryptography is a new concept, various challenges are also surfacing, such as failure to obtain sufficient and relevant market information, failure to use the information properly, insufficient knowledge about the market, and the inability to locate local and international sales and distributions centers. Therefore, proper customer awareness of quantum cryptography solutions is needed to implement the deployment of quantum cryptography solutions successfully. These are some of the major challenging factors that are affecting the market growth.

5.2.4.2 Technological and implementation challenges

The developed and known application of quantum cryptography is QKD. QKD systems are becoming a reality and play an important role in some technological and implementation challenges. These challenges are due to the fact that algorithms cannot be implemented easily. Moreover, it is very expensive to deploy quantum cryptography on normal computers. In addition to this, quantum cryptography has some technological issues. Issues such as a change in polarization, where a presence of paper clip can also disturb the information; need for dedicated channels; distance, where increasing distance would increase the deployment cost and operational risk. Many national government and funding agencies support quantum computing research to develop quantum computers for both civilian and national security

purposes. Companies such as Microsoft, IBM, NEC, Toshiba, Infineon, and HP are in their research phase to build quantum computers, which, in turn, would increase the demand for quantum cryptography. For instance, the Microsoft Quantum Research group in 2017 started understanding about post-quantum cryptography and securing communication techniques for quantum computing. Furthermore, HP Quantum Information Processing (QIP) group is focusing on three areas of quantum, namely, quantum computation, quantum cryptography, and quantum teleportation and communication. Quantum computers are expected to offer great market opportunities and are expected to increase the demand for quantum cryptography during the forecast period.

5.3 COVID-19 PANDEMIC-DRIVEN MARKET DYNAMICS AND FACTOR ANALYSIS

TABLE 4 DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES: QUANTUM CRYPTOGRAPHY MARKET

DRIVERS AND OPPORTUNITIES	REASON	YEAR	COVID-19 IMPACT
Growing cyber attacks	Threat vectors are increasingly trying to exploit vulnerabilities as organizations are adopting the work from home culture to fight against the COVID-19 pandemic. With the increase in access to sensitive data through VPN and lack of multifactor authentication systems in place, cybercriminals can easily leverage social engineering to enter organizations' networks.	2020	(+) 9-10%
	Cybercriminals and hackers are taking advantage of lack of awareness among the employees on security policies and best practices to extract the sensitive and properties of the organization.	2021	(+) 8-9%
	Due to this threat landscape of cyberattacks, there is a huge adoption of quantum cryptography hardware and services.		
Rising demand for next-generation security solutions for cloud	Cloud services have become important due to the rise in high-capacity networks as well as the decreased cost of computers and data storage devices. The rise in the adoption and migration toward cloud services is raising a key concern with respect to security tools.	2020	(+) 7-8%
	Most of the researcher agree that cloud infrastructure requires new ways of providing security. Quantum cryptography offers unprecedented secure communication using quantum mechanics	2021	(+)8-9%
Evolving next-generation wireless network technologies	Advancements in high-speed wireless network technologies, such as 4G, 5G, and Wi-Fi, have become ubiquitous. The upcoming 5G technology is expected to contribute to the increased adoption of quantum cryptography solutions and provide an opportunity for telecom companies to increase their revenue.	2020	(+) 8-9%
		2021	(+) 8.5-9.5%

In the age of 5G technology, as secure communication will become a crucial aspect of connectivity, the theory of quantum cryptography becomes vital. Researchers have devised a way to integrate quantum cryptography into 5G networks, protecting the next-generation telecoms infrastructure from cyber attack

RESTRAINTS AND CHALLENGES	REASON	YEAR	COVID-19 IMPACT
High implementation cost	Although the implementation of quantum cryptography solutions increases the safety of communication, these solutions are expensive and time-consuming to implement, and this results in less adoption of quantum cryptography solutions.	2020	(-) 3.5-4.5%
	Currently, quantum cryptography solutions are expensive. Industries are willing to spend less than 2% of the total cost on security, the remaining 98% is generally spent on features and functionality.	2021	(-) 4.0-5.0%
Quantum cryptography commercialization	Thus, in-order for quantum cryptography solutions to be universally deployed across multiple industry verticals, the cost of implementing these security solutions must be lower by approximately 90% from current pricing.		
	Commercializing quantum cryptography is a challenge due to the lack of customer awareness, as knowledge is one of the key drivers for economic growth. For the commercialization of high-technology innovations, the enterprise commercialization team must identify, obtain, combine, and manage the required technological knowledge.	2020	(-) 2.0-3.0%
		2021	(-) 2.0-3.0%

Source: Secondary Literature, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

5.3.1 COVID-19 DYNAMICS IMPACTING QUANTUM CRYPTOGRAPHY MARKET GROWTH

COVID-19 DYNAMICS IMPACTING QUANTUM CRYPTOGRAPHY MARKET GROWTH	YEAR	IMPACT
CUMULATIVE GROWTH FROM DRIVERS AND OPPORTUNITIES	2020	24.0-27.0%
	2021	24.5-27.5%
CUMULATIVE DECLINE FROM RESTRAINTS AND CHALLENGES	2020	5.5-7.5%
	2021	6.0-8.0%
CUMULATIVE IMPACT FROM DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES	2020	18.5-19.5%
	2021	18.5-19.5%

Source: MarketsandMarkets Analysis

5.4 INDUSTRY TRENDS

5.4.1 QUANTUM CRYPTOGRAPHY PROTOCOLS

5.4.1.1 BB84 protocol

The BB84 was proposed by Charles Bennett and Gilles Brassard in 1984, and is based on Heisenberg's uncertainty principle. BB84 uses the transmission of single-polarized photons to perform the encryption process, known as polarization. There are 4 types of polarization, which are grouped in 2 different non-orthogonal basis, namely, rectilinear basis (vertical/horizontal polarizations) and diagonal basis. These polarized photons can be used to send information by allocating the value '0' or '1' to each polarization. Furthermore, this value states that one photon can carry one quantum bit (qubit) of information, and these encodings are agreed between Client A and Client B before they attempt to exchange quantum states.

5.4.1.2 B92 protocol

The B92 protocol was proposed by Charles Bennett, which is the next version of the BB84 protocol and uses polarized photons with non-orthogonal quantum states for encoding information. Two quantum states are used, instead of the 4 required in BB84. Client A randomly chooses one or other of these quantum states and sends them to Client B via a quantum channel. Client B has 2 methods to measure the arriving photons, which will either register "detection" or "no detection." During the key sifting stage, Client A tells the receiver which photons he "detected," but not his actual measurement; all other photons are discarded. Error correction and privacy amplification continue as normal, to verify that the secret key is the same for both Client A and Client B.

5.4.1.3 SARG04 protocol

SARG04 protocol was proposed in 2004 by Scarani et al. in Physical Review Letters. The SARG04 protocol shares the exact same first phase as BB84. In the second phase when Client A and Client B determine for which bits their bases match, Client A does not directly announce her bases, rather Client A announces a pair of non-orthogonal states one of which she used to encode her bit. If Client B used the correct basis, he would measure the correct state. If he chooses incorrectly, he will not be able to determine the bit. If there are no errors, then the length of the key remaining after the sifting stage would be 1/4th of the raw key.

5.4.1.4 E91 protocol

E91 protocol was proposed by Artur Ekert in 1991. Ekert proposed a method, harnessing Bell's inequalities, to perform key distribution via entangled polarized photons in a quantum channel. These entangled photons can be created by Client A, Client B, or a Trusted Third Party (TTP). Each pair is separated in a way that results in Client A and Client B receiving one of each pair. This is arguably a more secure method of using polarized photons, as there is no information to be eavesdropped. Client A and Client B both independently and randomly choose from 2 different orientations of their analyzers to measure the polarization of the photons, and the choice of analyzer is the basis for the publicly discussed key sifting stage. The measurements are divided into 2 groups: the first is when different orientations of the analyzer were used, and the second when the same analyzer orientation was employed. Any photons that were not registered are discarded. Client A and Client B then reveal the results of the first group only, and check that they correspond to the value expected from Bell's inequality ($=-2\sqrt{2}$), and if this is so, then Client A and Client B can be sure that the results they obtained in the second group are anti-correlated and can be used to produce a secret key string. Moreover, any eavesdropper cannot obtain any information from the photons when they are in transit. Information is only present once the authorized users perform their analyzer measurements and key sifting. Eavesdropper's only hope is to inject their own data for Client A and Client B, but as they do not know their analyzer orientations, they will always be detected.

5.4.1.5 SSP protocol

Six-State Protocol (SSP) was proposed by Pasquucci and Gisin in 1999. SSP is a discrete-variable protocol for QKD that permits tolerating a noisier channel than the BB84 protocol. Furthermore, SSP produces a higher rate of errors during attempted eavesdropping, thus making it easier to detect errors. Additionally, SSP can be implemented without a quantum computer, using only optical technologies.

5.4.2 QUANTUM CRYPTOGRAPHY TYPES

Quantum cryptography describes the use of quantum mechanical effects to perform cryptographic tasks to offer quantum-safe communication. The developed and known application of quantum cryptography is QKD, which offers quantum communication between 2 parties (Alice and Bob) without a third party learning anything about that key. Quantum cryptography notifies the parties (Alice and Bob) when a third party tries to eavesdrop any information.

5.4.2.1 Position-based quantum cryptography

Position-based cryptography offers new cryptographic methods, ensuring that certain tasks can only be performed at a particular geographical position. For example, a country could send a message in such a way that it can only be deciphered at the location of its embassy in another country.

5.4.2.2 Device-independent quantum cryptography

A quantum cryptographic protocol is device-independent if its security does not depend on believing that the quantum device utilized is trustworthy. The implementation of such protocols is challenging, as it requires the observation of Bell inequality violation between 2 distant users while performing encryption.

5.4.2.3 Bounded- and noisy-quantum-storage model

Bounded- and noisy-quantum-storage model can build an entire secure quantum commitment and quantum oblivious transfer protocol. In this model, the amount of quantum data that an adversary can store is limited by some known constant Q .

5.4.3 QUANTUM CRYPTOGRAPHY MARGINS

5.4.3.1 Quantum repeaters

A quantum repeater is one of the limitations, as there is no existence of quantum repeaters in quantum cryptography. Quantum repeater is the solution to improve the signal-to-noise ratio of the communication medium to ensure that the signal can travel larger distances, for which quantum computation principles are required to construct a quantum repeater. Many companies are in the research phase and studying to extend the range and increase the data rate of QKD systems. Moreover, these companies are ready to integrate quantum cryptography systems with existing information security solutions. For instance, in February 2018, phys.org illustrated a new quantum repeater scheme for long-distance quantum communication.

5.4.3.2 Point-to-point communication

Quantum cryptography is a point-to-point technique, which is enabled by standard optical amplifiers or repeaters, used in long fiber cables, switches, and routers.

5.4.3.3 Vulnerable to attacks

Quantum cryptography does not guarantee safe communication; it only notifies the parties (Alice and Bob) about the interference of third parties. A crucial part of QKD systems is single-photon detectors, used in the QKD receiver, to detect weak optical signals sent by the sender through the communication channel. These signals are vulnerable to different attacks. For instance, Trojan-horse attack is where the third party can exploit communication by sending bright-light into the receiver module to probe and retrieve information about the encoding devices. Other attacks such as Denial of Service (DoS), bright-light, back-flash, and efficiency mismatch and time-shift attacks are some of the threatening issues of quantum cryptography implementation.

5.4.4 FUTURE SCOPE OF QUANTUM CRYPTOGRAPHY

The next 5 years is going to be significant for the quantum cryptography market. The market is expected to see new business models, value-added services, emerging technology solutions, and different variants of workaround solutions and non-quantum key exchange systems. Additionally, the emergence of cloud technologies, such as CloudQKDP (QKD Protocol for Cloud Computing), has leveraged quantum mechanisms to secure cloud storage and data dynamics or QKD as a service.

A new dedicated Quantum Safe Communication Infrastructure, built by the telecom industry vertical, is expected to be introduced in the near future and will be offered as premium services. This is evident from the fact that many telecom companies are collaborating with quantum cryptography vendors to offer enhanced and innovative quantum communication. Furthermore, chip-based quantum-safe IoT security system, which is an emerging solution, can also become a part of the future quantum solution.

Furthermore, the existing encryptions deployed across the globe would need an upgrade, from the present key exchange mechanisms based on Diffie-Hellman or Rivest-Shamir-Adleman (RSA) to post-quantum cryptography (PQC) ones including QKD. This would spur the demand for a changeover, from existing encryption systems to post cryptographic ones.

5.4.5 USE CASES

TABLE 5 INDUSTRIAL USE CASES: TELECOM VERTICAL

COMPANY	CLIENT	VERTICAL	CASE	BENEFIT
ID Quantique	A private asset and wealth management company	BFSI	One of the world's leading private asset and wealth management company had shifted its headquarters, which provided an opportunity to the security team to re-assess the firm's long-term data protection procedures and architecture, rationalize and update the communications network for leading-edge performance, and ensure legal and regulatory compliance. Furthermore, the company also had to deal with multiple protocols for network and storage and required a fiber channel-over-Ethernet protocol for the link between the headquarters and Data Recovery Center (DRC).	The company chooses ID Quantique's (IDQ's) hybrid encryption solution, incorporating state-of-the-art layer 2 encryption with QKD. IDQ's 10 Gigabit Ethernet encryptors were chosen to secure the bank's extended backbone – between the headquarters and DRC (nearly 100 kilometers away) – with an additional layer of security provided by the Cerberis QKD server.

Quantum Xchange	A private oil and gas company	Energy and utilities	One of the leaders in the energy and utilities sector needed high security for the data generated and shared by various devices in the connected environment. Safety and monetization of the proprietary data was the sole purpose for the adoption of next-generation encryption methodologies.	The company selected Quantum Xchange's quantum cryptography solution to secure and transmit the sensor data. Moreover, the data generated from this connected device cannot be shared as it is safeguarded with quantum encryption. Moreover, the receiver of the encrypted data needs the exact key, as the key changes for transmission of every single bit. Moreover, the eavesdropper in the network can be easily identified through quantum cryptography.
Post-Quantum	North Atlantic Treaty Organization (NATO)	Military and defense	NATO is a political and military alliance based around cooperation between 28 nations for mutual security. The government body demanded a robust, intuitive, and reliable military environment with all their communication tools secure and capable of resisting a determined adversary. Moreover, there was the need for an interference detection mechanism without causing any destruction to NATO's security systems in the present as well as in the future.	NATO chose Post-Quantum Solutions as it provided a full-featured quantum-secure communication app for trial. Compatible with iOS and Android smartphones and designed in the style of popular consumer messaging apps, the system offered instant familiarity and ease of use. It also incorporated the core features users would expect from such an app: messaging, voice calling, image sharing, and video and audio messaging. Furthermore, the trial successfully demonstrated that instant messaging and communication can be made quantum-safe, and that this capability can be deployed in the present to guard against future threats. Furthermore, the trial enabled NATO to demonstrate its capabilities and intent in the quantum cryptography space.

Source: company websites

6 QUANTUM CRYPTOGRAPHY MARKET, BY COMPONENT

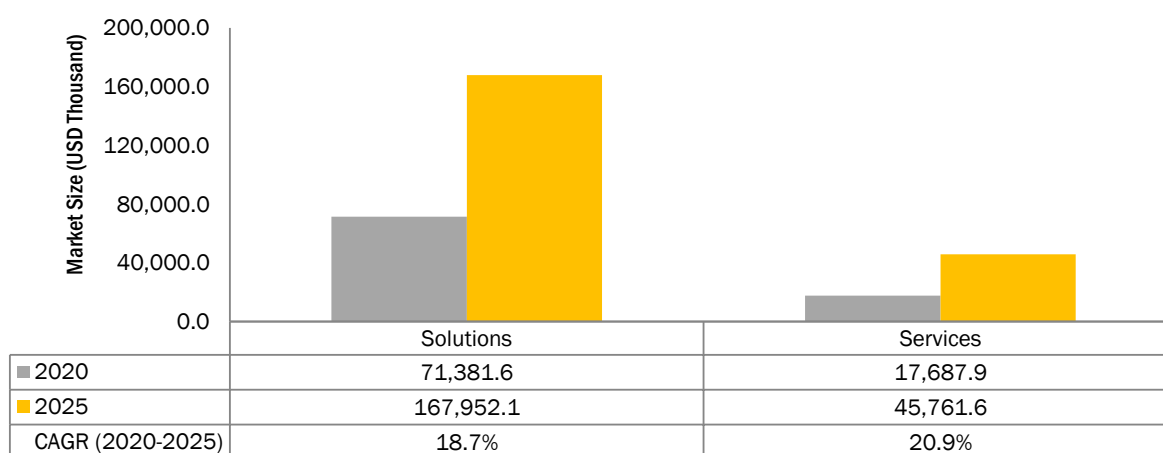
KEY FINDINGS

- The overall quantum cryptography market size is expected to grow from USD 89,069.4 thousand in 2020 to USD 2,13,713.7 thousand by 2025, at a CAGR of 19.1% during the forecast period.
- The overall quantum cryptography market size by solution is expected to grow from USD 71,381.6 thousand in 2020 to USD 1,67,952.1 thousand by 2025, at a CAGR of 18.7% during the forecast period.
- The overall quantum cryptography market size by service is expected to grow from USD 17,687.9 thousand in 2020 to USD 45,761.6 thousand by 2025, at a CAGR of 20.9% during the forecast period.
- For the implementation of quantum cryptography solutions, the QKD and QRNG methodologies are used. The QKD method holds around 80% of the implementations.
- The key generation and distribution capability eliminate the reliance on external key servers and provide a robust, fault-tolerant security mechanism.

6.1 INTRODUCTION

With the constant evolution of virtualization, mobility, regulatory mandates, and advanced threats, enterprises across the globe are adopting data-centric approaches to protect sensitive and critical business information. Quantum cryptography vendors offer a variety of solutions and services to protect enterprises' sensitive information from breaches and thefts. Quantum cryptography solutions protect the data at rest, data in transit, and data in use through quantum-safe techniques. The services segment includes consulting and advisory, deployment and integration, and support and maintenance services. Quantum cryptography hardware and services help organizations in combating advanced threats and vulnerabilities and securing the enterprise data.

FIGURE 15 SOLUTIONS SEGMENT TO ACCOUNT FOR A HIGHER MARKET SHARE DURING THE FORECAST PERIOD



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 6 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, 2017–2019 (USD THOUSAND)

Component	2017	2018	2019	CAGR (2017–2019)
Solutions	42,343.3	50,590.6	60,469.7	19.5%
Services	9,656.7	11,809.4	14,441.5	22.3%
Total	52,000.0	62,400.0	74,911.2	20.0%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 7 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Solutions	60,469.7	72,340.5	86,651.9	1,03,925.1	1,24,849.0	1,50,270.8	1,81,253.6	20.2%
Services	14,441.5	17,665.3	21,625.1	26,494.5	32,502.3	39,934.7	49,156.1	22.7%
Total	74,911.2	90,005.8	1,08,277.0	1,30,419.6	1,57,351.3	1,90,205.5	2,30,409.8	20.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 8 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Solutions	60,469.7	71,381.6	84,578.6	1,00,316.2	1,19,130.0	1,41,465.5	1,67,952.1	18.7%
Services	14,441.5	17,687.9	21,369.4	25,762.0	31,029.2	37,524.2	45,761.6	20.9%
Total	74,911.2	89,069.4	1,05,948.1	1,26,078.2	1,50,159.1	1,78,989.7	2,13,713.7	19.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the global quantum cryptography market size by component. The overall market size is expected to grow from USD 89,069.4 thousand in 2020 to USD 2,13,713.7 thousand by 2025, at a CAGR of 19.1% during the forecast period. The advent of quantum computers is expected to generate a high level of computing power because of which traditional cryptography algorithms can be infiltrated. Hence, quantum cryptography solutions play a major role in achieving a zero-trust network.

6.2 SOLUTIONS

6.2.1 INCREASING NEED TO ENCRYPT CRITICAL INFORMATION WITHOUT ANY LEAKAGE

The quantum cryptography solutions segment includes the revenue generated from QKD platforms, QKD servers, virtual encryptors, QKD distributors, key and policy managers, SDK to integrate cryptography, quantum cryptography communication device, quantum security gateway, QRNG, hackerbox, quantum secure communication network products, and repeaters, along with factors essential for the implementation of quantum cryptography. This solution helps in encrypting the critical information without being leaked midway. With the help of quantum-safe cryptography solutions, fraudulent cannot read the encrypted data, as it alerts the sender and receiver about it. These solutions play a vital role in offering point-to-point security and storage networks for long-term data protection. In addition to this, the R&D platform is gaining traction among the scientific community, as the need for quantum-safe encryption is increasing to counter the threat of a quantum computer. The R&D platform comprises two stations, namely, the transmitter unit and the receiver unit. Each station consists of optical and electronic platforms, which are controlled by an external computer, linked to the station through an Ethernet connector. With the R&D platform, a user can experiment with different parameters and study various setups. It is well-documented in scientific publications and has been extensively tested and characterized. ID Quantique is a leading provider of R&D platform for quantum cryptography.

TABLE 9 SOLUTIONS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	21,481.2	26,182.7	31,914.2	21.9%
Europe	11,214.3	13,237.2	15,629.5	18.1%
APAC	7,374.1	8,607.4	10,045.6	16.7%
MEA	2,273.7	2,563.2	2,880.3	12.6%
Total	42,343.3	50,590.6	60,469.7	19.5%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 10 SOLUTIONS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	31,914.2	38,716.7	46,808.3	56,489.0	68,155.1	82,302.2	99,572.7	20.8%
Europe	15,629.5	18,528.0	22,053.4	26,332.1	31,530.8	37,851.9	45,542.0	19.7%
APAC	10,045.6	11,808.0	13,977.1	16,630.2	19,869.2	23,817.0	28,620.4	19.4%
MEA	2,880.3	3,287.8	3,813.0	4,473.8	5,293.9	6,299.7	7,518.4	18.0%
Total	60,469.7	72,340.5	86,651.9	1,03,925.1	1,24,849.0	1,50,270.8	1,81,253.6	20.2%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 11 SOLUTIONS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	31,914.2	38,023.2	45,600.7	54,659.2	65,488.5	78,249.4	93,160.3	19.6%
Europe	15,629.5	18,634.6	22,225.5	26,417.1	31,186.1	36,563.1	42,933.2	18.2%
APAC	10,045.6	11,712.5	13,675.4	15,991.0	18,734.9	21,993.1	25,891.8	17.2%
MEA	2,880.3	3,011.3	3,077.0	3,249.0	3,720.5	4,659.9	5,966.8	14.7%
Total	60,469.7	71,381.6	84,578.6	1,00,316.2	1,19,130.0	1,41,465.5	1,67,952.1	18.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3 SERVICES

6.3.1 GROWING NEED FOR EXPERT ASSISTANCE FOR A CUSTOMIZED SOLUTION

Quantum cryptography service providers render a variety of services, such as consulting and advisory, deployment and integration, and support and maintenance services. These services offer advisory consulting, professional and technical design and implementation services to assist different industry verticals, to build security for a quantum-safe future. Based on the extensive experience in the industry-specific quantum cryptography sector, experts assist organizations from consultation to the implementation of customized solutions. These service provider's teams are focused on helping the organization to increase their efficiency and productivity while ensuring the long-term security of data.

TABLE 12 SERVICES: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	5,038.8	6,265.3	7,788.7	24.3%
Europe	2,461.7	2,986.8	3,622.7	21.3%
APAC	1,673.9	2,000.6	2,389.7	19.5%
MEA	482.3	556.8	640.5	15.2%
Total	9,656.7	11,809.4	14,441.5	22.3%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 13 SERVICES: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	7,788.7	9,634.6	11,874.5	14,605.5	17,956.4	22,090.9	27,223.1	23.1%
Europe	3,622.7	4,409.0	5,385.0	6,594.1	8,093.9	9,955.2	12,266.4	22.7%
APAC	2,389.7	2,873.6	3,478.5	4,230.7	5,165.0	6,324.1	7,759.9	22.0%
MEA	640.5	748.0	887.2	1,064.3	1,287.0	1,564.5	1,906.7	20.6%
Total	14,441.5	17,665.3	21,625.1	26,494.5	32,502.3	39,934.7	49,156.1	22.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 14 SERVICES: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	7,788.7	9,718.0	11,823.1	14,305.6	17,249.2	20,910.9	25,664.5	21.4%
Europe	3,622.7	4,434.4	5,427.0	6,615.4	8,005.4	9,616.3	11,563.8	21.1%
APAC	2,389.7	2,850.4	3,403.4	4,068.1	4,870.2	5,839.8	7,020.1	19.8%
MEA	640.5	685.1	716.0	772.9	904.4	1,157.3	1,513.2	17.2%
Total	14,441.5	17,687.9	21,369.4	25,762.0	31,029.2	37,524.2	45,761.6	20.9%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

7 QUANTUM CRYPTOGRAPHY MARKET, BY SERVICE

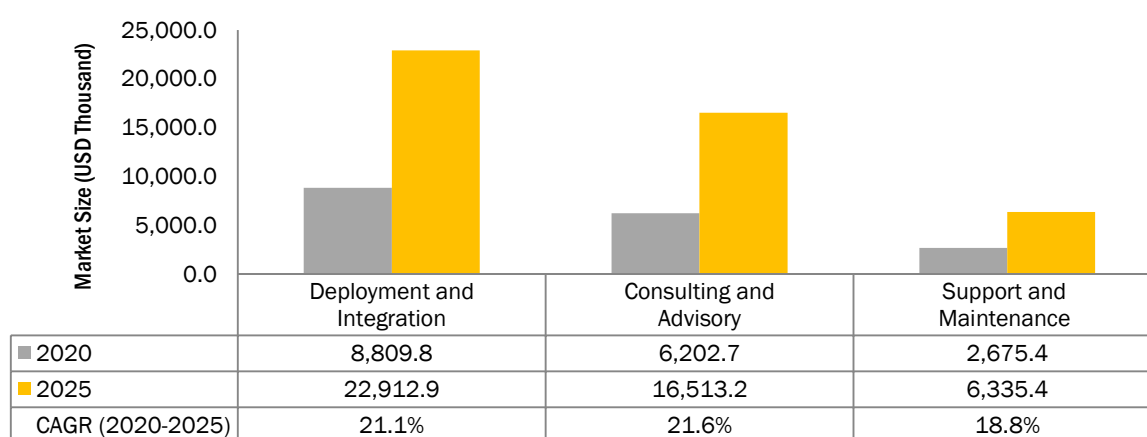
KEY FINDINGS

- The overall quantum cryptography market size by service is expected to grow from USD 17,687.9 thousand in 2020 to USD 45,761.6 thousand by 2025, at a CAGR of 20.9% during the forecast period.
- Among quantum cryptography services, the consulting and advisory services segment is expected to grow from USD 6,202.7 thousand in 2020 to USD 16,513.2 thousand by 2025, at a CAGR of 21.6% during the forecast period.
- The deployment and integration segment is projected to grow at a CAGR of 21.1% during the forecast period.
- Quantum cryptography services help understand the exact needs of the customers, thereby suggesting the correct quantum cryptography appliance and providing the needed support.
- Quantum cryptography consultants help enterprises devise security strategies, minimize risks, understand the maturity of their quantum cryptography solutions, and reduce compliance costs.

7.1 INTRODUCTION

The quantum cryptography services market is divided into consulting and advisory, deployment and integration, and support and maintenance services. Most of the organizations lack technical expertise in implementing and maintaining quantum cryptography solutions, for which they avail services from the service providers. The services offered by the quantum cryptography vendors, help in managing the operational challenges of customers belonging to different industry verticals. The overall services segment has a major influence on the market, as these services help in reducing costs, ensuring data security, and increasing overall business performance.

FIGURE 16 CONSULTING AND ADVISORY SEGMENT TO GROW AT THE HIGHEST CAGR DURING THE FORECAST PERIOD



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 15 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, 2017–2019 (USD THOUSAND)

Service	2017	2018	2019	CAGR (2017–2019)
Consulting and Advisory	3,352.8	4,109.0	5,035.3	22.5%
Deployment and Integration	4,719.7	5,821.3	7,179.6	23.3%
Support and Maintenance	1,584.1	1,879.1	2,226.7	18.6%
Total	9,656.7	11,809.4	14,441.5	22.3%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 16 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	5,035.3	6,171.7	7,570.1	9,292.9	11,422.4	14,061.9	17,342.8	23.0%
Deployment and Integration	7,179.6	8,855.3	10,928.5	13,496.7	16,688.7	20,667.3	25,640.8	23.7%
Support and Maintenance	2,226.7	2,638.3	3,126.6	3,705.0	4,391.2	5,205.5	6,172.5	18.5%
Total	14,441.5	17,665.3	21,625.1	26,494.5	32,502.3	39,934.7	49,156.1	22.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 17 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	5,035.3	6,202.7	7,536.5	9,137.4	11,068.6	13,462.7	16,513.2	21.6%
Deployment and Integration	7,179.6	8,809.8	10,660.4	12,868.7	15,514.8	18,775.4	22,912.9	21.1%
Support and Maintenance	2,226.7	2,675.4	3,172.6	3,755.8	4,445.7	5,286.1	6,335.4	18.8%
Total	14,441.5	17,687.9	21,369.4	25,762.0	31,029.2	37,524.2	45,761.6	20.9%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the global quantum cryptography market size by service. The deployment and integration segment is estimated to hold the largest market size of USD 8,809.8 thousand in 2020 and is projected to reach USD 22,912.9 thousand by 2025, at a CAGR of 21.1% during the forecast period.

7.2 CONSULTING AND ADVISORY

7.2.1 INCREASING DEMAND FOR CONSULTATION TO RESOLVE HINDRANCES FOR COMMERCIAL CLIENTS' BUSINESS GROWTH

Quantum cryptography consulting and advisory services play an important role in the implementation of the required set of solutions and techniques for securing the transmission of data. For this, qualified industry experts and security professionals are involved as consultants. Consultants help enterprises devise security strategies, minimize risks, understand the maturity of their quantum cryptography solutions, and reduce compliance costs. Advisory services assist organizations in solving a problem that hinders their business growth, the experts guide the path or directs the clients on where to go for help. Advisory service is provided by experienced experts, who offer hands-on, practical help, advice, and support in particular business areas that will help make a difference to the business. Consulting and advisory services are vital in managing the complete lifecycle of a solution until its deployment. They are an integral part of professional services, which majorly strengthen business operations by increasing productivity and performance and streamlining the business revenue and processes. Consulting and advisory services in quantum cryptography are crucial, as they deal with the transformation of ideas into solutions and infrastructures by providing advanced security solutions to the enterprises.

TABLE 18 CONSULTING AND ADVISORY: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	1,763.6	2,193.6	2,728.0	24.4%
Europe	837.0	1,019.5	1,241.5	21.8%
APAC	585.9	702.9	842.9	19.9%
MEA	166.4	192.9	222.9	15.7%
Total	3,352.8	4,109.0	5,035.3	22.5%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 19 CONSULTING AND ADVISORY: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	2,728.0	3,375.7	4,162.0	5,121.0	6,298.2	7,751.1	9,555.3	23.1%
Europe	1,241.5	1,517.1	1,860.3	2,287.1	2,818.4	3,480.4	4,305.5	23.2%
APAC	842.9	1,017.5	1,236.4	1,509.7	1,850.2	2,274.2	2,801.3	22.5%
MEA	222.9	261.4	311.4	375.1	455.5	556.1	680.7	21.1%
Total	5,035.3	6,171.7	7,570.1	9,292.9	11,422.4	14,061.9	17,342.8	23.0%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 20 CONSULTING AND ADVISORY: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	2,728.0	3,423.0	4,188.0	5,096.0	6,179.3	7,533.4	9,298.3	22.1%
Europe	1,241.5	1,528.8	1,882.1	2,307.9	2,809.4	3,394.7	4,106.5	21.9%
APAC	842.9	1,011.2	1,214.3	1,459.9	1,757.8	2,120.0	2,563.2	20.4%
MEA	222.9	239.8	252.1	273.7	322.1	414.6	545.3	17.9%
Total	5,035.3	6,202.7	7,536.5	9,137.4	11,068.6	13,462.7	16,513.2	21.6%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

7.3 DEPLOYMENT AND INTEGRATION

7.3.1 RISING NEED FOR SECURE DATA TRANSMISSION

Deployment and integration offer a series of services to simplify the deployment process, accelerate the implementation stage, minimize service interruptions, and reduce overall deployment costs. While deploying quantum cryptography, the regulatory and industry compliances set by the regulatory bodies and organizations have to be considered. The effective integration of these services requires a detailed understanding of the business processes and security assessments. Integration services collect customer requirements, and then install, integrate, test, and roll out the quantum cryptography solutions. They help both quantum cryptography service providers and enterprise business clients in modernizing and deploying their solutions to meet the rising demand for secure data transmission.

TABLE 21 DEPLOYMENT AND INTEGRATION: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	2,519.4	3,155.4	3,951.1	25.2%
Europe	1,206.2	1,475.2	1,803.7	22.3%
APAC	786.7	949.0	1,144.1	20.6%
MEA	207.4	241.7	280.7	16.3%
Total	4,719.7	5,821.3	7,179.6	23.3%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 22 DEPLOYMENT AND INTEGRATION: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	3,951.1	4,923.0	6,111.6	7,571.8	9,376.6	11,619.4	14,422.8	24.0%
Europe	1,803.7	2,212.8	2,724.2	3,362.7	4,160.5	5,158.3	6,406.8	23.7%
APAC	1,144.1	1,388.5	1,696.3	2,082.3	2,565.7	3,170.6	3,926.5	23.1%
MEA	280.7	331.0	396.3	479.9	585.9	719.0	884.7	21.7%
Total	7,179.6	8,855.3	10,928.5	13,496.7	16,688.7	20,667.3	25,640.8	23.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 23 DEPLOYMENT AND INTEGRATION: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	3,951.1	4,932.6	6,004.5	7,269.3	8,770.0	10,637.7	13,063.2	21.5%
Europe	1,803.7	2,210.2	2,708.0	3,304.7	4,003.4	4,814.3	5,795.8	21.3%
APAC	1,144.1	1,366.3	1,633.3	1,954.7	2,342.9	2,812.8	3,385.5	19.9%
MEA	280.7	300.7	314.6	340.1	398.5	510.5	668.5	17.3%
Total	7,179.6	8,809.8	10,660.4	12,868.7	15,514.8	18,775.4	22,912.9	21.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

7.4 SUPPORT AND MAINTENANCE

7.4.1 DEMAND FOR SUPPORT AND MAINTENANCE SERVICES FOR THE IMPLEMENTATION AND USE OF HARDWARE

As the deployment of quantum cryptography is increasing, the demand for support and maintenance services is also gaining traction. Support and maintenance services offer assistance required for the implementation and use of the hardware provided by the quantum cryptography vendors. It also helps in solving product issues. These services also include training, in line with the business requirements of the commercial clients to help develop the requisite skill sets in the IT team for easy management of operations and for promoting the adoption of quantum cryptography in the market. Support and maintenance services range from pre-sales to post-sales, offered by the quantum cryptography vendors. These services also include timely updates of software and applications and help solve the problems related to quantum cryptography appliances.

TABLE 24 SUPPORT AND MAINTENANCE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	755.8	916.3	1,109.6	21.2%
Europe	418.5	492.0	577.5	17.5%
APAC	301.3	348.7	402.7	15.6%
MEA	108.5	122.2	136.9	12.3%
Total	1,584.1	1,879.1	2,226.7	18.6%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 25 SUPPORT AND MAINTENANCE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	1,109.6	1,335.9	1,600.9	1,912.6	2,281.6	2,720.4	3,245.0	19.4%
Europe	577.5	679.2	800.5	944.4	1,115.0	1,316.5	1,554.2	18.0%
APAC	402.7	467.6	545.7	638.7	749.1	879.3	1,032.1	17.2%
MEA	136.9	155.7	179.6	209.2	245.5	289.3	341.3	17.0%
Total	2,226.7	2,638.3	3,126.6	3,705.0	4,391.2	5,205.5	6,172.5	18.5%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 26 SUPPORT AND MAINTENANCE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	1,109.6	1,362.4	1,630.6	1,940.3	2,299.9	2,739.8	3,303.0	19.4%
Europe	577.5	695.4	836.9	1,002.9	1,192.6	1,407.2	1,661.5	19.0%
APAC	402.7	472.9	555.7	653.5	769.4	907.0	1,071.5	17.8%
MEA	136.9	144.7	149.3	159.1	183.8	232.1	299.4	15.7%
Total	2,226.7	2,675.4	3,172.6	3,755.8	4,445.7	5,286.1	6,335.4	18.8%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

8 QUANTUM CRYPTOGRAPHY MARKET, BY SECURITY TYPE

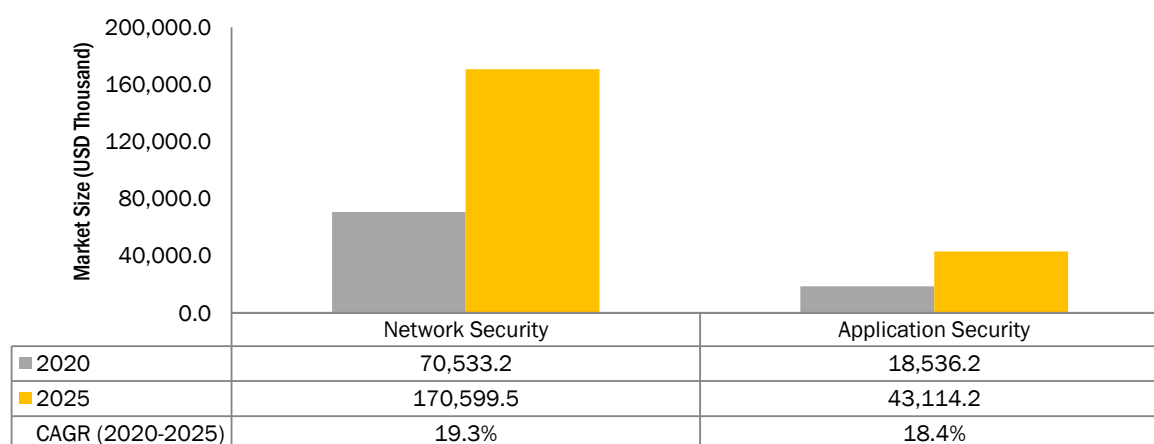
KEY FINDINGS

- The overall quantum cryptography market size by security type is expected to grow from USD 89,069.4 thousand in 2020 to USD 2,13,713.7 thousand by 2025, at a CAGR of 19.1% during the forecast period.
- Among quantum cryptography security types, the network security segment is expected to grow from USD 70,533.2 thousand in 2020 to USD 1,70,599.5 thousand by 2025, at a higher CAGR of 19.3% during the forecast period.
- Cyber attackers try to alter the data in transit; hence, network security has become the most important segment.

8.1 INTRODUCTION

In an era of digitalization, enterprises across the globe are aggressively expanding their businesses, thereby creating a surge in the number of endpoint nodes, data centers, and cloud storage. This, in turn, transfers large volumes of data from the sender to the recipient through the network that has to be monitored carefully to ensure protection against active and passive attackers. Therefore, the confidentiality, integrity, and availability of data remain the subject of concern. Organizations require highly protective encryption standards and algorithms for safeguarding their data from threats. This protection is provided by quantum cryptography technology. On the basis of security, the quantum cryptography market is segmented into network security and application security.

FIGURE 17 NETWORK SECURITY SEGMENT TO HOLD A LARGER MARKET SIZE DURING THE FORECAST PERIOD



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 27 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, 2017–2019 (USD THOUSAND)

Security Type	2017	2018	2019	CAGR (2017–2019)
Network Security	40,992.9	49,264.8	59,230.1	20.2%
Application Security	11,007.1	13,135.2	15,681.1	19.4%
Total	52,000.0	62,400.0	74,911.2	20.0%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 28 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	59,230.1	71,274.4	85,879.5	1,03,610.0	1,25,211.4	1,51,606.3	1,83,956.9	20.9%
Application Security	15,681.1	18,731.4	22,397.5	26,809.7	32,139.8	38,599.2	46,452.9	19.9%
Total	74,911.2	90,005.8	1,08,277.0	1,30,419.6	1,57,351.3	1,90,205.5	2,30,409.8	20.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 29 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	59,230.1	70,533.2	84,027.7	1,00,149.1	1,19,468.8	1,42,641.3	1,70,599.5	19.3%
Application Security	15,681.1	18,536.2	21,920.4	25,929.1	30,690.3	36,348.4	43,114.2	18.4%
Total	74,911.2	89,069.4	1,05,948.1	1,26,078.2	1,50,159.1	1,78,989.7	2,13,713.7	19.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the global quantum cryptography market size by security type. The network security segment is estimated to hold a market size of USD 70,533.2 thousand in 2020 and is projected to reach USD 1,70,599.5 thousand by 2025, growing at a higher CAGR of 19.3% during the forecast period. An increase in network complexity and a high volume of data transfer are the factors driving the growth of quantum cryptographic solutions.

8.2 NETWORK SECURITY

8.2.1 GROWING NEED TO ENSURE THE INTEGRITY AND USABILITY OF NETWORK AND INFORMATION

Network security is a process intended to ensure the integrity and usability of network and information. It incorporates both hardware and software technologies. Powerful network security focuses on an assortment of dangers and prevents them from entering or spreading on your system. Network security consists of wireless communication and remote access security, and gateway. Wireless communication is carried out with the help of various secure protocols, such as Low-Power Wide-Area Network (LPWAN), Zigbee, internet protocol version 6 (IPv6) over Low-Power Wireless Personal Area Networks (6LoWPAN), Bluetooth, Z-Wave, and Near Frequency Communication (NFC). Gateway antivirus and antispyware ensure the security of IoT/ mobile to mobile (M2M) gateway from intrusions, viruses, spyware, worms, Trojans, adware, keyloggers, and Malicious Mobile Code (MMC) by using mechanisms, such as access control lists (ACLs), intrusion detection system (IDS)/ intrusion prevention system (IPS), and filtering. This has tempted attackers to gain illegitimate access to information for various reasons. Using quantum cryptography, a secret key is generated using polarized photons in quantum systems. The data is encoded and decoded using a code generator, which makes the transmission of data through a channel less vulnerable to attacks.

TABLE 30 NETWORK SECURITY: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	20,685.6	25,347.4	31,061.2	22.5%
Europe	10,940.8	13,004.2	15,461.1	18.9%
APAC	7,147.9	8,396.4	9,861.6	17.5%
MEA	2,218.6	2,516.9	2,846.2	13.3%
Total	40,992.9	49,264.8	59,230.1	20.2%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 31 NETWORK SECURITY: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	31,061.2	37,883.8	46,047.6	55,870.4	67,773.2	82,284.6	1,00,092.6	21.4%
Europe	15,461.1	18,455.7	22,120.2	26,595.4	32,067.6	38,764.0	46,963.6	20.5%
APAC	9,861.6	11,665.4	13,896.0	16,638.7	20,005.6	24,133.0	29,184.3	20.1%
MEA	2,846.2	3,269.4	3,815.7	4,505.4	5,365.1	6,424.8	7,716.4	18.7%
Total	59,230.1	71,274.4	85,879.5	1,03,610.0	1,25,211.4	1,51,606.3	1,83,956.9	20.9%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 32 NETWORK SECURITY: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	31,061.2	37,405.8	45,059.7	54,196.8	65,117.8	78,160.0	93,800.3	20.2%
Europe	15,461.1	18,561.9	22,292.7	26,681.3	31,717.0	37,444.0	44,273.4	19.0%
APAC	9,861.6	11,571.0	13,596.1	15,999.2	18,863.5	22,284.8	26,401.9	17.9%
MEA	2,846.2	2,994.5	3,079.2	3,272.0	3,770.5	4,752.5	6,123.9	15.4%
Total	59,230.1	70,533.2	84,027.7	1,00,149.1	1,19,468.8	1,42,641.3	1,70,599.5	19.3%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

8.3 APPLICATION SECURITY

8.3.1 INCREASING DEMAND TO PROTECT APPLICATIONS FROM INTRUDERS

Application security is the use of hardware, software, and procedural methods to protect applications from intruders. It involves the security of web applications, mobile applications, cloud applications, and stand-alone applications that are used on remote devices, such as laptops, tablets, and smartphones. Public and private organizations are always the high-risk targets for attacks and penetration of the IT systems. The attacks can cause significant financial loss to the organizations, and the attackers can target areas of weaknesses in the applications, database, and infrastructure layers of the IT systems. The most commonly known attacks on the application layer are software tampering, input validation, brute force attack, and session hijacking. Application security is vital, as it prevents data loss and any data misuse in the network. Due to an increase in the demand for application security products and services by organizations, the need for quantum cryptography is increasing. Quantum cryptography helps provide constant data confidentiality, integrity, and authenticity, even in the face of an attack. Moreover, it ensures data privacy through the use of encryption.

TABLE 33 APPLICATION SECURITY: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	5,834.4	7,100.6	8,641.8	21.7%
Europe	2,735.2	3,219.8	3,791.1	17.7%
APAC	1,900.1	2,211.6	2,573.7	16.4%
MEA	537.4	603.1	674.6	12.0%
Total	11,007.1	13,135.2	15,681.1	19.4%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 34 APPLICATION SECURITY: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	8,641.8	10,467.5	12,635.2	15,224.0	18,338.4	22,108.5	26,703.2	20.6%
Europe	3,791.1	4,481.3	5,318.2	6,330.8	7,557.1	9,043.2	10,844.9	19.3%
APAC	2,573.7	3,016.3	3,559.6	4,222.2	5,028.6	6,008.2	7,196.0	19.0%
MEA	674.6	766.4	884.5	1,032.7	1,215.8	1,439.3	1,708.8	17.4%
Total	15,681.1	18,731.4	22,397.5	26,809.7	32,139.8	38,599.2	46,452.9	19.9%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 35 APPLICATION SECURITY: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	8,641.8	10,335.4	12,364.2	14,768.0	17,619.9	21,000.3	25,024.5	19.3%
Europe	3,791.1	4,507.0	5,359.7	6,351.2	7,474.5	8,735.3	10,223.6	17.8%
APAC	2,573.7	2,991.9	3,482.8	4,059.9	4,741.5	5,548.1	6,510.0	16.8%
MEA	674.6	701.9	713.8	749.9	854.4	1,064.7	1,356.1	14.1%
Total	15,681.1	18,536.2	21,920.4	25,929.1	30,690.3	36,348.4	43,114.2	18.4%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

9 QUANTUM CRYPTOGRAPHY MARKET, BY VERTICAL

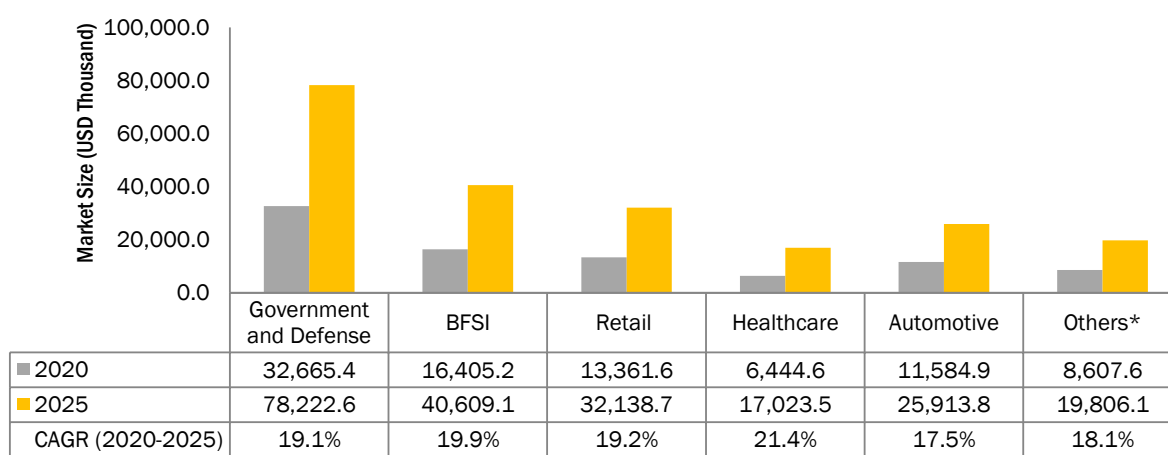
KEY FINDINGS

- The healthcare vertical is expected to grow from USD 6,444.6 thousand in 2020 to USD 17,023.5 thousand by 2025, at the highest CAGR of 21.4% during the forecast period.
- The government and defense vertical is expected to have the largest market size of USD 32,665.4 thousand in 2020, which is projected to grow to USD 78,222.6 thousand by 2025.
- The government and defense vertical is the leading adopter of quantum cryptography solutions, owing to the confidentiality of data.
- The use of quantum cryptography in BFSI and retail verticals is expected to grow rapidly, owing to the increase in the number of transactions.

9.1 INTRODUCTION

On the basis of verticals, the global quantum cryptography market has been segmented into government and defense, BFSI, retail, healthcare, automotive, and others (telecom, energy and utilities, logistics and transportations, and education). Enterprises from various verticals are adopting quantum cryptography hardware and services to ensure the safety of business data and transactions. Businesses are implementing their own security guidelines and techniques to protect their work environment. Furthermore, the adoption of cloud technology and extensive use of mobile devices in organizations have made the organizational data more vulnerable to cyber attacks. This has encouraged enterprises to adopt quantum cryptography hardware and services to ensure data security.

FIGURE 18 GOVERNMENT AND DEFENSE SEGMENT TO HOLD THE LARGEST MARKET SIZE DURING THE FORECAST PERIOD



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

*Others includes telecom, energy and utilities, logistics, and education

TABLE 36 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, 2017–2019 (USD THOUSAND)

Vertical	2017	2018	2019	CAGR (2017–2019)
Government and Defense	18,984.7	22,834.5	27,476.4	20.3%
BFSI	9,349.1	11,318.5	13,708.5	21.1%
Retail	7,835.4	9,376.9	11,226.3	19.7%
Healthcare	3,595.8	4,370.3	5,313.8	21.6%
Automotive	7,052.2	8,341.7	9,871.0	18.3%
Others*	5,182.8	6,158.1	7,315.2	18.8%
Total	52,000.0	62,400.0	74,911.2	20.0%

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 37 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	27,476.4	33,081.5	39,871.5	48,108.4	58,138.5	70,390.1	85,404.3	20.9%
BFSI	13,708.5	16,616.7	20,166.8	24,505.6	29,827.1	36,373.3	44,450.7	21.7%
Retail	11,226.3	13,451.3	16,136.8	19,382.2	23,318.6	28,107.7	33,952.6	20.3%
Healthcare	5,313.8	6,465.1	7,874.1	9,601.2	11,725.6	14,346.8	17,591.2	22.2%
Automotive	9,871.0	11,690.9	13,864.1	16,462.3	19,580.1	23,332.9	27,864.4	19.0%
Others*	7,315.2	8,700.4	10,363.7	12,360.0	14,761.4	17,654.7	21,146.6	19.4%
Total	74,911.2	90,005.8	1,08,277.0	1,30,419.6	1,57,351.3	1,90,205.5	2,30,409.8	20.7%

e: estimated; p: projected

Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 38 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	27,476.4	32,665.4	38,852.1	46,223.7	55,030.6	65,558.8	78,222.6	19.1%
BFSI	13,708.5	16,405.2	19,641.6	23,523.4	28,191.2	33,807.8	40,609.1	19.9%
Retail	11,226.3	13,361.6	15,910.3	18,948.9	22,578.6	26,916.5	32,138.7	19.2%
Healthcare	5,313.8	6,444.6	7,818.6	9,487.5	11,518.5	13,991.6	17,023.5	21.4%
Automotive	9,871.0	11,584.9	13,602.0	15,970.7	18,754.9	22,027.9	25,913.8	17.5%
Others*	7,315.2	8,607.6	10,123.5	11,924.0	14,085.4	16,687.1	19,806.1	18.1%
Total	74,911.2	89,069.4	1,05,948.1	1,26,078.2	1,50,159.1	1,78,989.7	2,13,713.7	19.1%

e: estimated; p: projected

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

9.2 GOVERNMENT AND DEFENSE

9.2.1 INCREASING NEED TO SECURE CONFIDENTIAL DATA FROM THEFTS AND UNAUTHORIZED ACCESS

The government vertical primarily consists of 3 institutional units, namely, central government, state government, and local government. Public utilities are those businesses that fulfill the everyday needs of citizens. With the increasing use of mobility, the government bodies across the globe have progressively started using mobile devices to enhance the productivity of workers and improve the functioning of public sector departments. They have to work on critical information, intelligence reports, and other confidential data. This highly confidential information is used by employees across various locations and is transferred through mobile devices and other networks such as social media and e-mails. Therefore, it is necessary to secure this confidential data from thefts and unauthorized access. Here, quantum cryptography plays a major role in ensuring security at the time of data transfer and reduce the risk of data loss.

9.2.2 GOVERNMENT AND DEFENSE: COVID-19 IMPACT

- Quantum cryptography helps improve the security of critical information and reduce the threat of unauthorized access in the government and defense vertical.
- During the COVID-19 scenario, access to sensitive government information through unsecured platforms has increased the risk of cyberattacks, such as data and identity thefts, among government and defense organizations, which is expected to drive the adoption of quantum cryptography in this vertical.

TABLE 39 GOVERNMENT AND DEFENSE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	10,077.6	12,347.8	15,130.0	22.5%
Europe	4,649.8	5,524.8	6,566.3	18.8%
APAC	3,348	3,931	4,614.4	17.4%
MEA	909.5	1,031.3	1,165.6	13.2%
Total	18,984.7	22,834.5	27,476.4	20.3%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 40 GOVERNMENT AND DEFENSE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	15,130.0	18,451.9	22,426.5	27,208.4	33,002.4	40,065.6	48,732.8	21.4%
Europe	6,566.3	7,835.4	9,387.9	11,283.2	13,600.0	16,434.2	19,903.4	20.5%
APAC	4,614.4	5,456	6,496	7,774	9,343	11,266	13,617	20.1%
MEA	1,165.6	1,338.3	1,561.1	1,842.4	2,192.9	2,624.7	3,150.8	18.7%
Total	27,476.4	33,081.5	39,871.5	48,108.4	58,138.5	70,390.1	85,404.3	20.9%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 41 GOVERNMENT AND DEFENSE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	15,130.0	18,181.3	21,854.4	26,229.4	31,446.9	37,664.1	45,103.6	19.9%
Europe	6,566.3	7,862.4	9,417.6	11,241.6	13,327.8	15,692.6	18,505.5	18.7%
APAC	4,614.4	5,400.2	6,328.9	7,428.3	8,735.5	10,293.2	12,163.3	17.6%
MEA	1,165.6	1,221.6	1,251.3	1,324.5	1,520.4	1,908.9	2,450.2	14.9%
Total	27,476.4	32,665.4	38,852.1	46,223.7	55,030.6	65,558.8	78,222.6	19.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

9.3 BANKING, FINANCIAL SERVICES, AND INSURANCE

9.3.1 NEED TO CATER TO FINANCIAL SECURITY, ONLINE FUND TRANSFER SECURITY, AND DATA SECURITY

To secure the transactional data, banks and financial institutions must ensure real-time data availability and protect the client and proprietary sensitive information. Quantum cryptography enables banking firms to implement quantum-safe encryption with easily manageable security policies that support all networks. Quantum random number generators are used to provide encryption keys for critical security applications, as well as, to deliver the randomness input for advanced financial simulations, such as Monte Carlo simulations.

The development of mobile and other online facilities with ubiquitous connectivity has compelled enterprises to provide advanced data security services to their customers. The increasing use of mobile technology has enhanced the adoption of quantum cryptography hardware and services in financial organizations. Quantum cryptography is used in the BFSI vertical to control and secure sensitive data of customers and provide protection to organizations from internal and external theft risks. The pressure from regulatory standards, such as Advanced Encryption Standard (AES) and Payment Card Industry Data Security Standard (PCI DSS), has led companies in this vertical to deliver security solutions with competent cryptography. Quantum cryptography helps banking and financial companies in catering financial security, online fund transfer security, and data security by encrypting the files, data, and emails with unbreakable quantum mechanisms.

9.3.2 BANKING, FINANCIAL SERVICES, AND INSURANCE: COVID-19 IMPACT

- Quantum cryptography helps financial institutions effectively secure sensitive information and ensure data privacy.
- Due to COVID-19, the BFSI vertical is moving toward high technological adoption, making it a prime target for cybercriminals to exploit sensitive information, which is expected to drive the adoption of quantum cryptography during this pandemic.

TABLE 42 BANKING, FINANCIAL SERVICES, AND INSURANCE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	4,773.6	5,892.1	7,273.0	23.4%
Europe	2,324.9	2,783.8	3,334.2	19.8%
APAC	1,809.6	2,138.5	2,526.8	18.2%
MEA	441.0	504.1	574.5	14.1%
Total	9,349.1	11,318.5	13,708.5	21.1%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 43 BANKING, FINANCIAL SERVICES, AND INSURANCE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	7,273.0	8,935.2	10,940.0	13,370.6	16,337.5	19,980.3	24,481.9	22.3%
Europe	3,334.2	4,009.4	4,840.9	5,863.3	7,121.9	8,672.6	10,584.7	21.4%
APAC	2,526.8	3,007.0	3,603.6	4,340.9	5,250.8	6,372.3	7,752.7	20.9%
MEA	574.5	665.1	782.2	930.8	1,117.0	1,348.0	1,631.5	19.7%
Total	13,708.5	16,616.7	20,166.8	24,505.6	29,827.1	36,373.3	44,450.7	21.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 44 BANKING, FINANCIAL SERVICES, AND INSURANCE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	7,273.0	8,800.3	10,651.5	12,872.4	15,540.0	18,741.3	22,598.7	20.8%
Europe	3,334.2	4,021.7	4,852.6	5,835.2	6,969.0	8,266.0	9,819.5	19.5%
APAC	2,526.8	2,975.9	3,509.8	4,145.6	4,906.0	5,817.5	6,918.0	18.4%
MEA	574.5	607.4	627.7	670.2	776.1	983.1	1,272.9	15.9%
Total	13,708.5	16,405.2	19,641.6	23,523.4	28,191.2	33,807.8	40,609.1	19.9%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

9.4 RETAIL

9.4.1 INCREASING ADOPTION OF QUANTUM-SAFE ENCRYPTION HARDWARE TO ENCRYPT AND PROTECT CUSTOMERS' CONFIDENTIAL AND PRIVATE INFORMATION

The retail vertical is different from other verticals, such as BFSI and government and defense, as it has diversified networks that connect the retailers to their customers. The advancement in mobile technologies has encouraged this vertical to adopt mobile devices with real-time access, in order to provide better quality services to customers. However, the use of mobile devices and unsecured networks has led to an increase in data thefts and data loss among retailers, who have direct access to the customers' private information. Furthermore, the growing instances of cyber attacks and data theft activities have increased the pressure on retailers to secure their processes. This has resulted in the increased adoption of quantum-safe encryption hardware to encrypt and protect the confidential and private information of the customers.

Additionally, the regulatory compliances, such as AES and PCI DSS, have assisted in ensuring the privacy of the customer-sensitive data. The attacks on POS systems are also encouraging the use of data encryption, file encryption, and email encryption software. However, IT budget constraints are hampering the adoption trend of quantum cryptography hardware and services.

9.4.2 RETAIL: COVID-19 IMPACT

- Retail companies are expected to deploy quantum cryptography solutions for safeguarding customers' sensitive information.
- Due to COVID-19 pandemic, the use of mobile devices has led to an increase in data thefts and data losses among retailers with direct access link to customers' private information. This is expected to drive the adoption of quantum cryptography.

TABLE 45 RETAIL: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	3,978.0	4,853.2	5,921.3	22.0%
Europe	2,051.4	2,426.6	2,871.3	18.3%
APAC	1,447.7	1,692.7	1,979.0	16.9%
MEA	358.3	404.3	454.7	12.7%
Total	7,835.4	9,376.9	11,226.3	19.7%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 46 RETAIL: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	5,921.3	7,190.5	8,701.9	10,512.2	12,696.1	15,347.4	18,587.5	20.9%
Europe	2,871.3	3,411.1	4,068.8	4,868.7	5,842.4	7,028.6	8,474.7	20.0%
APAC	1,979.0	2,330.2	2,763.1	3,293.2	3,941.5	4,732.8	5,697.2	19.6%
MEA	454.7	519.5	603.0	708.1	838.7	998.9	1,193.2	18.1%
Total	11,226.3	13,451.3	16,136.8	19,382.2	23,318.6	28,107.7	33,952.6	20.3%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 47 RETAIL: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	5,921.3	7,123.4	8,571.9	10,299.3	12,361.6	14,821.9	17,769.2	20.1%
Europe	2,871.3	3,442.1	4,127.8	4,933.2	5,855.6	6,902.7	8,149.6	18.8%
APAC	1,979.0	2,318.5	2,720.3	3,196.3	3,762.9	4,438.7	5,250.9	17.8%
MEA	454.7	477.6	490.3	520.2	598.5	753.2	969.0	15.2%
Total	11,226.3	13,361.6	15,910.3	18,948.9	22,578.6	26,916.5	32,138.7	19.2%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

9.5 HEALTHCARE

9.5.1 NEED TO IMPROVE THE PROFICIENCY OF CLINICAL AND IT STAFF, AND PROVIDE REAL-TIME INFORMATION ACCESS DURING EMERGENCIES SUCH AS PANDEMIC

The healthcare vertical covers the personal health information and critical data about patients. The patients' data needs high security; thus, data security is of utmost importance in this vertical. Healthcare organizations use quantum cryptography to prevent theft of the medical records of patients. Quantum cryptography hardware and services help healthcare professionals secure patients' personal health information on their personal devices, such as laptops and smartphones. Furthermore, the encryption of data, files, and folders help healthcare professionals to ensure the security and privacy of confidential data, financial transactions, and medical records related to the patients. Moreover, the use of the quantum-safe encryption technology improves the proficiency of clinical and IT staff, and provides real-time information access, in case of emergencies. Additionally, the implementation of regulatory compliances, such as Health Insurance Portability and Accountability Act of 1996 (HIPAA), is increasing the adoption rate of quantum cryptography among healthcare organizations and clinical staff. These compliance regulations also help secure the sensitive data, protect the identities by providing digital signatures, secure the network gateways, and encrypt the databases. Due to the rising use of this technology and increasing awareness about regulatory compliance, the adoption trend for quantum cryptography is expected to grow among the developing countries.

9.5.2 HEALTHCARE: COVID-19 IMPACT

- During the COVID-19, healthcare facilities are more prone to becoming a target of cyberattacks. The healthcare sector has become the epicenter of this unexpected global pandemic. Increased investments by the government in the healthcare vertical to tackle COVID-19 as well as being prepared for any such future scenarios, is expected to make the healthcare vertical adopt these solutions in the near future.
- The growing need for securing patients' data is expected to boost the adoption of quantum cryptography in the healthcare vertical. The acceleration in digital technologies and connectivity in healthcare departments have increased the exposure to cyberattacks and contributed to the growth of the quantum cryptography market.

TABLE 48 HEALTHCARE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	1,856.4	2,298.6	2,846.2	23.8%
Europe	1,094.1	1,311.6	1,572.8	19.9%
APAC	452.4	539.2	642.5	19.2%
MEA	192.9	221.0	252.4	14.4%
Total	3,595.8	4,370.3	5,313.8	21.6%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 49 HEALTHCARE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	2,846.2	3,507.6	4,308.1	5,281.8	6,474.0	7,942.5	9,762.4	22.7%
Europe	1,572.8	1,893.6	2,289.1	2,775.9	3,375.8	4,115.8	5,029.3	21.6%
APAC	642.5	771.1	931.9	1,132.1	1,381.0	1,690.1	2,073.7	21.9%
MEA	252.4	292.8	345.1	411.5	494.8	598.4	725.7	19.9%
Total	5,313.8	6,465.1	7,874.1	9,601.2	11,725.6	14,346.8	17,591.2	22.2%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 50 HEALTHCARE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	2,846.2	3,486.9	4,273.2	5,228.7	6,391.2	7,804.2	9,528.1	22.3%
Europe	1,572.8	1,916.0	2,334.9	2,835.5	3,420.2	4,097.0	4,915.4	20.7%
APAC	642.5	771.7	928.3	1,118.4	1,349.9	1,632.6	1,980.1	20.7%
MEA	252.4	270.0	282.3	304.9	357.3	457.8	599.8	17.3%
Total	5,313.8	6,444.6	7,818.6	9,487.5	11,518.5	13,991.6	17,023.5	21.4%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

9.6 AUTOMOTIVE

9.6.1 INCREASING DEMAND TO PROVIDE PHYSICAL ACCESS AND PROTECTION TO CONFIDENTIAL INFORMATION AND CRITICAL SAFETY SYSTEMS

The automotive vertical is also set to gain traction due to the increase in security demand generated by the data coming from connected vehicles as well as connected devices in the automobile industry. The increasing connectivity of vehicles via mobile networks enables a lot of new services and interactions between drivers and end users. Cryptography plays a major role in securing cars against potential threats. Moreover, quantum cryptography provides not only physical access and protection of confidential information but also considers critical safety systems. Furthermore, its security features can prevent “chipping” altogether or can detect the presence of any unauthorized modification and take action to mitigate or eliminate the effects of the modification so that the vehicle is still safe to drive while indicating the need for corrective attention.

9.6.2 AUTOMOTIVE: COVID-19 IMPACT

- The automotive industry is a primary victim of the COVID-19 outbreak, and the pandemic is affecting the industry in multiple ways. However, automotive connectivity is a rapidly evolving segment of the IoT world with ever-increasing complexities.
- The exponential growth in data usage by connected cars has opened numerous doors for cyberattacks to impact the integrity of the vehicle, thereby leading to increased adoption of quantum cryptography solutions.

TABLE 51 AUTOMOTIVE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	3,447.6	4,155.9	5,009.8	20.5%
Europe	2,188.2	2,565.0	3,007.5	17.2%
APAC	1,085.8	1,252.5	1,444.6	15.3%
MEA	330.7	368.4	409.0	11.2%
Total	7,052.2	8,341.7	9,871.0	18.3%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 52 AUTOMOTIVE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	5,009.8	6,010.9	7,187.4	8,578.7	10,237.1	12,226.9	14,631.1	19.5%
Europe	3,007.5	3,540.6	4,185.1	4,962.4	5,900.9	7,034.8	8,405.3	18.9%
APAC	1,444.6	1,678.1	1,963.1	2,308.3	2,725.6	3,228.8	3,834.5	18.0%
MEA	409.0	461.3	528.6	612.8	716.5	842.4	993.4	16.6%
Total	9,871.0	11,690.9	13,864.1	16,462.3	19,580.1	23,332.9	27,864.4	19.0%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 53 AUTOMOTIVE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	5,009.8	5,935.0	7,033.0	8,321.4	9,835.5	11,613.3	13,710.4	18.2%
Europe	3,007.5	3,561.0	4,217.9	4,978.7	5,836.9	6,796.0	7,924.8	17.4%
APAC	1,444.6	1,664.5	1,920.6	2,219.3	2,569.5	2,980.9	3,467.9	15.8%
MEA	409.0	424.5	430.5	451.3	513.0	637.8	810.6	13.8%
Total	9,871.0	11,584.9	13,602.0	15,970.7	18,754.9	22,027.9	25,913.8	17.5%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

9.7 OTHERS

The other verticals in the quantum cryptography market include education, telecom, energy and utilities, and logistics. The information in the above-given industries is accessible to engineers, production departments, contractors, and executives, which in turn, increases the risk of data loss and theft. Thus, companies in this vertical have started using quantum cryptography to protect their data. This will reduce the cost and complexity while securing data. Additionally, in today's education system, schools and universities have centralized and localized the administrative departments that handle accounting processes, purchase of educational materials, financials, and student information for the entire organization. This means that large volumes of highly sensitive data are exchanged every day, often through fax machines, emails, and wired networks. With security threats becoming a growing concern, protecting this sensitive data has become important. Quantum cryptography enables educational institutes to store and share private information of the students, without the risk of any theft. Moreover, quantum cryptography hardware and services are helping telecommunication companies to enhance their services and provide secured information to the customers while complying with regulations, such as AES.

TABLE 54 OTHERS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	2,386.8	2,900.5	3,522.6	21.5%
Europe	1,367.6	1,612.2	1,900.0	17.9%
APAC	904.8	1,054.5	1,228.0	16.5%
MEA	523.6	590.9	664.6	12.7%
Total	5,182.8	6,158.1	7,315.2	18.8%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 55 OTHERS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	3,522.6	4,255.1	5,119.0	6,142.8	7,364.5	8,830.3	10,600.0	20.0%
Europe	1,900.0	2,247.0	2,666.6	3,172.8	3,783.7	4,521.1	5,410.9	19.2%
APAC	1,228.0	1,439.4	1,697.9	2,011.8	2,392.1	2,851.5	3,405.2	18.8%
MEA	664.6	758.9	880.2	1,032.5	1,221.1	1,451.7	1,730.5	17.9%
Total	7,315.2	8,700.4	10,363.7	12,360.0	14,761.4	17,654.7	21,146.6	19.4%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 56 OTHERS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	3,522.6	4,214.4	5,039.9	6,013.6	7,162.4	8,515.5	10,114.8	19.1%
Europe	1,900.0	2,265.9	2,701.7	3,208.4	3,782.0	4,425.0	5,182.1	18.0%
APAC	1,228.0	1,432.0	1,671.0	1,951.3	2,281.2	2,670.1	3,131.7	16.9%
MEA	664.6	695.3	710.9	750.8	859.7	1,076.4	1,377.4	14.7%
Total	7,315.2	8,607.6	10,123.5	11,924.0	14,085.4	16,687.1	19,806.1	18.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

10 QUANTUM CRYPTOGRAPHY MARKET, BY REGION

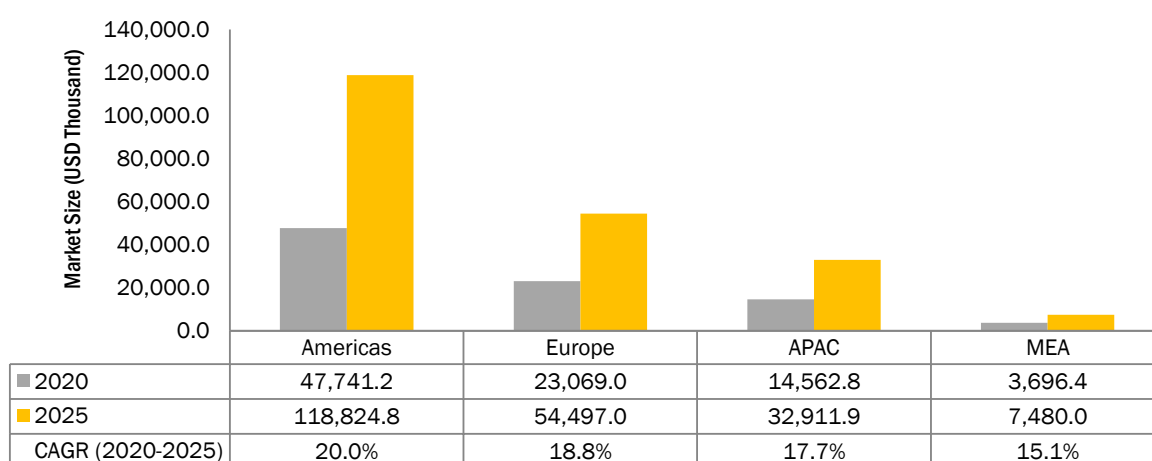
KEY FINDINGS

- The Americas region is estimated to hold the largest market size of USD thousand 47,741.2 in 2020 and projected to grow at a CAGR of 20.0% during the forecast period to reach USD 1,18,824.8 thousand by 2025.
- Europe is estimated to be the second-largest contributor to the overall quantum cryptography market in 2020 and is projected to grow at a CAGR of 18.8% during the forecast period.
- APAC is expected to grow from USD 14,562.8 thousand in 2020 to USD 32,911.9 thousand by 2025, at a CAGR of 17.7% during the forecast period.
- The growing IT management complexities and increasing need for data privacy and security are important factors driving the quantum cryptography market in North America.
- Countries such as the US and UK are at the forefront in the development of quantum cryptographic solutions.

10.1 INTRODUCTION

This chapter includes the segmentation of the quantum cryptography market on the basis of regions. The quantum cryptography market is broadly classified into 4 major regional markets, namely, the Americas, Europe, APAC, and MEA. This chapter provides insights about these regional markets in terms of market size, growth rate, and future trends. The widespread adoption of PCs and internet services for business purpose and storage of critical data across the world has increased the propensity of data breach and theft. In addition, the major growth factors that would drive the adoption of quantum cryptography include the growing need for data privacy and security and an increasing number of cyber attacks. The Americas and Europe regions are the largest revenue contributors in the quantum cryptography market, as the market is being driven by the early adoption of advanced technologies such as cloud and IoT, high level of IT capabilities, and higher IT investments. The other regions namely APAC and MEA are expected to witness significant growth in the quantum cryptography market as traction is being observed across these regions.

FIGURE 19 THE AMERICAS TO HOLD THE LARGEST MARKET SIZE DURING THE FORECAST PERIOD



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 57 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, 2017–2019 (USD THOUSAND)

Region	2017	2018	2019	CAGR (2017–2019)
Americas	26,520.0	32,448.0	39,702.9	22.4%
Europe	13,676.0	16,224.0	19,252.2	18.6%
APAC	9,048.0	10,608.0	12,435.3	17.2%
MEA	2,756.0	3,120.0	3,520.8	13.0%
Total	52,000.0	62,400.0	74,911.2	20.0%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 58 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	39,702.9	48,351.3	58,682.8	71,094.4	86,111.6	1,04,393.0	1,26,795.8	21.3%
Europe	19,252.2	22,937.0	27,438.4	32,926.3	39,624.7	47,807.2	57,808.5	20.3%
APAC	12,435.3	14,681.7	17,455.6	20,860.9	25,034.2	30,141.1	36,380.4	19.9%
MEA	3,520.8	4,035.8	4,700.2	5,538.1	6,580.9	7,864.1	9,425.2	18.5%
Total	74,911.2	90,005.8	1,08,277.0	1,30,419.6	1,57,351.3	1,90,205.5	2,30,409.8	20.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 59 QUANTUM CRYPTOGRAPHY MARKET SIZE, BY REGION, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Region	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Americas	39,702.9	47,741.2	57,423.9	68,964.8	82,737.7	99,160.3	1,18,824.8	20.0%
Europe	19,252.2	23,069.0	27,652.4	33,032.5	39,191.5	46,179.3	54,497.0	18.8%
APAC	12,435.3	14,562.8	17,078.8	20,059.0	23,605.0	27,832.9	32,911.9	17.7%
MEA	3,520.8	3,696.4	3,792.9	4,021.9	4,624.9	5,817.2	7,480.0	15.1%
Total	74,911.2	89,069.4	1,05,948.1	1,26,078.2	1,50,159.1	1,78,989.7	2,13,713.7	19.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

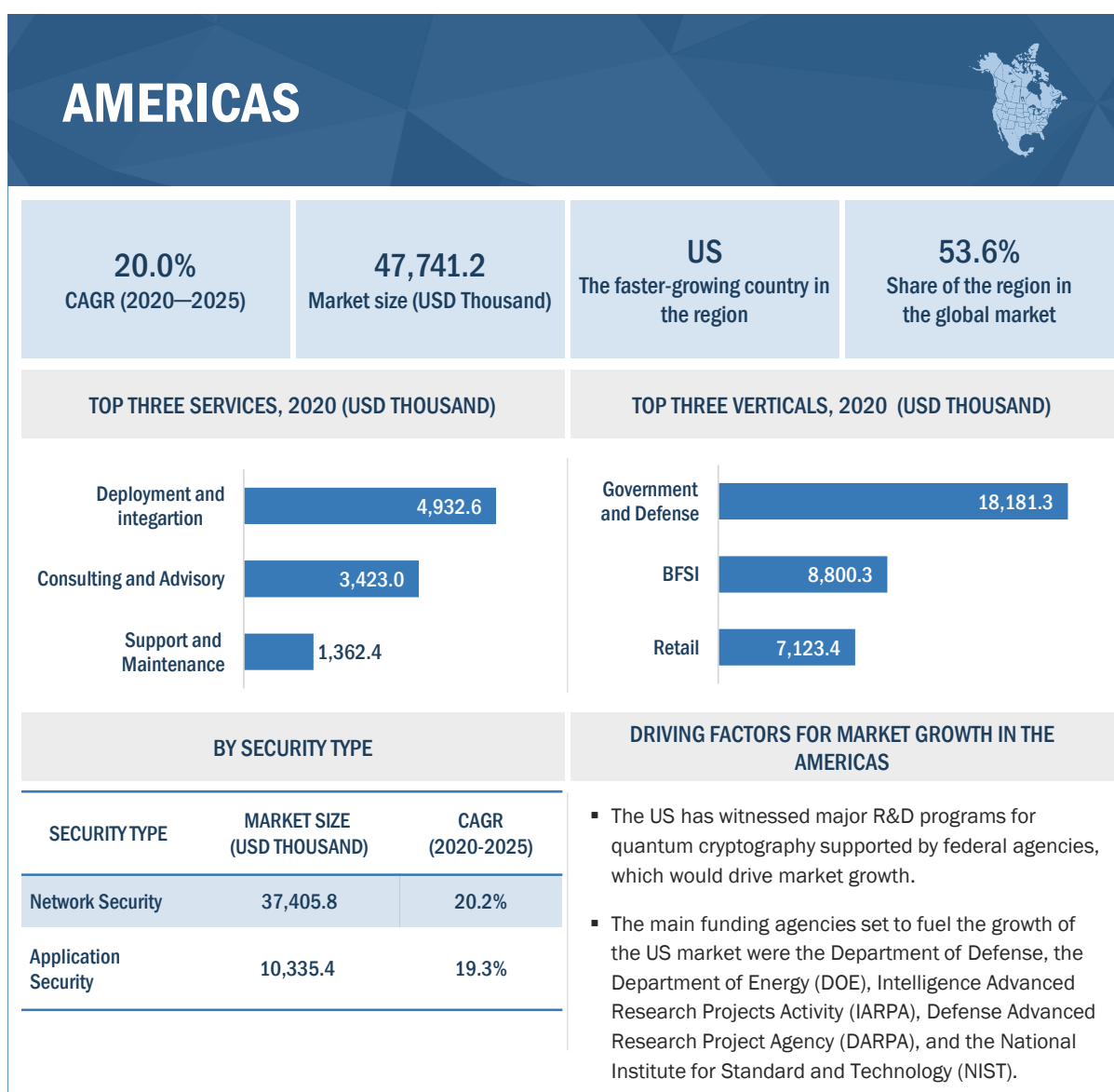
10.2 THE AMERICAS

The Americas are expected to be the largest contributor to the quantum cryptography market in 2019. The rise in cyber attacks in this region is one of the major factors responsible for the growth of the market. Due to the high adoption of encryption-based applications in the Americas, the attacks are increasing dramatically and becoming more sophisticated. Furthermore, the growing IT management complexity, along with the increasing need for data privacy and security is an important factor driving the quantum cryptography market in the region. Countries such as the US, Canada, Brazil, and Mexico are generating significant revenue in this market. The organizations in this region are focusing on securing their applications and networks using various quantum cryptography appliances. The other drivers for the increasing demand for quantum cryptography appliances include the evolution of next-generation wireless networks and an increase in the growth of R&D investments in the quantum cryptography market.

10.2.1 AMERICAS: COVID-19 IMPACT

- With the increasing demand for digitalization, there is a surge in the demand for quantum cryptography. Companies know that data recovery is very costly, and with the outbreak of COVID-19, there is rise in the number of cyberattacks for which companies are buying quantum cryptography solutions to protect their data.
- Consulting services for deploying quantum cryptography enable organizations to reduce the time and cost needed to deploy quantum cryptography solutions while reducing other business risks amid the pandemic. Region-wide regulations, such as the HIPAA in the US and the industry-wide standards such as PCI DSS standards are also driving the quantum cryptography market growth.

FIGURE 20 THE AMERICAS: MARKET SNAPSHOT



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 60 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, 2017–2019 (USD THOUSAND)

Component	2017	2018	2019	CAGR (2017–2019)
Solutions	21,481.2	26,182.7	31,914.2	21.9%
Services	5,038.8	6,265.3	7,788.7	24.3%
Total	26,520.0	32,448.0	39,702.9	22.4%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 61 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Solutions	31,914.2	38,716.7	46,808.3	56,489.0	68,155.1	82,302.2	99,572.7	20.8%
Services	7,788.7	9,634.6	11,874.5	14,605.5	17,956.4	22,090.9	27,223.1	23.1%
Total	39,702.9	48,351.3	58,682.8	71,094.4	86,111.6	1,04,393.0	1,26,795.8	21.3%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 62 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Solutions	31,914.2	38,023.2	45,600.7	54,659.2	65,488.5	78,249.4	93,160.3	19.6%
Services	7,788.7	9,718.0	11,823.1	14,305.6	17,249.2	20,910.9	25,664.5	21.4%
Total	39,702.9	47,741.2	57,423.9	68,964.8	82,737.7	99,160.3	1,18,824.8	20.0%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 63 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, 2017–2019 (USD THOUSAND)

Service	2017	2018	2019	CAGR (2017–2019)
Consulting and Advisory	1,763.6	2,193.6	2,728.0	24.4%
Deployment and Integration	2,519.4	3,155.4	3,951.1	25.2%
Support and Maintenance	755.8	916.3	1,109.6	21.2%
Total	5,038.8	6,265.3	7,788.7	24.3%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 64 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	2,728.0	3,375.7	4,162.0	5,121.0	6,298.2	7,751.1	9,555.3	23.1%
Deployment and Integration	3,951.1	4,923.0	6,111.6	7,571.8	9,376.6	11,619.4	14,422.8	24.0%
Support and Maintenance	1,109.6	1,335.9	1,600.9	1,912.6	2,281.6	2,720.4	3,245.0	19.4%
Total	7,788.7	9,634.6	11,874.5	14,605.5	17,956.4	23,038.1	27,223.1	24.5%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 65 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	2,728.0	3,423.0	4,188.0	5,096.0	6,179.3	7,533.4	9,298.3	22.1%
Deployment and Integration	3,951.1	4,932.6	6,004.5	7,269.3	8,770.0	10,637.7	13,063.2	21.5%
Support and Maintenance	1,109.6	1,362.4	1,630.6	1,940.3	2,299.9	2,739.8	3,303.0	19.4%
Total	7,788.7	9,718.0	11,823.1	14,305.6	17,249.2	20,910.9	25,664.5	21.4%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 66 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, 2017–2019 (USD THOUSAND)

Security Type	2017	2018	2019	CAGR (2017–2019)
Network Security	20,685.6	25,347.4	31,061.2	22.5%
Application Security	5,834.4	7,100.6	8,641.8	21.7%
Total	26,520.0	32,448.0	39,702.9	22.4%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 67 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	31,061.2	37,883.8	46,047.6	55,870.4	67,773.2	82,284.6	1,00,092.6	21.4%
Application Security	8,641.8	10,467.5	12,635.2	15,224.0	18,338.4	22,108.5	26,703.2	20.6%
Total	39,702.9	48,351.3	58,682.8	71,094.4	86,111.6	1,04,393.0	1,26,795.8	21.3%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 68 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	31,061.2	37,405.8	45,059.7	54,196.8	65,117.8	78,160.0	93,800.3	20.2%
Application Security	8,641.8	10,335.4	12,364.2	14,768.0	17,619.9	21,000.3	25,024.5	19.3%
Total	39,702.9	47,741.2	57,423.9	68,964.8	82,737.7	99,160.3	1,18,824.8	20.0%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 69 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, 2017–2019 (USD THOUSAND)

Vertical	2017	2018	2019	CAGR (2017–2019)
Government and Defense	10,077.6	12,347.8	15,130.0	22.5%
BFSI	4,773.6	5,892.1	7,273.0	23.4%
Retail	3,978.0	4,853.2	5,921.3	22.0%
Healthcare	1,856.4	2,298.6	2,846.2	23.8%
Automotive	3,447.6	4,155.9	5,009.8	20.5%
Others*	2,386.8	2,900.5	3,522.6	21.5%
Total	26,520.0	32,448.0	39,702.9	22.4%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 70 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	15,130.0	18,451.9	22,426.5	27,208.4	33,002.4	40,065.6	48,732.8	21.4%
BFSI	7,273.0	8,935.2	10,940.0	13,370.6	16,337.5	19,980.3	24,481.9	22.3%
Retail	5,921.3	7,190.5	8,701.9	10,512.2	12,696.1	15,347.4	18,587.5	20.9%
Healthcare	2,846.2	3,507.6	4,308.1	5,281.8	6,474.0	7,942.5	9,762.4	22.7%
Automotive	5,009.8	6,010.9	7,187.4	8,578.7	10,237.1	12,226.9	14,631.1	19.5%
Others*	3,522.6	4,255.1	5,119.0	6,142.8	7,364.5	8,830.3	10,600.0	20.0%
Total	39,702.9	48,351.3	58,682.8	71,094.4	86,111.6	1,04,393.0	1,26,795.8	21.3%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 71 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	15,130.0	18,181.3	21,854.4	26,229.4	31,446.9	37,664.1	45,103.6	19.9%
BFSI	7,273.0	8,800.3	10,651.5	12,872.4	15,540.0	18,741.3	22,598.7	20.8%
Retail	5,921.3	7,123.4	8,571.9	10,299.3	12,361.6	14,821.9	17,769.2	20.1%
Healthcare	2,846.2	3,486.9	4,273.2	5,228.7	6,391.2	7,804.2	9,528.1	22.3%
Automotive	5,009.8	5,935.0	7,033.0	8,321.4	9,835.5	11,613.3	13,710.4	18.2%
Others*	3,522.6	4,214.4	5,039.9	6,013.6	7,162.4	8,515.5	10,114.8	19.1%
Total	39,702.9	47,741.2	57,423.9	68,964.8	82,737.7	99,160.3	1,18,824.8	20.0%

e: estimated; p: projected

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 72 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, 2017–2019 (USD THOUSAND)

Country	2017	2018	2019	CAGR (2017–2019)
US	19,624.8	24,086.2	29,562.8	22.7%
Canada	5,834.4	7,122.3	8,694.9	22.1%
Rest of Americas*	1,060.8	1,239.5	1,445.2	16.7%
Total	26,520.0	32,448.0	39,702.9	22.4%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 73 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Country	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
US	29,562.8	36,113.6	43,965.1	53,427.5	64,910.9	78,931.6	96,161.9	21.6%
Canada	8,694.9	10,564.8	12,792.8	15,463.0	18,686.2	22,601.1	27,387.9	21.0%
Rest of Americas*	1,445.2	1,673.0	1,924.8	2,203.9	2,514.5	2,860.4	3,246.0	14.2%
Total	39,702.9	48,351.3	58,682.8	71,094.4	86,111.6	1,04,393.0	1,26,795.8	21.3%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 74 AMERICAS: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Country	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
US	29,562.8	35,677.0	43,067.9	51,909.8	62,500.0	75,173.4	90,401.9	20.4%
Canada	8,694.9	10,407.6	12,461.0	14,896.4	17,788.6	21,220.3	25,309.7	19.5%
Rest of Americas*	1,445.2	1,656.6	1,895.0	2,158.6	2,449.0	2,766.6	3,113.2	13.4%
Total	39,702.9	47,741.2	57,423.9	68,964.8	82,737.7	99,160.3	1,18,824.8	20.0%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

10.2.2 UNITED STATES

10.2.2.1 Increase in funding and adoption of quantum technology

The US has witnessed major R&D programs for quantum cryptography supported by federal agencies. This support is federally funded basic and applied research in Quantum Information Sciences (QIS). The main funding agencies were Department of Defense, the Department of Energy (DOE), Intelligence Advanced Research Projects Activity (IARPA), Defense Advanced Research Project Agency (DARPA), National Institute for Standard and Technology (NIST), and the National Science Foundation. Moreover, under National Photonics Initiative, the US government provided USD 500 million over 5 years to set up some Quantum Innovations Labs, which will accelerate the development of commercially available quantum-based technologies to facilitate growth in the US economy.

10.2.3 CANADA

10.2.3.1 Growing need for a developed quantum ecosystem

The Canadian government has also developed quantum ecosystem under the national quantum strategy. In particular, the Ontario regional government has a well-established program aiming at the commercialization of quantum technologies, in collaboration with private investors such as the Lazaridis and Fregin's Quantum Valley Investments fund. Moreover, Canada has invested over USD 1 billion in quantum R&D over the last decade and is currently a global leader in this emerging field. As quantum technologies increasingly disrupt current approaches, Canada is well-positioned to leverage its strengths in quantum R&D and its industrial base for global research leadership and long-term economic prosperity.

10.2.4 REST OF AMERICAS

The quantum cryptography market in South America is set to succeed due to the growing usage and sharing of digital information across various channels. Furthermore, it is backed by growing IoT adoption, inclusive of Industrial IoT, smart grids, smart cities, smart consumer wearables, smart appliances and smartphones, which generates huge data that needs to be protected. Moreover, concerns related to data security owing to increasing incidences of data theft and cyberattacks, growing cybersecurity market and booming fiber optics industry are anticipated to positively influence the region's quantum cryptography market during the forecast period.

10.3 EUROPE

The region comprises major growing economies such as the UK, Germany, and France. Europe exhibits a progressive growth in the adoption of quantum cryptography. The growing number of mobile user expansively accessing business data across the region has resulted in concerns regarding data security and privacy. Furthermore, the growing trend of large-scale adoption of cloud has resulted in the simplified accessibility of data across the enterprise network. However, the proliferation in the adoption of cloud services has attracted sophisticated cyber attacks on cloud data. These concerns have led to the substantial adoption of quantum cryptography solutions among businesses. Factors such as data security and data sovereignty are compelling enterprises in this region to adopt quantum cryptography solutions. Furthermore, the cohesive involvement of the European government to strengthen the data security of organizations and its people has fortified the growth of security solutions in this region. In addition, for the security of end-users' personal and work-related data, and to avert privacy breaches, the government has largely encouraged the enforcement of stringent laws for data security that the encryption solution providers and cloud providers have to comply with. Presently, the organizations in the UK, Germany, and France are investing in cryptography solutions for preventing privacy breaches and to have accountability for complete data security.

10.3.1 EUROPE: COVID-19 IMPACT

- A rise in the use of cloud-based software across different industries has increased the use of quantum cryptography solutions. Enterprises are witnessing the increasing demand for training and consulting services to ensure the effective deployment of quantum cryptography solutions in the region. The increasing establishment of security companies in the database and the services offered are also driving the quantum encryption market.
- Europe's data protection rules are the strongest in the world, which is also a factor for the growing adoption of quantum cryptography solutions. The growing need to assist customers in making the right long-term architectural decisions, prior to implementing quantum cryptography solutions is a major driver expected to fuel the quantum cryptography market growth in this region.

TABLE 75 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, 2017-2019 (USD THOUSAND)

Component	2017	2018	2019	CAGR (2017-2019)
Solutions	11,214.3	13,237.2	15,629.5	18.1%
Services	2,461.7	2,986.8	3,622.7	21.3%
Total	13,676.0	16,224.0	19,252.2	18.6%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 76 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, PRE-COVID-19 SCENARIO, 2019-2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020-2025)
Solutions	15,629.5	18,528.0	22,053.4	26,332.1	31,530.8	37,851.9	45,542.0	19.7%
Services	3,622.7	4,409.0	5,385.0	6,594.1	8,093.9	9,955.2	12,266.4	22.7%
Total	19,252.2	22,937.0	27,438.4	32,926.3	39,624.7	47,807.2	57,808.5	20.3%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 77 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Solutions	15,629.5	18,634.6	22,225.5	26,417.1	31,186.1	36,563.1	42,933.2	18.2%
Services	3,622.7	4,434.4	5,427.0	6,615.4	8,005.4	9,616.3	11,563.8	21.1%
Total	19,252.2	23,069.0	27,652.4	33,032.5	39,191.5	46,179.3	54,497.0	18.8%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 78 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, 2017–2019 (USD THOUSAND)

Service	2017	2018	2019	CAGR (2017–2019)
Consulting and Advisory	837.0	1,019.5	1,241.5	21.8%
Deployment and Integration	1,206.2	1,475.2	1,803.7	22.3%
Support and Maintenance	418.5	492.0	577.5	17.5%
Total	2,461.7	2,986.8	3,622.7	21.3%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 79 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	1,241.5	1,517.1	1,860.3	2,287.1	2,818.4	3,480.4	4,305.5	23.2%
Deployment and Integration	1,803.7	2,212.8	2,724.2	3,362.7	4,160.5	5,158.3	6,406.8	23.7%
Support and Maintenance	577.5	679.2	800.5	944.4	1,115.0	1,316.5	1,554.2	18.0%
Total	3,622.7	4,409.0	5,385.0	6,594.1	8,093.9	10,306.8	12,784.5	23.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 80 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	1,241.5	1,528.8	1,882.1	2,307.9	2,809.4	3,394.7	4,106.5	21.9%
Deployment and Integration	1,803.7	2,210.2	2,708.0	3,304.7	4,003.4	4,814.3	5,795.8	21.3%
Support and Maintenance	577.5	695.4	836.9	1,002.9	1,192.6	1,407.2	1,661.5	19.0%
Total	3,622.7	4,434.4	5,427.0	6,615.4	8,005.4	9,616.3	11,563.8	21.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 81 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, 2017–2019 (USD THOUSAND)

Security Type	2017	2018	2019	CAGR (2017–2019)
Network Security	10,940.8	13,004.2	15,461.1	18.9%
Application Security	2,735.2	3,219.8	3,791.1	17.7%
Total	13,676.0	16,224.0	19,252.2	18.6%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 82 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	15,461.1	18,455.7	22,120.2	26,595.4	32,067.6	38,764.0	46,963.6	20.5%
Application Security	3,791.1	4,481.3	5,318.2	6,330.8	7,557.1	9,043.2	10,844.9	19.3%
Total	19,252.2	22,937.0	27,438.4	32,926.3	39,624.7	47,807.2	57,808.5	20.3%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 83 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	15,461.1	18,561.9	22,292.7	26,681.3	31,717.0	37,444.0	44,273.4	19.0%
Application Security	3,791.1	4,507.0	5,359.7	6,351.2	7,474.5	8,735.3	10,223.6	17.8%
Total	19,252.2	23,069.0	27,652.4	33,032.5	39,191.5	46,179.3	54,497.0	18.8%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 84 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, 2017–2019 (USD THOUSAND)

Vertical	2017	2018	2019	CAGR (2017–2019)
Government and Defense	4,649.8	5,524.8	6,566.3	18.8%
BFSI	2,324.9	2,783.8	3,334.2	19.8%
Retail	2,051.4	2,426.6	2,871.3	18.3%
Healthcare	1,094.1	1,311.6	1,572.8	19.9%
Automotive	2,188.2	2,565.0	3,007.5	17.2%
Others*	1,367.6	1,612.2	1,900.0	17.9%
Total	13,676.0	16,224.0	19,252.2	18.6%

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 85 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	6,566.3	7,835.4	9,387.9	11,283.2	13,600.0	16,434.2	19,903.4	20.5%
BFSI	3,334.2	4,009.4	4,840.9	5,863.3	7,121.9	8,672.6	10,584.7	21.4%
Retail	2,871.3	3,411.1	4,068.8	4,868.7	5,842.4	7,028.6	8,474.7	20.0%
Healthcare	1,572.8	1,893.6	2,289.1	2,775.9	3,375.8	4,115.8	5,029.3	21.6%
Automotive	3,007.5	3,540.6	4,185.1	4,962.4	5,900.9	7,034.8	8,405.3	18.9%
Others*	1,900.0	2,247.0	2,666.6	3,172.8	3,783.7	4,521.1	5,410.9	19.2%
Total	19,252.2	22,937.0	27,438.4	32,926.3	39,624.7	47,807.2	57,808.5	20.3%

e: estimated; p: projected

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 86 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	6,566.3	7,862.4	9,417.6	11,241.6	13,327.8	15,692.6	18,505.5	18.7%
BFSI	3,334.2	4,021.7	4,852.6	5,835.2	6,969.0	8,266.0	9,819.5	19.5%
Retail	2,871.3	3,442.1	4,127.8	4,933.2	5,855.6	6,902.7	8,149.6	18.8%
Healthcare	1,572.8	1,916.0	2,334.9	2,835.5	3,420.2	4,097.0	4,915.4	20.7%
Automotive	3,007.5	3,561.0	4,217.9	4,978.7	5,836.9	6,796.0	7,924.8	17.4%
Others*	1,900.0	2,265.9	2,701.7	3,208.4	3,782.0	4,425.0	5,182.1	18.0%
Total	19,252.2	23,069.0	27,652.4	33,032.5	39,191.5	46,179.3	54,497.0	18.8%

e: estimated; p: projected

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 87 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, 2017–2019 (USD THOUSAND)

Country	2017	2018	2019	CAGR (2017–2019)
UK	7,385.0	8,696.1	10,242.2	17.8%
Germany	2,461.7	2,969.0	3,580.9	20.6%
Rest of Europe*	3,829.3	4,558.9	5,429.1	19.1%
Total	13,676.0	16,224.0	19,252.2	18.6%

*Rest of Europe includes France, Italy, and others

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 88 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Country	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
UK	10,242.2	12,110.7	14,377.7	17,121.7	20,446.3	24,477.3	29,366.7	19.4%
Germany	3,580.9	4,335.1	5,268.2	6,420.6	7,845.7	9,609.2	11,792.9	22.2%
Rest of Europe*	5,429.1	6,491.2	7,792.5	9,384.0	11,332.7	13,720.7	16,648.8	20.7%
Total	19,252.2	22,937.0	27,438.4	32,926.3	39,624.7	47,807.2	57,808.5	20.3%

e: estimated; p: projected

*Rest of Europe includes France, Italy, and others

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 89 EUROPE: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Country	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
UK	10,242.2	12,205.8	14,550.7	17,285.9	20,395.3	23,897.8	28,044.2	18.1%
Germany	3,580.9	4,371.6	5,336.9	6,490.9	7,838.3	9,397.5	11,280.9	20.9%
Rest of Europe*	5,429.1	6,491.6	7,764.8	9,255.7	10,958.0	12,884.0	15,172.0	18.5%
Total	19,252.2	23,069.0	27,652.4	33,032.5	39,191.5	46,179.3	54,497.0	18.8%

e: estimated; p: projected

*Rest of Europe includes France, Italy, and othersSource: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

10.3.2 UNITED KINGDOM

10.3.2.1 The need to deliver quantum encryption systems that enable secure data transactions and transmissions

The UK is the major contributor to the quantum cryptography market share in Europe as it has witnessed major investments in the quantum cryptography space. This is evident from the fact that in 2015 the UK government announced more than USD 400 million investment on quantum technologies over five years. Furthermore, a quantum communications hub has been set up which comprise universities, industries, and public-sector bodies, with the aim to deliver quantum encryption systems that will enable secure transactions and transmissions of data across a wide range of users in real-world applications: from government agencies and industrial set-ups to commercial establishments. The project is part of a major national initiative, the UK National Quantum Technologies Programme, which aims to ensure the successful transition of quantum technologies from the laboratory to industries.

10.3.3 GERMANY

10.3.3.1 Need to explore the development of highly secure communication links to recognize every attack

Germany also actively participated in the R&D of quantum cryptography, as its Leopoldina National Academy of Science published research on 'Quantum Technology', which highlighted the potential for communication security. Moreover, a national programme named Quantum technology - Fundamentals and Applications (QUTEGA) has also been formed to explore the development of highly secure communication links in which every attack can be recognized, and the encryption systems of which, cannot be broken even with high-power quantum computers. Additionally, the German government has proposed funding of USD 350 million for the coming 10 years in the quantum cryptography space.

10.3.4 REST OF EUROPE

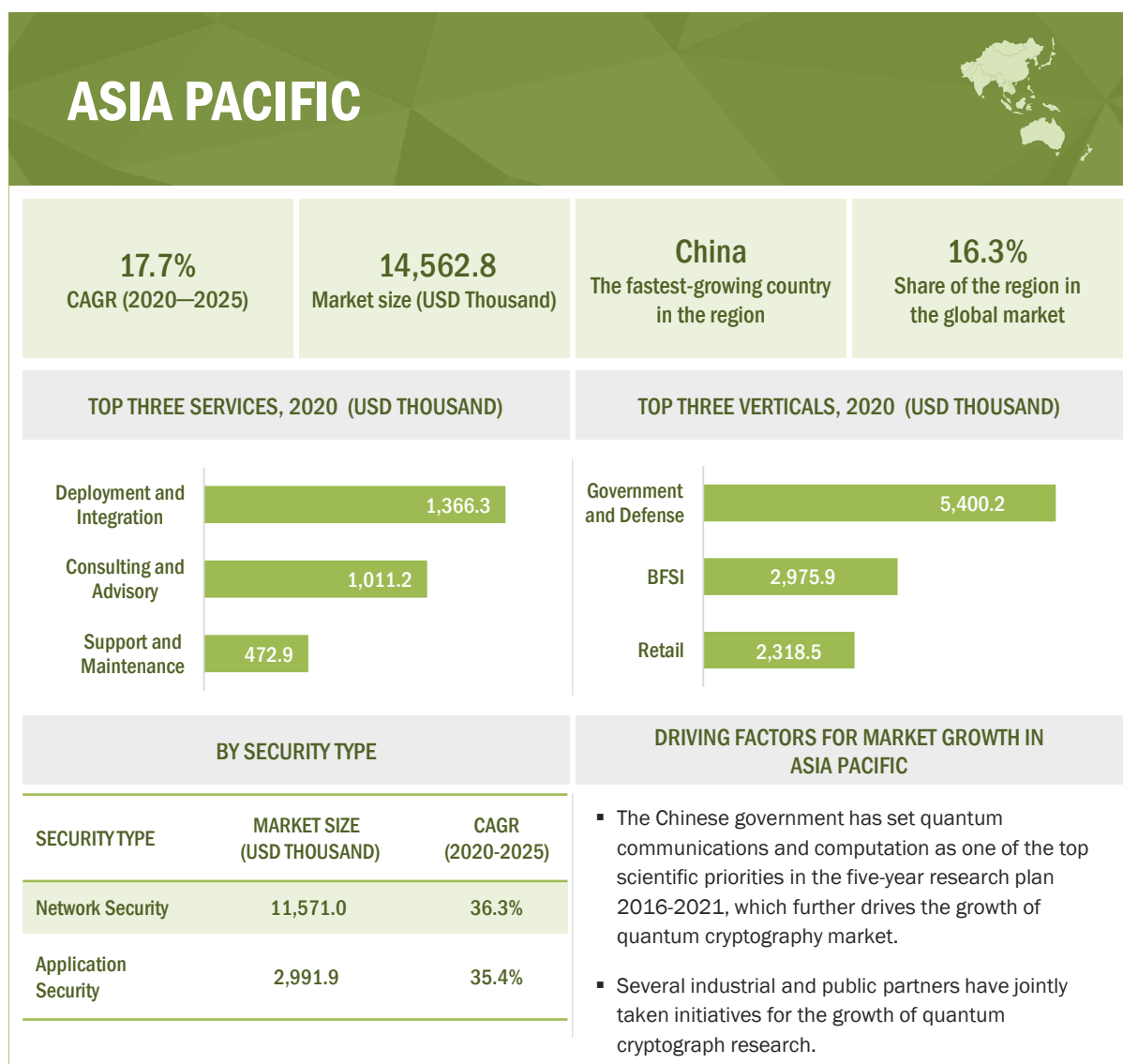
France has also noticed significant adoption and funding for quantum cryptography. For instance, in October 2017, the National Research Agency (ANR) has started a specific call on Quantum Technologies, with a budget of USD 12 million per year, for consecutive 3 years. Moreover, in Italy, a national quantum backbone is currently being developed by the Istituto Nazionale di Ricerca Metrologica (INRiM) that will provide QKD, alongside time stamping for financial applications and frequency/time distribution to scientific institutions which have already been implemented. Furthermore, the Swiss National Science Foundation is funding a National Centre of Competence in Research devoted to Quantum Science and Technology, and one of its aims is to promote applications of quantum cryptography.

10.4 ASIA PACIFIC

Across the APAC region, there are a number of prosperous economies with well-developed cyber ecosystems such as India, China, Australia, Singapore, and Japan. APAC has specifically been one of the fastest growing markets in terms of mobile workforce expansion, propagated with the rising adoption of mobile devices in this region. The enterprises and SMEs in the APAC countries are admitting the significance of data security, and are receptive toward adopting dedicated quantum cryptography solutions to protect their critical and sensitive business data from commercial espionage, cyber threats, and misuse of data for monetary gains of computer hackers. In addition, the increasing advancements in mobility and cloud adoption in this region, and the mandated compliance with government regulations to resolve data security issues have enforced companies to adopt efficient cryptography solutions. Moreover, with the rising intensity and complexity of cyber attacks, the data security concerns of business in this region have been escalating. The populous countries of India and China are now gradually advancing toward considering quantum encryption solutions.

10.4.1 ASIA PACIFIC: COVID-19 IMPACT

- The Government of India proposed internet intermediaries to actively filter online content and name sources of private messages that involves end-to-end encryption. The high growth is due to increasing advancements in innovative technologies across financial markets and growing stringent regulations in the country.
- Lack of skilled cybersecurity professionals evokes the need for consulting services segment in APAC, which ultimately drives the quantum cryptography market to grow at a significant CAGR during the forecast period. There are several market players that offer consulting services to effectively deploy quantum cryptography solutions in the region.

FIGURE 21 ASIA PACIFIC: MARKET SNAPSHOT, 2020 (E)

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 90 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, 2017–2019 (USD THOUSAND)

Component	2017	2018	2019	CAGR (2017–2019)
Solutions	7,374.1	8,607.4	10,045.6	16.7%
Services	1,673.9	2,000.6	2,389.7	19.5%
Total	9,048.0	10,608.0	12,435.3	17.2%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 91 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Solutions	10,045.6	11,808.0	13,977.1	16,630.2	19,869.2	23,817.0	28,620.4	19.4%
Services	2,389.7	2,873.6	3,478.5	4,230.7	5,165.0	6,324.1	7,759.9	22.0%
Total	12,435.3	14,681.7	17,455.6	20,860.9	25,034.2	30,141.1	36,380.4	19.9%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 92 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Solutions	10,045.6	11,712.5	13,675.4	15,991.0	18,734.9	21,993.1	25,891.8	17.2%
Services	2,389.7	2,850.4	3,403.4	4,068.1	4,870.2	5,839.8	7,020.1	19.8%
Total	12,435.3	14,562.8	17,078.8	20,059.0	23,605.0	27,832.9	32,911.9	17.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 93 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, 2017–2019 (USD THOUSAND)

Service	2017	2018	2019	CAGR (2017–2019)
Consulting and Advisory	585.9	702.9	842.9	19.9%
Deployment and Integration	786.7	949.0	1,144.1	20.6%
Support and Maintenance	301.3	348.7	402.7	15.6%
Total	1,673.9	2,000.6	2,389.7	19.5%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 94 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	842.9	1,017.5	1,236.4	1,509.7	1,850.2	2,274.2	2,801.3	22.5%
Deployment and Integration	1,144.1	1,388.5	1,696.3	2,082.3	2,565.7	3,170.6	3,926.5	23.1%
Support and Maintenance	402.7	467.6	545.7	638.7	749.1	879.3	1,032.1	17.2%
Total	2,389.7	2,873.6	3,478.5	4,230.7	5,165.0	6,556.5	8,080.2	23.0%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 95 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	842.9	1,011.2	1,214.3	1,459.9	1,757.8	2,120.0	2,563.2	20.4%
Deployment and Integration	1,144.1	1,366.3	1,633.3	1,954.7	2,342.9	2,812.8	3,385.5	19.9%
Support and Maintenance	402.7	472.9	555.7	653.5	769.4	907.0	1,071.5	17.8%
Total	2,389.7	2,850.4	3,403.4	4,068.1	4,870.2	5,839.8	7,020.1	19.8%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 96 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, 2017–2019 (USD THOUSAND)

Security Type	2017	2018	2019	CAGR (2017–2019)
Network Security	7,147.9	8,396.4	9,861.6	17.5%
Application Security	1,900.1	2,211.6	2,573.7	16.4%
Total	9,048.0	10,608.0	12,435.3	17.2%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 97 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	9,861.6	11,665.4	13,896.0	16,638.7	20,005.6	24,133.0	29,184.3	20.1%
Application Security	2,573.7	3,016.3	3,559.6	4,222.2	5,028.6	6,008.2	7,196.0	19.0%
Total	12,435.3	14,681.7	17,455.6	20,860.9	25,034.2	30,141.1	36,380.4	19.9%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 98 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	9,861.6	11,571.0	13,596.1	15,999.2	18,863.5	22,284.8	26,401.9	17.9%
Application Security	2,573.7	2,991.9	3,482.8	4,059.9	4,741.5	5,548.1	6,510.0	16.8%
Total	12,435.3	14,562.8	17,078.8	20,059.0	23,605.0	27,832.9	32,911.9	17.7%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 99 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, 2017–2019 (USD THOUSAND)

Vertical	2017	2018	2019	CAGR (2017–2019)
Government and Defense	3,348	3,931	4,614	17.4%
BFSI	1,810	2,138	2,527	18.2%
Retail	1,448	1,693	1,979	16.9%
Healthcare	452	539	642	19.2%
Automotive	1,086	1,252	1,445	15.3%
Others*	905	1,055	1,228	16.5%
Total	9,048	10,608	12,435	17.2%

*Others includes telecom, energy and utilities, logistics, and education

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 100 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	4,614	5,456	6,496	7,774	9,343	11,266	13,617	20.1%
BFSI	2,527	3,007	3,604	4,341	5,251	6,372	7,753	20.9%
Retail	1,979	2,330	2,763	3,293	3,941	4,733	5,697	19.6%
Healthcare	642	771	932	1,132	1,381	1,690	2,074	21.9%
Automotive	1,445	1,678	1,963	2,308	2,726	3,229	3,834	18.0%
Others*	1,228	1,439	1,698	2,012	2,392	2,852	3,405	18.8%
Total	12,435	14,682	17,456	20,861	25,034	30,141	36,380	19.9%

e: estimated; p: projected

*Others includes telecom, energy and utilities, logistics, and education

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 101 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, POST-COVID-19 SCENARIO, 2020–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	4,614.4	5,400.2	6,328.9	7,428.3	8,735.5	10,293.2	12,163.3	17.6%
BFSI	2,526.8	2,975.9	3,509.8	4,145.6	4,906.0	5,817.5	6,918.0	18.4%
Retail	1,979.0	2,318.5	2,720.3	3,196.3	3,762.9	4,438.7	5,250.9	17.8%
Healthcare	642.5	771.7	928.3	1,118.4	1,349.9	1,632.6	1,980.1	20.7%
Automotive	1,444.6	1,664.5	1,920.6	2,219.3	2,569.5	2,980.9	3,467.9	15.8%
Others*	1,228.0	1,432.0	1,671.0	1,951.3	2,281.2	2,670.1	3,131.7	16.9%
Total	12,435.3	14,562.8	17,078.8	20,059.0	23,605.0	27,832.9	32,911.9	17.7%

e: estimated; p: projected

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 102 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, 2017–2019 (USD THOUSAND)

Country	2017	2018	2019	CAGR (2017–2019)
China	3,619.2	4,264.4	5,023.8	17.8%
Japan	2,352.5	2,766.6	3,253.1	17.6%
India	1,673.9	1,939.1	2,245.8	15.8%
Rest of APAC*	1,402.4	1,637.9	1,912.5	16.8%
Total	9,048.0	10,608.0	12,435.3	17.2%

*Rest of APAC includes Taiwan, Hong Kong, and other countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 103 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Country	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
China	5,023.8	5,960.8	7,121.9	8,553.0	10,314.1	12,478.4	15,134.2	20.5%
Japan	3,253.1	3,852.5	4,594.3	5,507.3	6,629.0	8,005.5	9,691.7	20.3%
India	2,245.8	2,619.2	3,075.7	3,629.8	4,300.9	5,111.9	6,090.1	18.4%
Rest of APAC*	1,912.5	2,249.2	2,663.7	3,170.9	3,790.2	4,545.3	5,464.3	19.4%
Total	12,435.3	14,681.7	17,455.6	20,860.9	25,034.2	30,141.1	36,380.4	19.9%

e: estimated; p: projected

*Rest of APAC includes Taiwan, Hong Kong, and other countries

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 104 ASIA PACIFIC: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Country	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
China	5,023.8	5,861.5	6,848.6	8,013.6	9,394.8	11,035.7	13,000.2	17.3%
Japan	3,253.1	3,834.4	4,524.2	5,345.7	6,328.5	7,506.5	8,929.0	18.4%
India	2,245.8	2,608.2	3,033.2	3,532.4	4,121.4	4,817.9	5,647.7	16.7%
Rest of APAC*	1,912.5	2,258.7	2,672.8	3,167.3	3,760.3	4,472.7	5,335.0	18.8%
Total	12,435.3	14,562.8	17,078.8	20,059.0	23,605.0	27,832.9	32,911.9	17.7%

e: estimated; p: projected

*Rest of APAC includes Taiwan, Hong Kong, and other countriesSource: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

10.4.2 CHINA

10.4.2.1 China to gather quantum technology researchers to enhance its cybersecurity space

China is the major contributor of quantum cryptography market share in APAC. In 2012, Hefei completed its first quantum communications network linking of 40 telephones and 16 video cameras installed at city government agencies, military units, financial institutions, and health care offices, at the cost approximately USD 9 million for the Hefei government. China is a leading the market with its presence in quantum technologies, this is evident from the fact that in August 2016, China sent its first quantum satellite into space and in 2017, scientists used the satellite to connect a quantum-encrypted video call between Chinese and Austrian scientists and demonstrated its intercontinental video call record for cryptography by the use of entangled photons. Along with this, China has gathered 120 researchers for its research institute as part of a push to develop military applications for artificial intelligence and quantum technology.

10.4.3 JAPAN

10.4.3.1 Upsurge demand for quantum communication

Japan counts as one of the richest and innovative countries across the globe. In Japan, several industrial and public partners have jointly developed an extensive quantum network in Tokyo in October 2010 under the Project Updating Quantum Cryptography and Communications (UQCC). The Tokyo QKD network is constructed in a part of the NICT open test-bed network Japan Giga Bit Network 2 plus. The network operations include video transmission, eavesdropping detection, and rerouting to secondary secure links. NICT, NEC, Mitsubishi Electric, NTT, Toshiba Research Europe, ID Quantique, the Austrian Institute of Technology, the Institute of Quantum Optics and Quantum Information and the University of Vienna members of this project demonstrated to the public QKD network operations. The Project UQCC aims at promoting R&D on quantum cryptography and communications and disseminating outcomes for the benefit of society.

10.4.4 INDIA

10.4.4.1 The need to revolutionize future computation and communication systems

Institutes in India such as Raman Research Institute, conducting cutting-edge research in the fields of quantum information, quantum computing and quantum communication using quantum optics. Other institutes such as IISc set up a Centre for Quantum Information and Quantum Computation (CQIQC) in 2010 as a 5-year Research Project sanctioned by the Department of Science and Technology, Government of India. India is improving its research in quantum information science, including quantum computers and QKD. This is evident from the fact that in December 2017, Department of Science and Technology (DST), Government of India has initiated a new directed research programme on Quantum Information Science and Technology (QuST) a scheme under Interdisciplinary Cyber Physical Systems (ICPS) programme. Some of the important developments covered under this program are quantum computers, fast quantum algorithms, quantum teleportation, super dense coding, remote state preparation, quantum cryptography, and several quantum communication protocols. QuST promises to revolutionize the future computation and communication systems to build huge impact on the Nation.

10.4.5 REST OF ASIA PACIFIC

Singapore has been actively involved in quantum research from a long time. In September 2018, Singapore's National Research Foundation (NRF) announced to invest USD 25 million over 5 years in a new Quantum Engineering Programme (QEP). The program will connect Singapore researchers in quantum information, photonics devices, system engineering together with industry partners from the onset of the translation, and development process to advance quantum technologies. The Australian government is implementing a Government Quantum Network for intra-governmental communications in Canberra with its CQC2T program a Centre for Quantum Computation and Communication Technologies (CQC2T). The CQC2T main goal is to demonstrate Quantum Cryptography in the Australian Parliamentary Triangle. QuintessenceLabs partnered with the University of Queensland and Lockheed Martin (Australia) to provide the highest level of information security for intra-governmental communications in Canberra.

10.5 MIDDLE EAST AND AFRICA

The MEA region includes economies such as Israel, the United Arab Emirates (UAE), Kenya, Egypt, South Africa, and Libya. The region of MEA is relatively cautious about the adoption of quantum cryptography; however, the quantum cryptography market is expected to grow at a fast pace during the forecast period. With the growth in demands for a wide range of cryptography to defend the corporate network and valuable data from cyber attacks, misuse, and loss, this region is gaining momentum to enable the implementation of policy-adhered data security solutions. Currently, the expansive investments are being made in data security adoption across financial services and healthcare sectors in this region, which are boosting the demand for quantum cryptography.

10.5.1 MIDDLE EAST AND AFRICA: COVID-19 IMPACT

- The growing adoption of quantum cryptography solutions can be attributed to the increasing demand for these solutions to secure cloud data in the region.
- The MEA region has witnessed steady growth and less awareness among people about advanced technologies that can be another factor, which would drive the quantum cryptography market toward adopting better security and breaches.

TABLE 105 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, 2017–2019 (USD THOUSAND)

Component	2017	2018	2019	CAGR (2017–2019)
Solutions	2,273.7	2,563.2	2,880.3	12.6%
Services	482.3	556.8	640.5	15.2%
Total	2,756.0	3,120.0	3,520.8	13.0%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 106 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Solutions	2,880.3	3,287.8	3,813.0	4,473.8	5,293.9	6,299.7	7,518.4	18.0%
Services	640.5	748.0	887.2	1,064.3	1,287.0	1,564.5	1,906.7	20.6%
Total	3,520.8	4,035.8	4,700.2	5,538.1	6,580.9	7,864.1	9,425.2	18.5%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 107 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COMPONENT, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Component	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Solutions	2,880.3	3,011.3	3,077.0	3,249.0	3,720.5	4,659.9	5,966.8	14.7%
Services	640.5	685.1	716.0	772.9	904.4	1,157.3	1,513.2	17.2%
Total	3,520.8	3,696.4	3,792.9	4,021.9	4,624.9	5,817.2	7,480.0	15.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 108 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, 2017–2019 (USD THOUSAND)

Service	2017	2018	2019	CAGR (2017–2019)
Consulting and Advisory	166.4	192.9	222.9	15.7%
Deployment and Integration	207.4	241.7	280.7	16.3%
Support and Maintenance	108.5	122.2	136.9	12.3%
Total	482.3	556.8	640.5	15.2%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 109 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	222.9	261.4	311.4	375.1	455.5	556.1	680.7	21.1%
Deployment and Integration	280.7	331.0	396.3	479.9	585.9	719.0	884.7	21.7%
Support and Maintenance	136.9	155.7	179.6	209.2	245.5	289.3	341.3	17.0%
Total	640.5	748.0	887.2	1,064.3	1,287.0	1,564.5	1,906.7	20.6%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 110 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SERVICE, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Service	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Consulting and Advisory	222.9	239.8	252.1	273.7	322.1	414.6	545.3	17.9%
Deployment and Integration	280.7	300.7	314.6	340.1	398.5	510.5	668.5	17.3%
Support and Maintenance	136.9	144.7	149.3	159.1	183.8	232.1	299.4	15.7%
Total	640.5	685.1	716.0	772.9	904.4	1,157.3	1,513.2	17.2%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 111 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, 2017–2019 (USD THOUSAND)

Security Type	2017	2018	2019	CAGR (2017–2019)
Network Security	2,218.6	2,516.9	2,846.2	13.3%
Application Security	537.4	603.1	674.6	12.0%
Total	2,756.0	3,120.0	3,520.8	13.0%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 112 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY SECURITY TYPE, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	2,846.2	3,269.4	3,815.7	4,505.4	5,365.1	6,424.8	7,716.4	18.7%
Application Security	674.6	766.4	884.5	1,032.7	1,215.8	1,439.3	1,708.8	17.4%
Total	3,520.8	4,035.8	4,700.2	5,538.1	6,580.9	7,864.1	9,425.2	18.5%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 113 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, POST-COVID-19 SCENARIO, BY SECURITY TYPE, 2019–2025 (USD THOUSAND)

Security Type	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Network Security	2,846.2	2,994.5	3,079.2	3,272.0	3,770.5	4,752.5	6,123.9	15.4%
Application Security	674.6	701.9	713.8	749.9	854.4	1,064.7	1,356.1	14.1%
Total	3,520.8	3,696.4	3,792.9	4,021.9	4,624.9	5,817.2	7,480.0	15.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 114 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, 2017–2019 (USD THOUSAND)

Vertical	2017	2018	2019	CAGR (2017–2019)
Government and Defense	909.5	1,031.3	1,165.6	13.2%
BFSI	441.0	504.1	574.5	14.1%
Retail	358.3	404.3	454.7	12.7%
Healthcare	192.9	221.0	252.4	14.4%
Automotive	330.7	368.4	409.0	11.2%
Others*	523.6	590.9	664.6	12.7%
Total	2,756.0	3,120.0	3,520.8	13.0%

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 115 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	1,165.6	1,338.3	1,561.1	1,842.4	2,192.9	2,624.7	3,150.8	18.7%
BFSI	574.5	665.1	782.2	930.8	1,117.0	1,348.0	1,631.5	19.7%
Retail	454.7	519.5	603.0	708.1	838.7	998.9	1,193.2	18.1%
Healthcare	252.4	292.8	345.1	411.5	494.8	598.4	725.7	19.9%
Automotive	409.0	461.3	528.6	612.8	716.5	842.4	993.4	16.6%
Others*	664.6	758.9	880.2	1,032.5	1,221.1	1,451.7	1,730.5	17.9%
Total	3,520.8	4,035.8	4,700.2	5,538.1	6,580.9	7,864.1	9,425.2	18.5%

e: estimated; p: projected

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 116 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY VERTICAL, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Vertical	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Government and Defense	1,165.6	1,221.6	1,251.3	1,324.5	1,520.4	1,908.9	2,450.2	14.9%
BFSI	574.5	607.4	627.7	670.2	776.1	983.1	1,272.9	15.9%
Retail	454.7	477.6	490.3	520.2	598.5	753.2	969.0	15.2%
Healthcare	252.4	270.0	282.3	304.9	357.3	457.8	599.8	17.3%
Automotive	409.0	424.5	430.5	451.3	513.0	637.8	810.6	13.8%
Others*	664.6	695.3	710.9	750.8	859.7	1,076.4	1,377.4	14.7%
Total	3,520.8	3,696.4	3,792.9	4,021.9	4,624.9	5,817.2	7,480.0	15.1%

e: estimated; p: projected

*Others includes education, telecom, energy and utilities, and logistics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 117 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, 2017–2019 (USD THOUSAND)

Country	2017	2018	2019	CAGR (2017–2019)
Middle East	2,315.0	2,623.9	2,964.5	13.2%
Africa	441.0	496.1	556.3	12.3%
Total	2,756.0	3,120.0	3,520.8	13.0%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 118 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, PRE-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Country	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Middle East	2,964.5	3,402.2	3,967.0	4,679.7	5,567.4	6,660.9	7,992.5	18.6%
Africa	556.3	633.6	733.2	858.4	1,013.5	1,203.2	1,432.6	17.7%
Total	3,520.8	4,035.8	4,700.2	5,538.1	6,580.9	7,864.1	9,425.2	18.5%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 119 MIDDLE EAST AND AFRICA: QUANTUM CRYPTOGRAPHY MARKET SIZE, BY COUNTRY, POST-COVID-19 SCENARIO, 2019–2025 (USD THOUSAND)

Country	2019	2020-e	2021	2022	2023	2024	2025-p	CAGR (2020–2025)
Middle East	2,964.5	3,127.1	3,224.0	3,434.7	3,968.2	5,014.4	6,477.7	15.7%
Africa	556.3	569.2	568.9	587.2	656.7	802.8	1,002.3	12.0%
Total	3,520.8	3,696.4	3,792.9	4,021.9	4,624.9	5,817.2	7,480.0	15.1%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

10.5.2 MIDDLE EAST

10.5.2.1 Growing demand for securing business content

In the countries of Saudi Arabia and UAE, the adoption of cryptography solutions has primarily been used for securing the business content, residing in the cloud and locally placed PCs, from prevalent perils of privacy breaches and losses.

10.5.3 AFRICA

10.5.3.1 Increasing number of security initiatives

The African countries such as South Africa, Egypt, and Kenya have incorporated cryptography solutions to ensure the security of data traveling through emails. In South Africa, quantum communication security solution has been deployed starting from 2008 in Durban's municipal fiber-optic network. Moreover, the country's QuantumCity project aims at helping the city with quantum security solutions.

11 COMPETITIVE LANDSCAPE

11.1 OVERVIEW

The competitive landscape describes the market analysis of the key players in the quantum cryptography market based on their offerings. Moreover, it explains the key growth strategies adopted by the top market players during 2016–2020 to expand their market shares and improve their offerings in the quantum cryptography market.

11.2 COMPETITIVE LEADERSHIP MAPPING (START-UP/SMSE)

The competitive leadership mapping provides information about the major players who offer quantum cryptography solutions and associated services to its commercial clients. It outlines the findings and analysis on how well each market vendor performs within the predefined criteria. The vendor evaluations are based on 2 broad categories: product excellence and business excellence. Each category carries various criteria based on which the vendors have been evaluated. The evaluation criteria considered under product excellence include the breadth and depth of product offering (based on the industries that the vendors cater to, and solution and service offerings), product feature and functionality, use cases, support, and patents. The evaluation criteria considered under business excellence include funding frequencies received till date, size of team, partner ecosystem, and customer presence across the globe.

11.2.1 PROGRESSIVE

ID Quantique, ISARA Corporation, and QitessenceLabs are the leaders in the quantum cryptography market and are recognized as such. They have a strong portfolio of solutions and services in the market. These companies performed exceedingly well in both product excellence and business excellence parameters. They have substantial partner ecosystem and well-developed marketing channels and received extensive funding for development of their product portfolio. These companies have been instrumental in achieving their growth prospects, thereby attained a high partner attractiveness.

11.2.2 RESPONSIVE

QuantumCTek Co. Ltd. and Qubitekk are recognized as the disruptors in the quantum cryptography market. These start-ups are technology disruptors with highly innovative product offerings as compared to their competitors. They are focusing on constant development of product/service portfolio and bringing new innovations in the market. However, they lag in business excellence parameter as compared to other start-ups. With appropriate marketing activities, opportune partnerships, and VC funding these start-ups will be known as Gazelles.

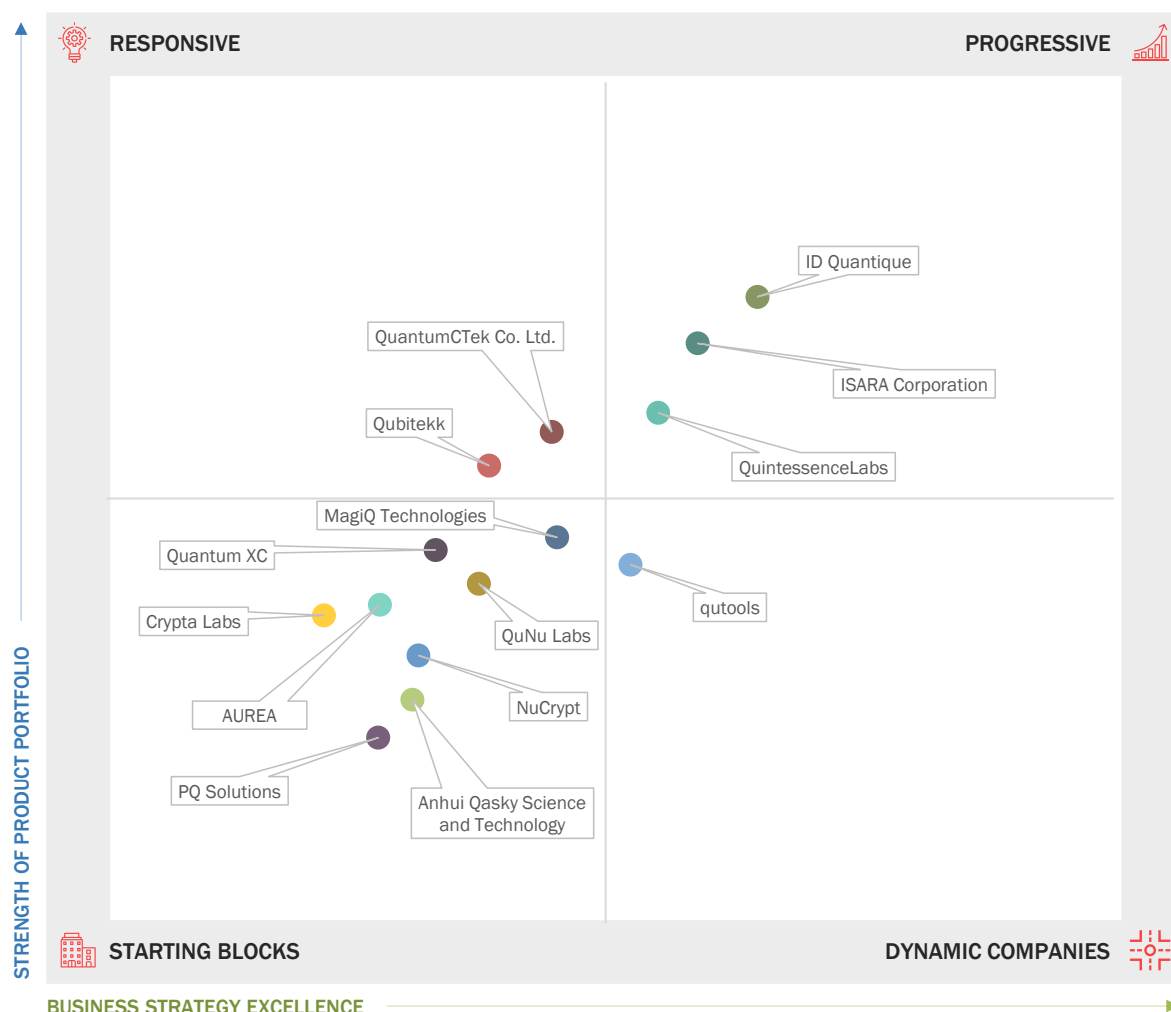
11.2.3 DYNAMIC COMPANIES

qutools is recognized as the dynamic company in the quantum cryptography market. These start-ups have performed substantially in business excellence parameter. However, they are slightly low in product excellence as compared to Gazelles and Disruptors. They generally focus on specific type of technology related to the product or service offering.

11.2.4 STARTING BLOCKS

MagiQTechnologies, Quantum XC, Aurea Technology, QuNu Labs, NuCrypt, Anhui Qasky Science and Technology Limited, PQ Solutions, and Crypta Labs are recognized as the starting blocks in the quantum cryptography market. They are startups with niche offerings and starting to gain their position in the market. They do not have much stronger business excellence and product excellence as compared to other established start-ups. However, they are always in an opportunity to grab a chance for consolidation of their market space. They might be new entrants in the market and require some more time before it gets significant traction in the market.

FIGURE 22 QUANTUM CRYPTOGRAPHY MARKET (START-UP/SMSE), COMPETITIVE LEADERSHIP MAPPING, 2020

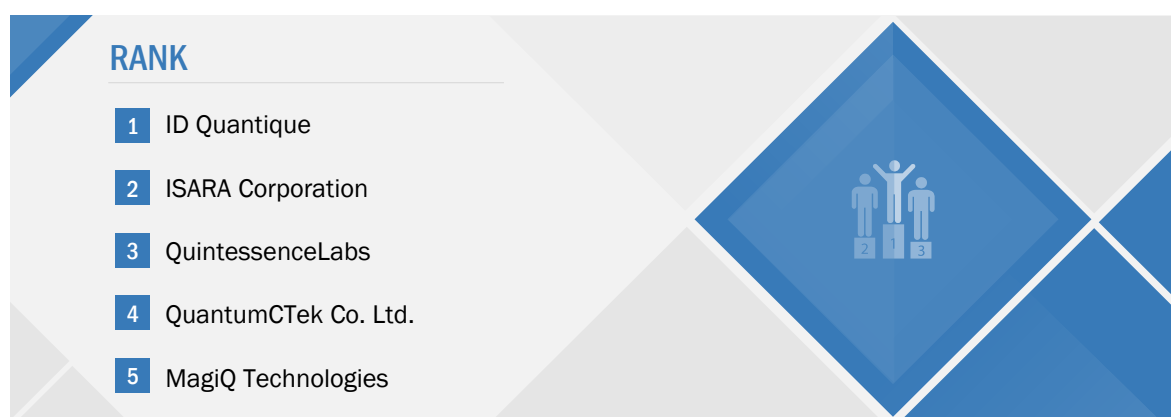


Note: Quantum XC and PQ Solutions are popularly known as Quantum Xchange and Post-Quantum respectively

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

11.3 MARKET RANKING

FIGURE 23 MARKET RANKING FOR THE QUANTUM CRYPTOGRAPHY MARKET, 2020



Note: The ranking has been estimated on the respective company's product offerings and business strategies. Secondary research, along with in-depth primary interviews conducted with key industry leaders, has also contributed to the analysis

Source: Annual Reports, Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

11.4 COMPETITIVE SCENARIO

The competitive scenario includes the recent developments of the key players in the global quantum cryptography market. The market share analysis of the key players has been done based on the following parameters: depth of product offerings and revenues generated. Additionally, the competitive landscape studies the key growth strategies adopted by the market players between 2016 and 2020 to expand their global presence and shares in the quantum cryptography market.

FIGURE 24 KEY DEVELOPMENTS BY LEADING PLAYERS IN THE QUANTUM CRYPTOGRAPHY MARKET DURING 2016–2020

COMPANY NAME	ORGANIC	INORGANIC
	New Product Launches/Product Enhancements	Partnerships, Collaborations, and Agreements
ID Quantique	In December 2019, ID Quantique announced that its product Cerberis3 Quantum Key Distribution (QKD) system can be deployed in any network configurations including point-to-point.	In December 2019, ID Quantique partnered with ADVA to provide its quantum-safe encryption solution. The solution integrates with ADVA's FSP 3000 platform to provide long-term security. 
Quintessence Labs		In November 2016, QuintessenceLabs collaborated with VMware to enhance the security of its virtualized applications. Moreover, customers would be able to securely integrate the cryptographic key and policy management solutions into their virtual applications, thereby securely encrypting their data while also retaining control over their keys.
QuantumCTek		In March 2018, QuantumCTek collaborated with University of Science and Technology in China, CAS Quantum Network Corporation, Jinan Institute of Quantum Technology, China Unicom, and Yangtze Optical Fibre and Cable Company, to introduce the integration of Quantum Key Distribution (QKD) with commercial backbone network carrying 3.6 Tbps classical data.
Crypta Labs	In April 2017, Crypta Labs introduced Quantum Random Number Generator (QRNG) prototype for secure encryption from a mobile phone or a photonic capable chip.	In May 2020, Crypta Labs collaborated with SSpace Research and Innovation Network for Technology (SPRINT) to develop its QRNGs for space applications
ISARA	In October 2018, ISARA announced the launch of ISARA Catalyst Agile Digital Certificate Technology, which enables organizations and manufacturers to secure their infrastructure and connected devices from encryption-breaking quantum compute threats. 	In August 2019, Thales collaborated with ISARA Corp. and ID Quantique for quantum-safe, crypto-agile solution designed to protect against the security threat of quantum computing

Note:  indicates events that would have a significant impact on the market

Source: Press Releases

11.4.1 NEW PRODUCT LAUNCHES AND PRODUCT ENHANCEMENTS

Companies adopted the strategy of new product launches and product enhancements to improve their product offerings and provide better solutions to enterprises. Crypta Labs, Qubitekk, and ISARA have adopted this strategy.

TABLE 120 NEW PRODUCT LAUNCHES AND PRODUCT ENHANCEMENTS, 2017–2019

DATE	COMPANY NAME	DESCRIPTION
December 2019	Product Enhancement	ID Quantique announced that its product Cerberis3 Quantum Key Distribution (QKD) system can be deployed in any network configurations including point-to-point.
February 2019	New Product Launch	ISARA launched new tools to test and implement crypto-agility and quantum-safe cryptography directly into existing systems.
October 2018	New Product Launch	ISARA announced the launch of ISARA Catalyst Agile Digital Certificate Technology, which enables organizations and manufacturers to secure their infrastructure and connected devices from encryption-breaking quantum compute threats.
July 2017	Product Enhancement	Qubitekk licensed an Oak Ridge National Laboratory developed a method to produce quantum light particles, known as photons, in a controlled, deterministic manner that guarantees improved speed and security when sharing the encrypted data.
April 2017	New Product Launch	Crypta Labs introduced Quantum Random Number Generator (QRNG) prototype for secure encryption from a mobile phone or a photonic capable chip.

Source: Press Releases

11.4.2 PARTNERSHIPS, COLLABORATIONS, AND AGREEMENTS

The strategy of partnerships, collaborations, and agreements accounted for a large share of the overall growth strategies implemented by the key players in the quantum cryptography market. Companies have partnered with other players to serve reputed clients across the globe and mark their presence in their respective regional markets. ID Quantique, QuintessenceLabs, Post-Quantum, ISARA, QuantumCTek, and Qasky have adopted this strategy in the global quantum cryptography market.

TABLE 121 PARTNERSHIPS, COLLABORATIONS, AND AGREEMENTS, 2016–2020

DATE	COMPANY NAME	DESCRIPTION
May 2020	Crypta Labs (UK) and SPRINT (UK)	Crypta Labs collaborated with SSpace Research and Innovation Network for Technology (SPRINT) to develop its QRNGs for space applications.
April 2020	CIP (UK) and ISARA (Canada)	Cybersec Innovation Partners Limited (CIP) partnered with ISARA to offer quantum-safe, crypto agile, and hybrid certificate offerings.
January 2020	Qubitekk (US) and QinetiQ's (UK)	Qubitekk acquired QinetiQ's Quantum Key Distribution (QKD) patent portfolio. The portfolio covers novel technological approaches in quantum science, including 57 patent filings, across 17 patent families.
December 2019	ID Quantique (Switzerland) and AVADA (Germany)	ID Quantique partnered with ADVA to provide a quantum-safe encryption solution. The solution integrates with ADVA's FSP 3000 platform to provide long-term security.

September 2019	ID Quantique (Switzerland) and Dacoso (Germany)	ID Quantique partnered with Dacoso to offer highly secure, agile, and flexible quantum-safe data encryption.
August 2019	Thales (France), ID Quantique (Switzerland) and ISARA (Canada)	Thales collaborated with ID Quantique and ISARA Corp. for a quantum-safe, crypto-agile solution designed to protect against security threats of quantum computing.
June 2019	ID Quantique (Switzerland) and Mt Pelerin (US)	ID Quantique partnered with Mt. Pelerin to provide quantum security for digital asset custody. The aim is to add an extra layer of quantum technologies on top of conventional custody solutions.
October 2018	ID Quantique (Switzerland) and InfoSec Global (Canada)	InfoSec Global collaborated with ID Quantique to provide a quantum-safe Virtual Private Network (VPN). The solution combines InfoSec Global's cryptographic agility with ID Quantique's Quantis quantum random number generator (QRNG) for encryption inside a single device.
September 2018	Digicert (US), Gemalto (Netherlands), and ISARA (Canada)	DigiCert, Gemalto, and ISARA announced a partnership to develop advanced quantum-safe digital certificates and secure key management for connected devices commonly referred to as IoT.
June 2018	ID Quantique (Switzerland) and Quantum Xchange (Canada)	ID Quantique announced its partnership with Quantum Xchange to bring quantum-safe cryptography as a service into the US market.
April 2018	ISARA (Canada) and CSS (US)	ISARA partnered with Certified Security Solutions (CSS) to introduce quantum-safe, full-stack Public Key Infrastructure (PKI) solution. The PKI solution can ensure its Over-The-Air (OTA) software updates are secure against attack from conventional systems and quantum computers in the future.
April 2018	ISARA (Canada) and Cisco (US)	Cisco and ISARA announced their collaboration to test digital certificates that operate in both classic and quantum-safe algorithm modes.
March 2018	QuantumCTek (China), University of Science and Technology (China), Jinan Institute of Quantum Technology (China), China Unicom (China), and Yangtze Optical Fibre and Cable Company (China)	QuantumCTek collaborated with University of Science and Technology of China, CAS Quantum Network, Jinan Institute of Quantum Technology, China Unicom, and Yangtze Optical Fibre and Cable Company, to introduce the integration of QKD with commercial backbone network carrying 3.6 Tbps classical data.
April 2017	Qasky (China) and Hentong Photoelectric (China)	Qasky and Hentong Photoelectric signed a Memorandum of Understanding (MoU) to establish Jiangsu Hengtong Qasky Information Institute (research institute), and expand cooperation in several aspects.
March 2017	AliCloud (China) and QuantumCTek (China)	AliCloud partnered with QuantumCTek to make AliCloud, a cloud computing company, which can provide quantum encrypted transmission of information.
December 2016	ID Quantique (Switzerland) and QTEC (China)	China Quantum Technologies (QTEC) and ID Quantique partnered to bring futuristic quantum encryption tools to the burgeoning Chinese industry.

November 2016	QuintessenceLabs (Australia) and VMware (US)	QuintessenceLabs collaborated with VMware to enhance the security of its virtualized applications. Moreover, customers would be able to securely integrate the cryptographic key and policy management solutions into their virtual applications, thereby securely encrypting their data while retaining control over their keys.
September 2016	QuintessenceLabs (Australia) and PKWARE (US)	QuintessenceLabs partnered with PKWARE to create an enhanced encryption solution, offering unique key management capabilities and control of data-security policies for global enterprises and government organizations.
August 2016	Post-Quantum (UK) and IPC (US)	Post-Quantum collaborated with IPC, to facilitate next-generation security solutions as part of IPC's Connexus Cloud community of services. This collaboration enabled Post-Quantum to integrate biometric, encryption, and blockchain technology-as-a-service, with the help of IPC's Connexus Cloud ecosystem.

Source: Press Releases

12 COMPANY PROFILES

12.1 ID QUANTIQUE

12.1.1 BUSINESS OVERVIEW

ID Quantique was founded in 2001 and is headquartered in Geneva, Switzerland. The company develops and commercializes network encryption, scientific instrumentation, and random number generator products. It operates through 3 business units, namely, quantum-safe security, single-photon systems, and random number generation.

COMPANY AT A GLANCE

Founded:	2001
Headquarters:	Geneva, Switzerland
Ownership:	Private
Specialties:	IT Services



Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

It offers the Quantum Key Distribution (QKD) solution to provide quantum-safe security for data protection and secure key exchange for encrypting security devices. QKD is practiced with the help of 2 different solutions, namely, Cerberis3 QKD System and Clavis3 QKD Platform. Cerberis3 QKD System is a modular QKD system and a secure key exchange based on QKD. It automates key exchange with continuous key renewal and integrates entropy source based on a quantum random number generator. On the other hand, Clavis3 QKD Platform was designed as a research platform, which supports both automated and manual operations. This platform enables users to experiment with different parameters, study various setups, and offer high-speed key generation and distribution up to 100 km. Furthermore, the company's Cerberis QKD system provides quantum-safe security, offering a secure key exchange for encryption and other security devices on storage networks. ID Quantique has established a strong global footprint with its customers in more than 60 countries and offers wide range of solutions and services to different industry verticals that include government and defense, finance, enterprises, and academic institutes.

12.1.2 SOLUTIONS AND SERVICES OFFERED

The company offers the following solutions:

CATEGORY	SOLUTION
Quantum Cryptography	▪ Cerberis3 QKD System ▪ Clavis3 QKD Platform ▪ Clavis300 Quantum Cryptography Platform

Source: Company Website

The company offers the following services:

CATEGORY	SERVICES
Quantum Cryptography	▪ Technology and IP licensing ▪ Professional Services ▪ Support Services

Source: Company Website

12.1.3 RECENT DEVELOPMENTS

DATE	NATURE OF DEVELOPMENT	DESCRIPTION
December 2019	Partnership	ID Quantique partnered with ADVA to provide a quantum-safe encryption solution. The solution integrates with ADVA's FSP 3000 platform to provide long-term security.
December 2019	Product Enhancement	ID Quantique product Cerberis3 Quantum Key Distribution (QKD) system can be deployed across any network configurations, including point-to-point.
September 2019	Partnership	ID Quantique partnered with Dacoso to offer highly secure, agile, and flexible quantum-safe data encryption.
August 2019	Collaboration	Thales collaborated with ID Quantique and ISARA Corp for a quantum-safe, crypto-agile solution designed to protect data against the security threat of quantum computing.
June 2019	Partnership	ID Quantique partnered with Mt Pelerin to provide quantum security for digital asset custody. The aim is to add an extra layer of quantum technologies on top of conventional custody solutions.
October 2018	Collaboration	InfoSec Global collaborated with ID Quantique to provide a quantum-safe VPN. The solution combines InfoSec Global's cryptographic agility with ID Quantique's Quantis QRNG for encryption inside a single device.
September 2018	Expansion	ID Quantique expanded its offerings to 2 new regions, namely, America and APAC with the opening of new offices in Washington and Seoul.
June 2018	Partnership	ID Quantique partnered with Quantum Xchange to bring quantum-safe cryptography-as-a-service in the US.
December 2016	Partnership	China Quantum Technologies (QTEC) and ID Quantique partnered to bring futuristic quantum encryption tools to the burgeoning Chinese industry.
November 2016	Expansion	ID Quantique expanded its offerings in the UK to support UK National Quantum Technologies Programme to develop quantum communications for safe and secure communication.

Source: Press Releases

12.1.4 MNM VIEW

ID Quantique is a prominent player in the field of quantum cryptography and provides enhanced features and functionalities to its commercial customers. The company provides high-performance, quantum-safe security solutions to help governments and commercial customers protect data in the era of large-scale quantum computers. The company's strategic intent lies in maintaining strong relationships with its channel partners, Value-Added Resellers (VARs), System Integrators (SI), and Independent Software Vendors (ISV). ID Quantique forms partnerships and collaborations to enhance its product portfolio and global presence. In October 2018, InfoSec Global collaborated with ID Quantique to deliver a Quantum-Safe VPN, including cryptographic agility and QRNG. This collaboration delivers a quantum-powered crypto agile Layer 3 VPN solution to customers for ensuring data integrity and confidentiality to evolve the data environment. In addition, June 2018, ID Quantique partnered with Quantum Xchange to bring quantum-safe cryptography-as-a-service in the US. The company plays a leading role in cutting-edge projects to drive innovation to the market. Moreover, in February 2018, ID Quantique announced a strategic investment plan of USD 65 million from investor SK Telecom. The investment would strengthen ID Quantique's position as a global leader in quantum-safe cryptography and quantum sensing solutions. Furthermore, in 2016, ID Quantique raised USD 4 million from SK Telecom to enhance its presence and product portfolio.

12.2 QUINTESSENCELABS

12.2.1 BUSINESS OVERVIEW

QuintessenceLabs was founded in 2008 and is headquartered in Deakin, Australia. The company offers security foundation for customers' data whether the data is on-premises, in the cloud, or in hybrid IT ecosystems. Its offering includes centralized enterprise key and policy management, an integrated hardware security module, a true random number generator, and secure encryption for data. The company's solutions integrate data protection capabilities with the latest quantum technology to deliver enhanced security to customers' data.

COMPANY AT A GLANCE

Founded:	2008
Headquarters:	Deakin, Australia
Ownership:	Private
Specialties:	Computer and Network Security

Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

The company's product portfolio addresses the most difficult security challenges and implements robust security strategies to protect the data. The offering includes qCrypt, qProtect, qRand, qClient, qStream, qOptica, and data security solutions. These offerings are instrumental in maximizing the security and reducing data security complexities. Its data protection solutions integrate unique quantum skills to offer enhanced security to customers. In addition, its quantum random number generators offer high-quality keys for all commercial applications, either as standalone cards or integrated into key management appliances and enable to securely exchange keys, even when quantum computers break current key exchange protocols.

The company provides its products and solutions in Australia and internationally, and has its geographical presence in Australia and North America.

12.2.2 SOLUTIONS OFFERED

The company offers the following solutions:

CATEGORY	SOLUTION
Quantum Cryptography	qProtect

Source: Company Website

12.2.3 RECENT DEVELOPMENTS

DATE	NATURE OF DEVELOPMENT	DESCRIPTION
November 2016	Collaboration	QuintessenceLabs collaborated with VMware to enhance the security of its virtualized applications. Moreover, customers would be able to securely integrate the cryptographic key and policy management solutions into their virtual applications, thereby securely encrypting their data while retaining control over their keys.
September 2016	Partnership	QuintessenceLabs partnered with PKWARE to create an enhanced encryption solution, offering unique key management capabilities and control of data-security policies for global enterprises and government organizations.

Source: Press Releases

12.2.4 MNM VIEW

QuintessenceLabs focuses on inorganic growth strategies and builds strong long-term partnerships with technology leaders and VARs across the globe to strengthen its product portfolio and market presence. The company is technologically updating its products and solutions to serve the needs for its clients. For instance, in November 2016, QuintessenceLabs collaborated with VMware, a leading provider of cloud computing and platform virtualization software and services. This collaboration helped QuintessenceLabs enhance the security of its virtualized applications and enable customers to securely encrypt their data while retaining the control of their keys.

In September 2016, QuintessenceLabs partnered with PKWARE, a provider of encryption and compression solutions. This partnership enabled the creation of an enhanced encryption solution, offering unique key management capabilities and control of data security policies for global enterprises and government organizations. In addition, the company has achieved a global footprint through its R&D partnerships with leading companies and R&D institutions. In July 2017, QuintessenceLabs received an investment of USD 3.26 million for QKD research from the Australian Department of Defense. The company used this investment to offer a sovereign Australian quantum key distribution solution. This solution assisted defense and Australian government systems from malicious cyber intrusion and disruption, and enhance the resilience of defense networks. Recently in February 2020, the company secured an investment of an undisclosed amount from In-Q-Tel, Inc, a not-for-profit venture capital firm.

12.3 CRYPTA LABS

12.3.1 BUSINESS OVERVIEW

Crypta Labs was founded in 2015 and is headquartered in Islington, UK. The company is a cybersecurity provider, which develops quantum-based encryption chips to secure IoT. Moreover, it offers hardware and software solutions. Crypta Labs' software solutions are used for mobile devices that are equipped with an image capturing device. Moreover, in hardware solutions, the company offers quantum random number generation encryption chip, harnessing the entropy of light photon reception to a sensor. Crypta Labs offers quantum cryptography under its flagship product, QRNG, a quantum random number encryption module. The company offers quantum cryptography to various industry verticals, such as transport, military, and medical. Crypta Labs' in the transport industry offers protection to the traffic management systems for security protection. In the military industry, cast iron encryption becomes vital, therefore Crypta Labs offers solutions to protect the industry from such threats. Furthermore, in the medical industry, the company provides solutions to safeguard holes in insulin pumps and pacemakers and ensures the safety of the overall medical system.

COMPANY AT A GLANCE

Founded:	2015
Headquarters:	Islington, UK
Ownership:	Private
Specialties:	Security and Investigations



Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

12.3.2 SOLUTIONS OFFERED

The company offers the following solutions:

CATEGORY	SOLUTION
Quantum Cryptography	QRNG

Source: Company Website

12.3.3 RECENT DEVELOPMENTS

DATE	NATURE OF DEVELOPMENT	DESCRIPTION
May 2020	Collaboration	Crypta Labs collaborated with SPRINT to develop its QRNGs for space applications.
April 2017	New Product Launch	Crypta Labs introduced QRNG prototype for secure encryption from a mobile phone or a photonic capable chip.

Source: Press Releases

12.3.4 MNM VIEW

The company is emerging significantly by offering reliable and enhanced products and solutions in the quantum cryptography market. Moreover, the company strengthened its base by enhancing its product portfolio and introducing innovative solutions. In line with its new product launches strategy, the company is working toward efficiently serving the security needs for its clients by technologically updating its products and solutions. For instance, in April 2017, Crypta Labs produced a prototype quantum random number generator microchip to secure communications for defense enterprise in the UK. This prototype helped Crypta Labs in proving that the result generated a random number is truly quantum in its nature and hence is capable of being used for secure encryption.

The company focuses on emerging technologies to enhance the existing range of product portfolio. For instance, in September 2017, Crypta Labs joined the Internet of Things Security Foundation (IoTSF). The IoTSF is an organization made up of over 100 companies that are dedicated to secure IoT. The foundation help companies increase the adoption and benefits of IoT. In July 2019, Crypta Labs received a funding of USD 3.3 million from Bloc Ventures. The aim was to further develop the company's QRNGs IP portfolio and to introduce the Quantum-enabled Hardware Security Module (QHSM) in the market.

12.4 QASKY

12.4.1 BUSINESS OVERVIEW

Qasky was founded in 2009 and is headquartered in Wuhu, China. The company was established by the joint investment of Wuhu Construction and Investment and the University of Science and Technology of China. It is a licensed seller and producer of commercial cipher products. The company provides point-to-point quantum cryptography communication technologies, quantum cryptography communication networking technologies, and quantum cryptography communication core devices.

COMPANY AT A GLANCE

Founded:	2009
Headquarters:	Wuhu, China
Ownership:	Private
Specialties:	Quantum Information Technology

Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

In the quantum cryptography arena, the company offerings include such as quantum cryptography communication device, quantum cryptography communication applications, quantum cryptography teaching and scientific research system, and key instrument of quantum and the derived technology. In addition, the company provides 2 different solutions for quantum cryptography, namely, scheme for quantum cryptography product networking and solutions for quantum cryptography product industry. Quantum cryptography communication devices offer quantum key distribution terminal and GHz quantum key distribution terminal. These terminals are dedicated devices to realize the quantum cryptography technology and are used as the most stable quantum cryptography communication solutions. Quantum cryptography communication applications offer quantum security gateway and quantum routing security based on the basic physical laws of quantum mechanics. Furthermore, the key instrument of quantum and derived technologies consist of infrared single photon detector, high-speed picosecond laser, and a high-speed clock generator.

12.4.2 SOLUTIONS OFFERED

The company offers the following solutions:

CATEGORY	SOLUTION
Quantum Cryptography	- Quantum cryptography communication device - Quantum cryptography communication applications - Key instrument of quantum and the derived technology

Source: Company Website

12.4.3 RECENT DEVELOPMENTS

DATE	NATURE OF DEVELOPMENT	DESCRIPTION
April 2017	Agreement	Qasky and Hentong Photoelectric signed a Memorandum of Understanding (MoU) to establish Jiangsu Hengtong Qasky Information Institute (research institute), and expand cooperation in several aspects.

Source: Press Releases

12.4.4 MNM VIEW

Qasky focuses on maintaining customer-focused and market-oriented solutions to provide users with advanced services and cost-effective products. Moreover, the company is constantly devoted to achieve the highest level of the information security system, R&D of equipment, and the active participation in the standardization construction to promote the maturity of the quantum security communication industry. For instance, in August 2016, Qasky signed MoU with Hentong Photoelectric for establishment of Jiangsu Hengtong Qasky Information Institute and expand cooperation in several aspects, such as construction and operation of application demonstration line of quantum security communication, application of quantum security communication in specific fields, relevant application technology of quantum security communication, and new optical fiber of quantum security communication and research of quantum photoelectric apparatus.

12.5 QUBITEKK

12.5.1 BUSINESS OVERVIEW

Qubitekk was founded in 2012 and is headquartered in California, US. The company offers quantum optic-based cybersecurity solutions, which enable interference-free communications within the electric grid automation equipment. Qubitekk offers various products, including QKD for Industrial Control Systems (ICS), QKD Demonstrator for R&D, and HackerBox. QKD ICS offers cutting-edge channel security by generating, maintaining, and distributing encryption keys, used on an ICS network. It uses quantum entangled photons to guarantee tamper detection and provide encryption security from quantum computing attacks. QKD Demonstrator for R&D is instrumental in demonstrating protocols and measurement methods involved in QKD. It is a flexible QKD platform, which is used to demonstrate polarization entangled QKD and reconfigure new quantum cryptography implementations.

COMPANY AT A GLANCE

Founded:	2012
Headquarters:	California, US
Ownership:	Private
Specialties:	Computer and Network Security

Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

HackerBox is an add-on device for the QKD Demonstrator, consisting a fiber optic switch, a polarizing beam splitter, 2 single photon counting detectors, and a polarized output. It helps intercept, measure, and reproduce a polarization-entangled single photon. Furthermore, the company is introducing QKD Quantum Repeater, which is in its developing phase and would be used to measure and amplify a quantum signal.

12.5.2 SOLUTIONS OFFERED

The company offers the following solutions:

CATEGORY	SOLUTION
Quantum Cryptography	- QKD for ICS - QKD Demonstrator for R&D - HackerBox

Source: Company Website

12.5.3 RECENT DEVELOPMENTS

DATE	NATURE OF DEVELOPMENT	DESCRIPTION
January 2020	Acquisition	Qubitekk acquired QinetiQ's Quantum Key Distribution (QKD) patent portfolio. The portfolio covers novel technological approaches in quantum science, including 57 patent filings, across 17 patent families.
July 2017	Product Enhancement	Qubitekk licensed an Oak Ridge National Laboratory developed a method to produce quantum light particles, known as photons, in a controlled, deterministic manner that guarantees improved speed and security while sharing the encrypted data.

Source: Press Releases

12.5.4 MNM VIEW

Qubitekk's strategic intent lies in creating a vibrant business ecosystem and maintaining long-term relationships with its partners. It majorly focuses on delivering customer-centric and industry-specific quantum cryptography solutions to ensure partner and customer success. The company has strengthened its base by enhancing its product portfolio and introducing innovative solutions. Moreover, in July 2017, Qubitekk licensed the single-photon source approach of Oak Ridge National Laboratory (ORNL) to improve the speed and security of the encrypted data. This approach specifies and controls every aspect of the photon's quantum state, and is expected to strengthen the technology through further development of ORNL's novel approach. In October 2018, the US Department of Energy awarded USD 28 million to support cybersecurity of the US energy infrastructure where Qubitekk would receive federal funding of USD 3 million to help protect the nation's power grid from cyber-attack. Furthermore, it is also expected to receive federal funding to introduce its quantum protected network Airforce depots.

12.6 ISARA

12.6.1 BUSINESS OVERVIEW

ISARA was founded in 2015 and is headquartered in Ontario, Canada. The company holds expertise in building security solutions and focuses on creating quantum-safe cryptography for computing ecosystems. In the area of quantum cryptography, the company offers ISARA Radiate Security Solution Suite, the first complete security solution to offer high-quality quantum-safe algorithms and integration tools made for developers. The suite offers end-to-end coverage across all areas required for quantum-safe algorithm development, such as lattice-based, hash-based, code-based, multivariate-based, and supersingular isogeny-based. Furthermore, the company provides an optional extension termed as ISARA Catalyst agile digital certificate technology. This technology is an optional X.509 digital certificate extension that enables users a seamless integration of quantum-safe security into user's certificate and Public Key Infrastructure (PKI) without impacting their legacy systems and offering future-proofed security solutions.

COMPANY AT A GLANCE

Founded:	2015
Headquarters:	Ontario, Canada
Ownership:	Private
Specialties:	Computer and Network Security

Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

12.6.2 SOLUTIONS OFFERED

The company offers the following solutions:

CATEGORY	SOLUTION
Quantum Cryptography	ISARA Radiate Security Solution Suite

Source: Company Website

12.6.3 RECENT DEVELOPMENTS

DATE	NATURE OF DEVELOPMENT	DESCRIPTION
April 2020	Partnership	Cybersec Innovation Partners Limited (CIP) partnered with ISARA to offer quantum-safe, crypto agile, and hybrid certificate offerings.
August 2019	Collaboration	Thales collaborated with ISARA Corp and ID Quantique for a quantum-safe, crypto-agile solution designed to protect data against the security threat of quantum computing.
February 2019	New Product Launch	ISARA launched new tools to test and implement crypto-agility and quantum-safe cryptography directly into existing systems.
October 2018	New Product Launch	ISARA announced ISARA Catalyst Agile Digital Certificate Technology, which enables organizations and manufacturers to secure their infrastructure and connected devices from encryption breaking quantum compute threats.
September 2018	Partnership	DigiCert, Gemalto, and ISARA partnered to develop advanced quantum-safe digital certificates and secure key management for connected devices commonly referred to as IoT.

April 2018	Partnership	ISARA partnered with Certified Security Solutions (CSS) to introduce quantum-safe, full-stack PKI solution. The PKI solution can ensure its OTA software updates are secure against attack from conventional systems and quantum computers in the future.
April 2018	Collaboration	Cisco ISARA collaborated to test digital certificates that operate in both classic and quantum-safe algorithm modes.

Source: Press Releases

12.6.4 MNM VIEW

ISARA accelerated its transformation as one of the leaders in the quantum cryptography market by launching innovative products for various industry verticals. The company's business strategic lies around its ability to partner and collaborate with various companies and co-develop products and solutions with them. For instance, In September 2018, DigiCert, Gemalto, and ISARA collaborated to build advanced quantum-safe digital certificates and secure key management for connected devices commonly referred to as IoT. The partnership offers significant advantages for enterprise security teams looking to secure connected devices with long-lasting product lifetimes. The partnership of DigiCert with ISARA and Gemalto would enable quantum-resistant certificates with the capability of hosted, on-premises, and hybrid deployment options. Moreover, in April 2018, Cisco collaborated with ISARA to test digital certificates that operate in both classic and quantum-safe algorithm modes. Furthermore, the company focuses on inorganic growth strategies, including new product launches and product enhancements. For instance, in October 2018, the company announced the launch of ISARA Catalyst Agile Digital Certificate Technology, a technology that enables enterprises and manufacturers to secure their infrastructure and connected devices from encryption breaking quantum computer threats and then in February 2019, the company launched new and updated toolkits to test and implement crypto-agility and quantum-safe cryptography directly into existing systems.

12.7 POST-QUANTUM

12.7.1 BUSINESS OVERVIEW

Post-Quantum was founded in 2009 and is headquartered in London, UK. The company provides data confidentiality, authenticity, and access control solutions. Moreover, it offers solutions for data-at-rest, data-in-transit, access to data, and business processes. These solutions are instrumental in providing future-proof encryption, cloud adoption, multiparty authorization, biometric signing, and regulation and transparency.

COMPANY AT A GLANCE

Founded:	2009
Headquarters:	London, UK
Ownership:	Private
Specialties:	Computer and Network Security

Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

In the quantum cryptography market, Post-Quantum offers quantum resistant encryption. This quantum resists encryption enables its commercial clientele to protect their data from code-breaking quantum computers. This solution can be deployed as a standalone encryption solution for providing security. Moreover, it offers a patented quantum-resistant encryption algorithm, which can be applied to the existing products and networks. The algorithm Never-The-Same (NTS) never generates the same ciphertext twice, even when the plaintext and encryption keys remain the same. The company offers solutions for physical and cyber threats to various industry verticals, such as finance, healthcare, defense, and government and banking.

12.7.2 SOLUTIONS OFFERED

The company offers the following solutions:

CATEGORY	SOLUTION
Quantum Cryptography	Quantum-Resistant Encryption

Source: Company Website

12.7.3 RECENT DEVELOPMENTS

DATE	NATURE OF DEVELOPMENT	DESCRIPTION
August 2016	Collaboration	Post-Quantum collaborated with IPC to facilitate next-generation security solutions as part of IPC's Connexus Cloud community of services.

Source: Press Releases

12.7.4 MNM VIEW

Post-Quantum's business strategy lies in maintaining stable long-term relationships with partners and association memberships. The company has been actively bagging a series of funding from various investment groups. For instance, in July 2016, Post-Quantum received funding of approximately USD 8.9 million from the Hong Kong-based VMS Investment Group and AM Partners. Moreover, in August 2016, the company collaborated with IPC, a leading global provider of secure and compliant communications and networking solutions. This collaboration enabled Post-Quantum to integrate biometric, encryption, and blockchain technology-as-a-service with the help of IPC's Connexus Cloud ecosystem.

12.8 QUANTUMCTEK

12.8.1 BUSINESS OVERVIEW

QuantumCTek was founded in 2009 and is headquartered in Hefei, China. The company is a Chinese pioneer in commercialized Quantum Information Technology (QIT), and provider of QIT-enabled ICT security products and services. It offers wide range of products, such as national commercial cipher products, quantum key distribution products, quantum key management products, quantum-secure communication networking products, quantum-secure communication network support system, and core devices.

COMPANY AT A GLANCE

Founded:	2009
Headquarters:	Hefei, China
Ownership:	Private
Specialties:	Quantum Information Technology

Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

In the quantum cryptography market, the company offers QKD-PHA300, QKD-POL1250, QKD-POL40, and QKD-G5 for quantum secured communication. The company has been providing long-term quantum safety to various industry verticals, such as government, financial, and energy. Since 2009, more than 1000 quantum secure products have been manufactured and running online, and securing communication links longer than 4000 km.

12.8.2 SOLUTIONS OFFERED

The company offers the following solutions:

CATEGORY	SOLUTION
Quantum Cryptography	- QKD-PHA300 - QKD-POL1250 - QKD-POL40 - QKD-G5

Source: Company Website

12.8.3 RECENT DEVELOPMENTS

DATE	NATURE OF DEVELOPMENT	DESCRIPTION
March 2018	Collaboration	QuantumCTek collaborated with University of Science and Technology of China, CAS Quantum Network, Jinan Institute of Quantum Technology, China Unicom, and Yangtze Optical Fibre and Cable Company, to introduce the integration of QKD with commercial backbone network carrying 3.6 Tbps classical data.
March 2017	Partnership	Alibaba partnered with QuantumCTek to make AliCloud, a cloud computing company, which can provide quantum encrypted transmission of information.

Source: Press Releases

12.8.4 MNM VIEW

The company's growth strategies lie in partnerships and collaborations with companies and R&D institutes to enhance its product portfolio. This fact is evident with ongoing innovations and collaborations of quantum science and technology, and offering QIT portfolio of quantum secure solutions in the telecom infrastructure, cloud computing, enterprise networks, and big data technology and services. For instance, in March 2018, QuantumCTek collaborated with University of Science and Technology of China, CAS Quantum Network, Jinan Institute of Quantum Technology, China Unicom, and Yangtze Optical Fibre and Cable Company, to introduce the integration of QKD with commercial backbone network carrying 3.6 Tbps classical data. Developments of the experiment started in July 2016, and with one year of preparation, the field experiment was carried out in June 2017. The collaboration is expected to make a key step for QuantumCTek and also toward building a quantum network that coexists with the existing classic fiber infrastructures.

12.9 MAGIQ TECHNOLOGIES

12.9.1 BUSINESS OVERVIEW

MagiQ Technologies was established in 1999 and is headquartered in Massachusetts, US. MagiQ Technologies is a technology development company, which develops and delivers advanced communication technologies for defense, telecommunications, energy exploration, and network security. The company is enhancing quantum innovation by developing proven, optical sensing applications for Radio-Frequency (RF) interference cancellation, quantum cryptography, and optical surveillance for advanced energy exploration.

COMPANY AT A GLANCE

Founded:	1999
Headquarters:	Massachusetts, US
Ownership:	Private
Specialties:	Quantum Information Solutions

Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

MagiQ Technologies offers quantum cryptography with the help of MagiQ QPN, a completely secured and future-proof communication channel against numerous cryptographic key distribution and management challenges. MagiQ QPN assists organizations to secure critical links to prevent data thefts. It supports a variety of network architectures and offers cryptography key exchange infrastructure to protect information channels. MagiQ Technologies offers complete quantum cryptography solutions as comprehensive packaged solutions to its commercial clientele, enabling them to gain a competitive edge in the market. The company possesses a significant range of offerings in the quantum cryptography arena, and its solutions provide enhanced features and functionalities to its commercial customers. The company serves various industry verticals, including aerospace and defense, government, telecom, and research.

12.10 NUCRYPT

12.10.1 BUSINESS OVERVIEW

NuCrypt was founded in 2003 and is headquartered in Illinois, US. The company develops products at the interface of photonic, electronic, and quantum technologies. The primary focus of the company is supplying photonic instrumentation to the research and development community for enhancing the productivity of end-users. The company offers a range of products, such as entangled photon source, correlated photon detection system, polarization analyzer, quantum system control software, programmable pulse generator, and deterministic fiber generator. In the quantum cryptography arena, NuCrypt is a leading provider of quantum optical instrumentation products, such as Entangled Photon Source (EPS), Correlated Photon Detection System (CPDS), polarization analyzer, and quantum system control software. The company provides a suite of products that generate, manipulate, and detect photons, including the fiber-coupled entangled pairs. The company includes a variety of features that enable its commercial clientele to break the quantum barrier in the field quickly and account for a variety of practical issues that make distributing quantum entanglement difficult.

COMPANY AT A GLANCE

Founded:	2003
Headquarters:	Illinois, US
Ownership:	Private
Specialties:	Telecommunications



Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

12.11 QUANTUM XCHANGE

12.11.1 BUSINESS OVERVIEW

Quantum Xchange was founded in 2018 and is headquartered in Maryland, US. The company offers commercial organizations and government agencies with the quantum-secured network in the US. Moreover, it secures the channel for critical communications, enabling organizations to send unbreakable data over any distance using quantum physics laws.

Quantum Xchange introduced Phio in June 2018, a fiber optic network, which helps transmit and receive quantum cryptographic keys between entities that are transferring the critical data. The technology used to build Phio is QKD and Quantum XChange's Trusted Node technology. Phio enables commercial enterprises and government agencies to securely exchange data between 2 entities at any distance using the secure QKD method and offering defense in keeping data safe and secure. The company with less than a year is growing significantly and increasing its network. For instance, in October 2018, Quantum Xchange announced an agreement with Zayo Group to offer 800 kilometers of quantum networks in the US.

COMPANY AT A GLANCE

Founded:	2018
Headquarters:	Maryland, US
Ownership:	Private
Specialties:	IT Services



Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

12.12 AUREA TECHNOLOGY

12.12.1 BUSINESS OVERVIEW

AUREA Technology was founded in 2010 and is headquartered in Besancon, France. It designs and manufactures optical instruments based on advanced single-photon avalanche photodiodes, ultrafast laser diodes, and fast timing electronics. One of the critical parts of QKD is the use of single-photon counting detectors, which are used to receive distributed secret keys and help detect fraudulent activities.

The photon counter, SPD_A series, is designed for QKD at telecom wavelengths. Its high detection performance (very low noise and high efficiency) enables government agencies, financial institutions, and data centers or companies with distributed offices to encrypt confidential communication with the highest level of security over distances exceeding 100 km. Furthermore, the company provides support to more than 200 customers across the globe directly or with the help of its professional distribution channels in North America, Europe, and Asia.

COMPANY AT A GLANCE

Founded:	2010
Headquarters:	Besancon, France
Ownership:	Private
Specialties:	Electrical/Electronic Manufacturing

Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

12.13 QUTOOLS

12.13.1 BUSINESS OVERVIEW

qutools was founded in 2005 and is headquartered in Munich, Germany. qutools is developing, manufacturing, and distributing tools for quantum physics experiments in science and the commercial industry. The company distributes products in the field of quantum information science, quantum metrology, quantum cryptography, and time-correlated single photon counting. The company offers quTAG, quTAU, and quRNG. quRNG, a quantum random number generator, produces true random numbers, with a maximum bit rate of 50 Mbit/s, and a readout via a USB 2.0 interface.

COMPANY AT A GLANCE

Founded:	2005
Headquarters:	Munich, Germany
Ownership:	Private
Specialties:	Research



Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

The source of randomness is based on the uncertainty of the detection time of photons emitted from a Light Emitting Diode (LED), and the photons are collected by a single photon detector at a high rate. It provides applications, such as cryptography, online gambling, numerical simulations, pin generation, and quantum cryptography. qutools provides the research and industrial version of quantum random number generator, and both the versions continuously run tests on the state of the random bit stream to ensure high quality.

12.14 QUNU LABS

12.14.1 BUSINESS OVERVIEW

QuNu Labs was founded in 2016 and is headquartered in Bangalore, India. It is an incubate company of Indian Institute of Technology (IIT), Madras, whose purpose is to collaborate with academia to conduct research on advanced versions of cybersecurity and create technology and products for secure communication. The company focuses on algorithms and high-precision optics to create a quantum encryption system, Quantum Random Number Generators (QRNG), which provides an enhanced secure optical link.

COMPANY AT A GLANCE

Founded:	2016
Headquarters:	Bangalore, India
Ownership:	Private
Specialties:	Telecommunications



Note: This is a privately held company. Hence, no financial information is available in the public domain.

Source: Company Website

In the arena of quantum cryptography, QuNu Labs is the provider of QKD, quantum safe network encryption, single-photon counters, and hardware random number generators. It offers 2 products, namely, QNL-X100 and QUTRIX 1.0, where the QNL-X100 is a basic QKD box providing key distillation based on differential phase shift protocol and a built-in AES-based encryption for 100 gigs data. On the other hand, QUTRIX 1.0 is a true random number generator based on quantum source of entropy.

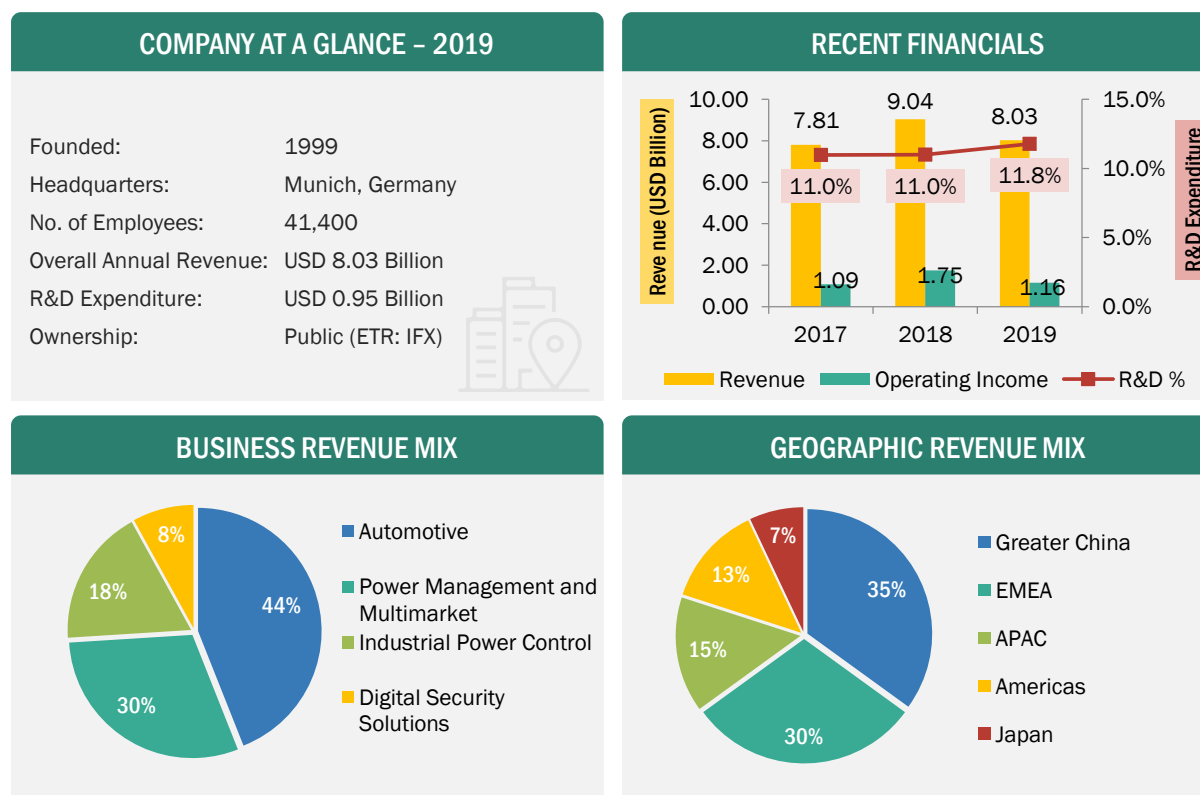
12.15 INFINEON

12.15.1 BUSINESS OVERVIEW

Infineon was founded in 1999 and is headquartered in Munich, Germany. The company designs, develops, and manufactures a wide range of semiconductors and system solutions. It operates through 4 segments: automotive, industrial power control, power management and multimarket, and digital security solutions. In the automotive segment, the company develops semiconductor solutions for a wide range of applications in combustion engine cars, and hybrid and electric vehicles. The industrial power control segment helps in converting the electric energy from the medium to a high power range. In the power management and multimarket segment, it develops components for mobile devices and is an expert in dealing with power semiconductors for power supplies. Under the digital security solutions segment, it helps address the security needs of the customers by offering payment cards and government IDs in the area of embedded security applications.

In the quantum cryptography market, the company is seen to emerge significantly by offering reliable and enhanced products and services. This is evident from the fact that, in 2017, Infineon demonstrated the first post-quantum cryptography on a contactless security chip, as used for electronic ID documents. The company continuously collaborates with the academic community, customers, and partners to offer improved and enhanced security solutions to customers. Furthermore, Infineon has established a strong global footprint, with its presence in the Americas, EMEA, and APAC.

FIGURE 25 INFINEON: COMPANY SNAPSHOT



Source: Company Website, Annual Reports, and SEC Filings

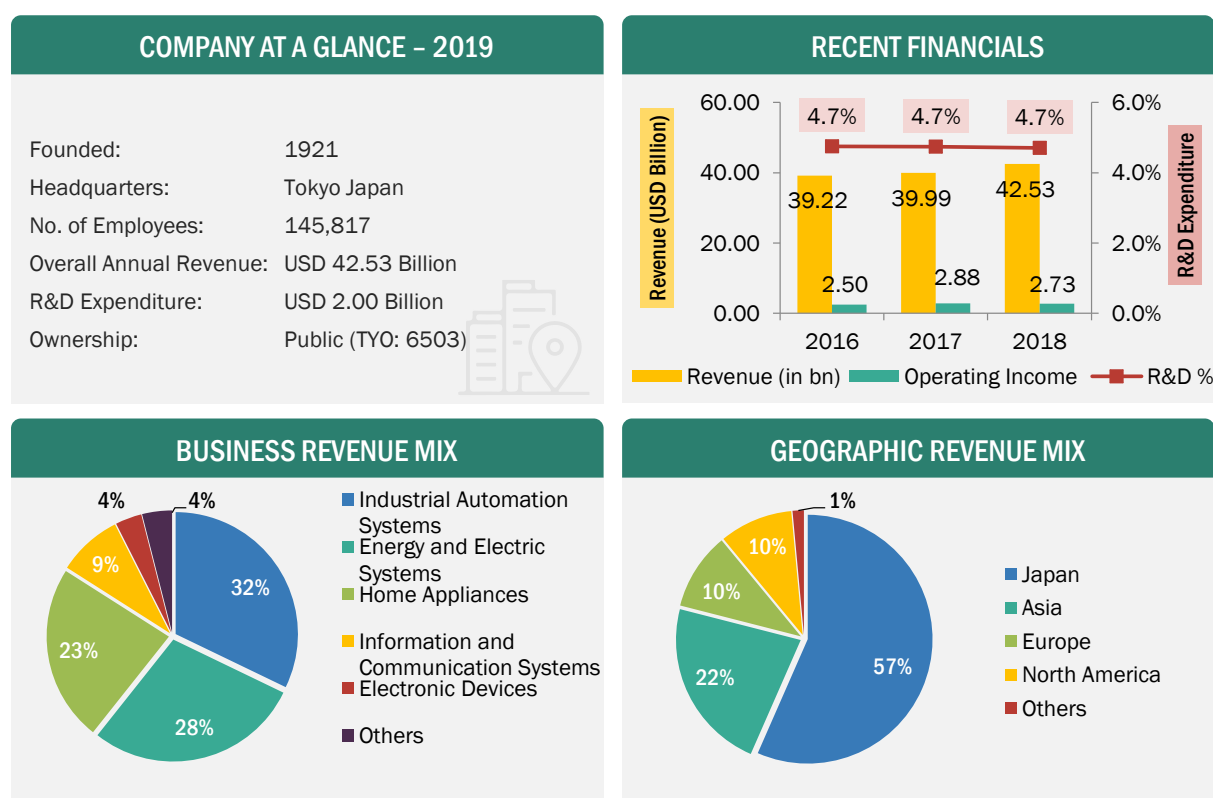
12.16 MITSUBISHI ELECTRIC

12.16.1 BUSINESS OVERVIEW

Mitsubishi Electric was founded in 1921 and is headquartered in Tokyo, Japan. The company develops, manufactures, distributes, and sells electrical and electronic equipment to its commercial clientele across the globe. The company has a dominant presence across all regions worldwide. It is positively investing in the R&D of quantum cryptography and quantum technologies. In October 2010, the National Institute of Information and Communications Technology (NICT) in cooperation with its commissioned research partners - NEC, Mitsubishi Electric, and NTT - started working on the QKD network using a part of NICT's open fiber testbed network, JGN2plus.

Cloud computing is gaining traction as a new way of data processing. This has initiated the need for optimum security measures to deal with concerns over the information that is leaked. In a collaborated research conducted with NTT, Mitsubishi developed functional encryption. It is a specific form of functional encryption, called attribute-based encryption, which incorporates the ciphertext and controls the access rights to the ciphertext. With an encryption algorithm, MISTY, and a quantum cryptographic communication system, Mitsubishi Electric aims at maintaining its prominent position in the cryptographic technology.

FIGURE 26 MITSUBISHI ELECTRIC: COMPANY SNAPSHOT



Source: Company Website, Annual Reports, and SEC Filings

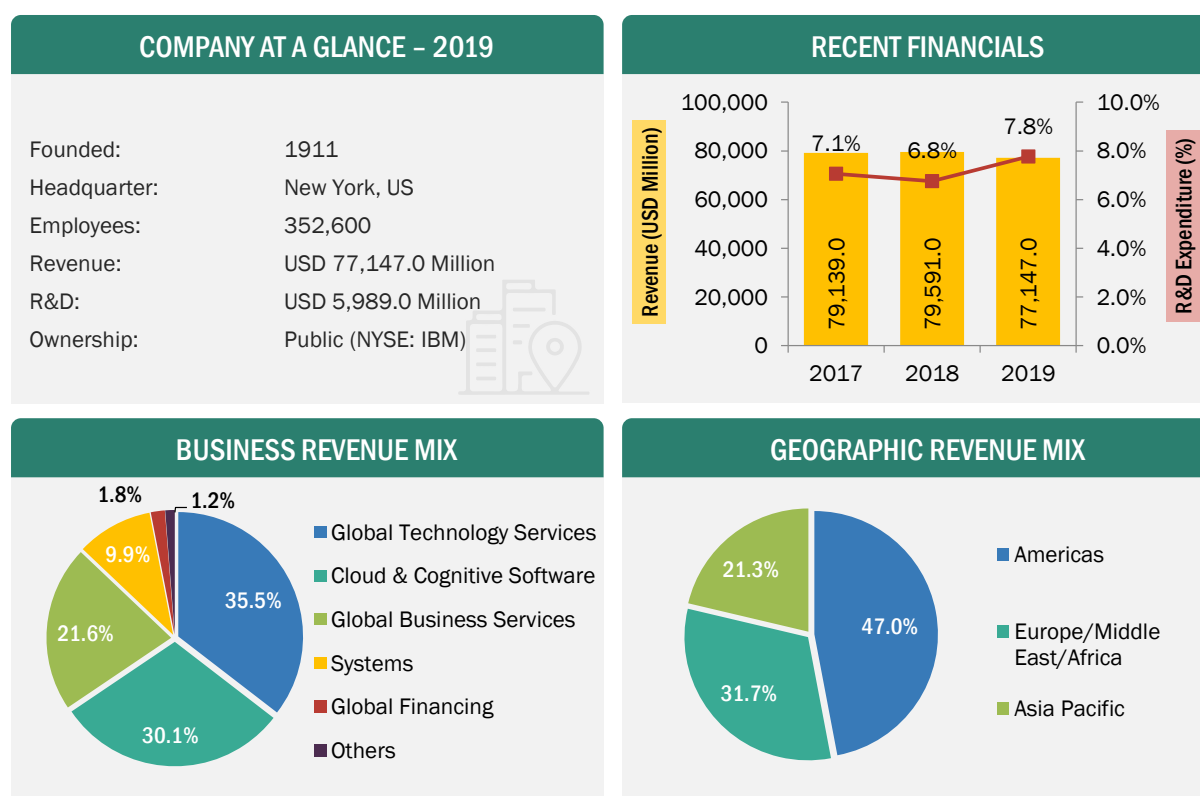
12.17 IBM

12.17.1 BUSINESS OVERVIEW

IBM was founded in 1911 and is headquartered in New York, US. The company has been instrumental in offering multiple technology products, solutions, and services across the globe. The company's major business segment includes technology services, cognitive solutions, global business services, systems, and global financing. The company also offers a robust product portfolio in various segments, such as analytics, cloud, IoT, mobile, and security. IBM offers its products and solutions to various verticals, such as automotive; aerospace and defense; government; healthcare; IT and telecom; Banking, Financial Services, and Insurance (BFSI); consumer products; and retail, in over 175 countries.

IBM has a cryptography research group at its T.J. Watson Research Center, where the theoretical, design, and implementation-level cryptographic protocols are discussed. IBM research activities are focused on developing practical cryptographic solutions that are resistant to the threats posed by quantum computers. Specifically, the company focuses on lattice-based cryptographic solutions. Lattice-based cryptography is an approach for creating security primitives.

FIGURE 27 IBM: COMPANY SNAPSHOT



Source: Company Website, Annual Reports, and SEC Filings

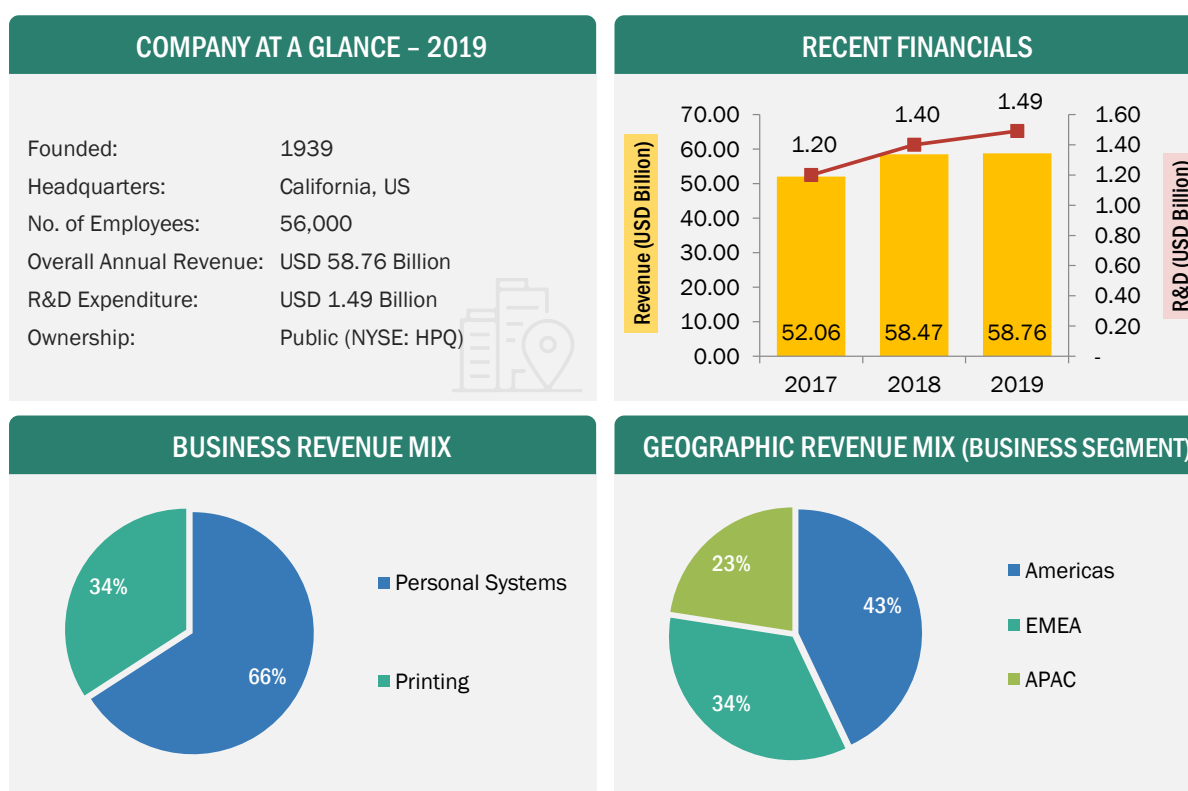
12.18 HP

12.18.1 BUSINESS OVERVIEW

HP was founded in 1939 and is headquartered in California, US. It is one of the global providers of personal computing, imaging and printing products, and associated solutions and services. It operates through 3 segments, namely, personal systems, printing, and corporate investments. The personal systems segment offers personal computers, workstations, tablets, software, and support and services to its customers. The printing segment comprises commercial printer hardware, supplies, media, and scanning devices. The corporate investments segment includes HP Labs and certain business incubation projects. The company caters to SMEs and large enterprises, including its clientele in the government, health, and education sectors.

The Quantum Information Processing (QIP) Group is focusing on quantum cryptography and is in the R&D phase. QIP Group is focusing on the 3 major areas of quantum, namely, quantum computation, quantum cryptography, and quantum teleportation and communication. In quantum cryptography, the QIP Group is working with collaborators to develop QKD technologies, including guaranteed random number generation, remote auctions, and authentication systems that prevent any possibility of imitation.

FIGURE 28 HP: COMPANY SNAPSHOT



Source: Company Website, Annual Reports, and SEC Filings

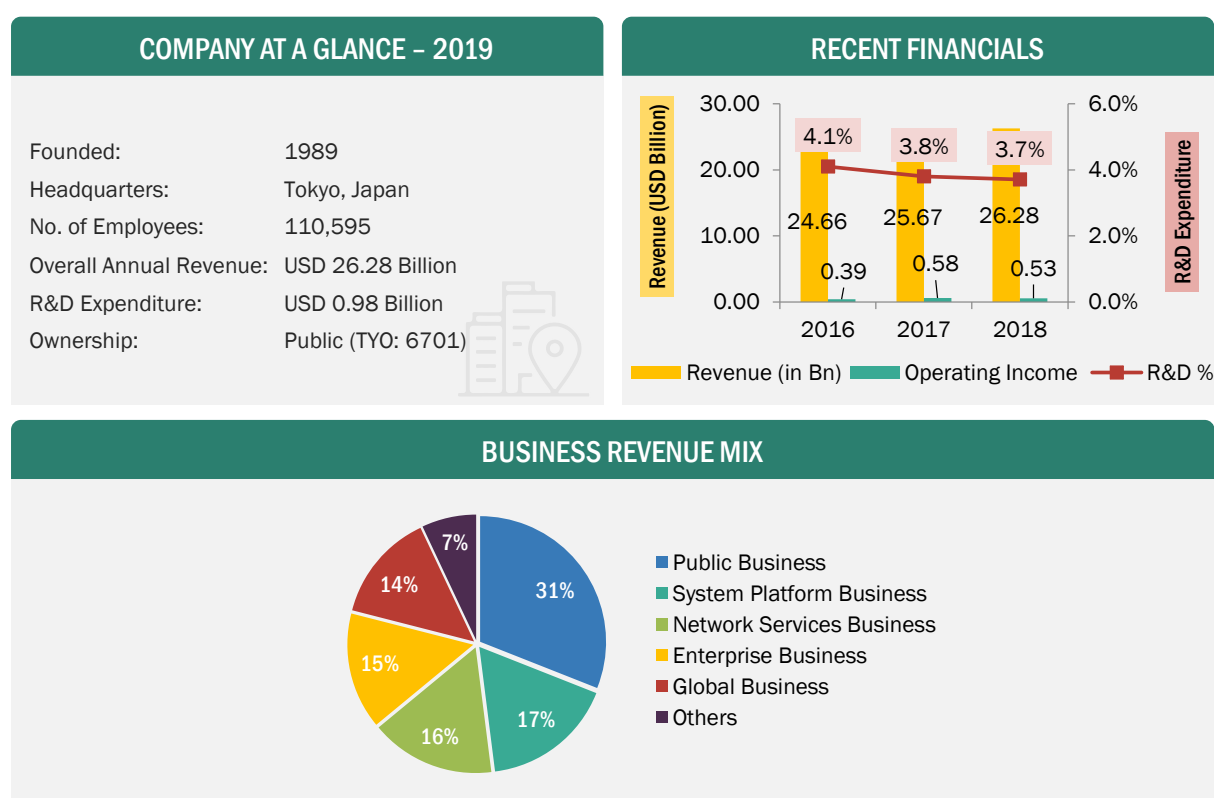
12.19 NEC

12.19.1 BUSINESS OVERVIEW

NEC was founded in 1989 and is headquartered in Tokyo, Japan. NEC is one of the leading providers of IT solutions, hardware, and networking products. It operates through different business segments, including public business, system platform business, global business, enterprise business, and network services business. The company caters its solutions and services to different industry verticals, such as finance, telecommunications, manufacturing, retail, healthcare, media, and government.

In 2015, NEC, along with the University of Tokyo and Fujitsu, successfully performed QKD using a single-photon emitter from a distance of 120 km. This success was attributed to the development of 2 new technologies: high-purity single-photon emitter operating in the 1.5-μm band and QKD system optimized for single-photon emitters using superconducting single-photon detectors. It is believed that the researchers at NEC are working on making single-photon QKD process more compact and faster, and are planning to bring a highly secure communication system for the major urban centers in the market.

FIGURE 29 NEC: COMPANY SNAPSHOT



Source: Company Website, Annual Reports, and SEC Filings

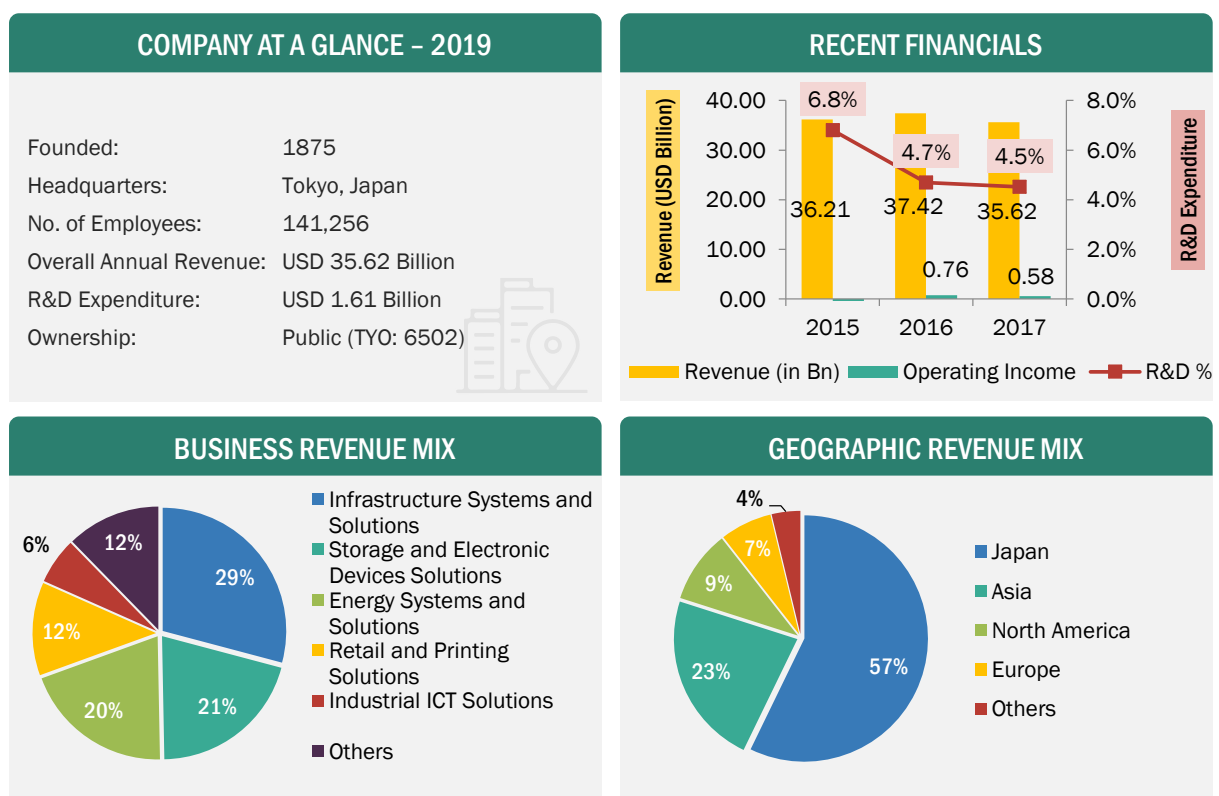
12.20 TOSHIBA

12.20.1 BUSINESS OVERVIEW

Toshiba was founded in 1875 and is headquartered in Tokyo, Japan. The company is one of the leading manufacturers and solution providers for advanced electronic products, electrical products, and systems, worldwide. Toshiba operates through 5 major business segments, namely, infrastructure systems and solutions, storage and electronic devices solutions, energy systems and solutions, retail and printing solutions, and industrial ICT solutions.

In October 2016, BT and Toshiba opened the secure quantum communication showcase at BT's research and development center in Ipswich. The 2 companies introduced cutting-edge quantum cryptography, which can be used to protect digital information transported within banks and financial services organizations from hacking. The company is aiming at offering quantum security solutions for optimum security systems. Toshiba is developing a new approach to IT using the fundamental laws of quantum physics to offer quantum-secure communications and computing. This is evident from the fact that in May 2018, Toshiba Research Europe announced a new protocol for QKD, which enables secure QKD over 500 km of optical fiber.

FIGURE 30 TOSHIBA: COMPANY SNAPSHOT



Source: Company Website, Annual Reports, and SEC Filings

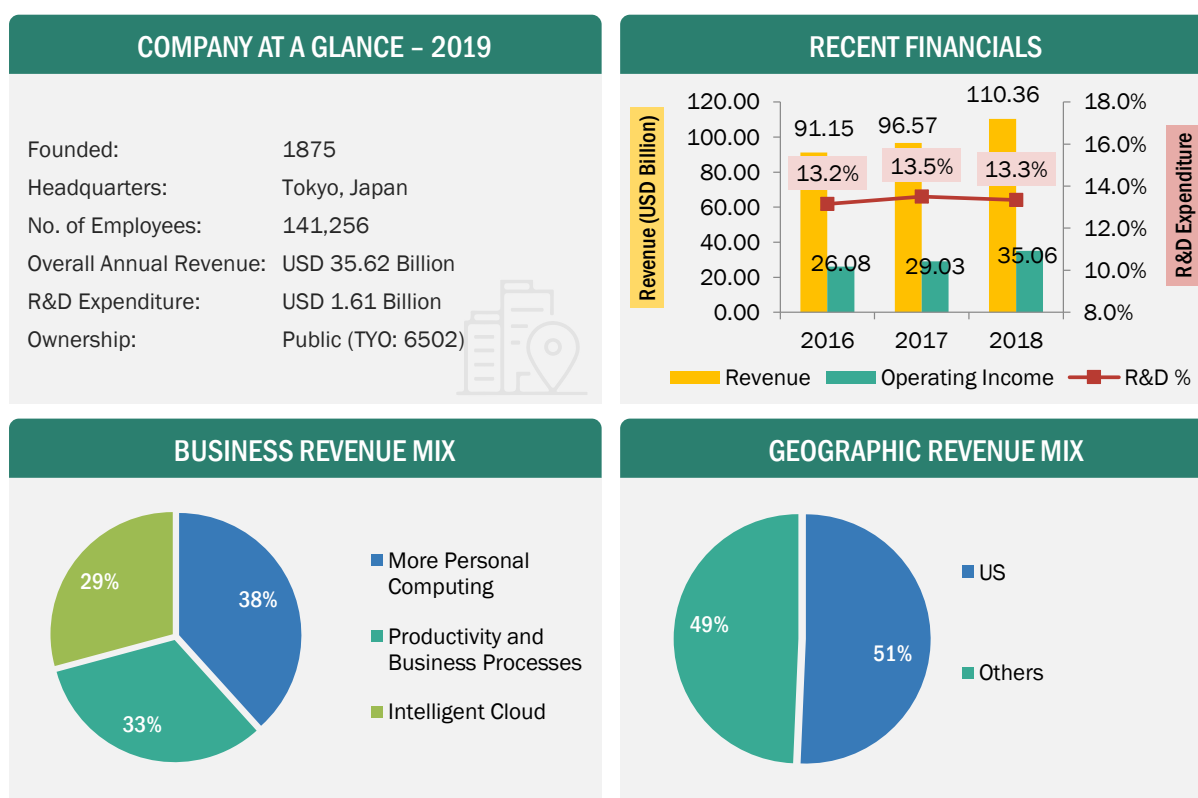
12.21 MICROSOFT

12.21.1 BUSINESS OVERVIEW

Microsoft was founded in 1975 and is headquartered in Washington, US. The company generates revenue by licensing and supporting a range of software products, offering a series of cloud-based and other services to customers across the globe. It operates through 3 business segments, namely, productivity and business processes, more personal computing, and intelligent cloud. Microsoft believes cybersecurity to be a challenge to the digital age, and focuses on innovating solutions to provide end-to-end security to customers with security operations at a global scale.

In the quantum cryptography market, Microsoft is in its research phase. In 2017, the company unveiled the vision for quantum algorithms for security challenges in the digital era. Microsoft Quantum Research Group is understanding post-quantum cryptography and focusing on how private information and communications can be protected when quantum computers, which can break that cryptography, are available.

FIGURE 31 MICROSOFT: COMPANY SNAPSHOT



Source: Company Website, Annual Reports, and SEC Filings

12.22 RAYTHEON TECHNOLOGIES

Founded	2020
Headquarters	Massachusetts, US
Ownership	Public
Business Overview	Raytheon Technologies is an aerospace and defense industrial company formed from the merger of United Technologies and Raytheon. The company operates through four business segments: Collins Aerospace Systems, Pratt and Whitney, Raytheon Intelligence and Space, and Raytheon Missiles and Defense. The company's quantum and engineering team offers next-generation technologies with new ways to apply it to computing, sensing, and communication. Its technical areas include quantum computing, classical cryogenic computing, quantum materials, quantum devices, and sensors and optics, and integrated photonics.
Solution Offered	Quantum Engineering and Computing
Geographic Presence	North America, APAC, and Europe

Source: Company Website

12.23 CRYPTO QUANTIQUE

Founded	2016
Headquarters	London, UK
Ownership	Private
Business Overview	Crypto !Quantique provides quantum-driven cybersecurity platforms. The company has combined quantum technologies with modern cryptography to offer unified end-to-end security for IoT. Its platform uses advanced techniques in cryptography and quantum physics and its IP generating unique, numerous cryptographic keys. Currently, the company has three patents for its core innovation.
Solutions Offered	- Quantum Driven Secure Chip (QDSC) - Cryptographic APIs
Geographic Presence	Europe

Source: Company Website

12.24 QRYPT

Founded	2017
Headquarters	New York, US
Ownership	Private
Business Overview	Qrypt offers encryption solutions capable of securing enterprises and private information networks. The company developed a cryptographic solution that can secure information against attacks by quantum computers, computational devices, or mathematical techniques. The company's first generation of hardware and datacenter appliances leverage multiple quantum technologies and are currently available via the cloud.
Solution Offered	Qrypt platform
Geographic Presence	North America

Source: Company Website

12.25 ONBOARD SECURITY (QUALCOMM)

Founded	2017
Headquarters	Massachusetts, US
Ownership	Public
Business Overview	OnBoard Security, a wholly-owned subsidiary of Qualcomm Technologies (QTI) offers cybersecurity solutions. The company provides security products that address automotive cybersecurity threats, critical security, and solutions for the IoT. Its target sectors are automotive and IoT organizations.
Solutions Offered	NTRU Cryptography
Geographic Presence	North America, APAC, and Europe

Source: Company Website

13 APPENDIX

13.1 DISCUSSION GUIDE

Q. 1. We have segmented the quantum cryptography market on the basis of components, so according to you,

- Which of these is the highest revenue contributor for 2019? Please rank on a scale of 1 – 10, (10 being the highest).
- How do you think will it grow in the next 5 years (2024)? Please rank on a scale of 1 – 10, (10 being the highest).

SR. NO.	BY COMPONENT	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 10, 10 BEING THE HIGHEST) – 2019	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 10, 10 BEING THE HIGHEST) – 2024
1	Solutions		
2	Services		

Note - Please suggest in case of any overlapping segments or technologies that you perceive is not relevant for the given market.*

Kindly mention the types of solutions such as Quantum key generation, Quantum random number generator, Quantum key distribution, and others

Q. 2. We have segmented the quantum cryptography market on the basis of services, so according to you,

- Which of these is the highest revenue contributor for 2019? Please rank on a scale of 1 – 10, (10 being the highest).
- How do you think will it grow in the next 5 years (2024)? Please rank on a scale of 1 – 10, (10 being the highest).

SR. NO.	BY SERVICE	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 10, 10 BEING THE HIGHEST) – 2019	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 10, 10 BEING THE HIGHEST) – 2024
1	Consulting		
2	Deployment and Integration		
3	Support and Maintenance		

Note - Please suggest in case of any overlapping segments or technologies that you perceive is not relevant for the given market.*

Q. 3. We have segmented the quantum cryptography market by vertical, so according to you,

- Which of these is the highest revenue contributor for 2019? Please rank on a scale of 1 – 10, (10 being the highest).
- How do you think will it grow in the next 5 years (2024)? Please rank on a scale of 1 – 10, (10 being the highest).

SR. NO.	BY VERTICAL	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 10, 10 BEING THE HIGHEST) – 2019	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 10, 10 BEING THE HIGHEST) – 2024
1	Government and Defense		
2	BFSI		
3	Retail		
4	Healthcare		
5	Automotive		
6	Others*		

Note - Please suggest in case of any overlapping segments or technologies that you perceive is not relevant for the given market.*

**Others include telecom, energy and utilities, logistics, and education*

Q. 4. Which Region, according to you, is the largest in terms of market opportunity the quantum cryptography now and in the future? Kindly give points on a scale of 1-10, with 10 being the biggest market.

- Which of these is the highest revenue contributor for 2019? Please rank on a scale of 1 – 10, (10 being the highest).
- How do you think will it grow in the next 5 years (2024)? Please rank on a scale of 1 – 10, (10 being the highest).

SR. NO.	BY REGION	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 10, 10 BEING THE HIGHEST) – 2019	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 10, 10 BEING THE HIGHEST) – 2024
1	North America		
2	Europe		
3	Latin America		
4	APAC		
5	MEA		

Note - Please suggest in case of any overlapping segments or technologies that you perceive is not relevant for the given market.*

13.2 KNOWLEDGE STORE: MARKETSANDMARKETS' SUBSCRIPTION PORTAL

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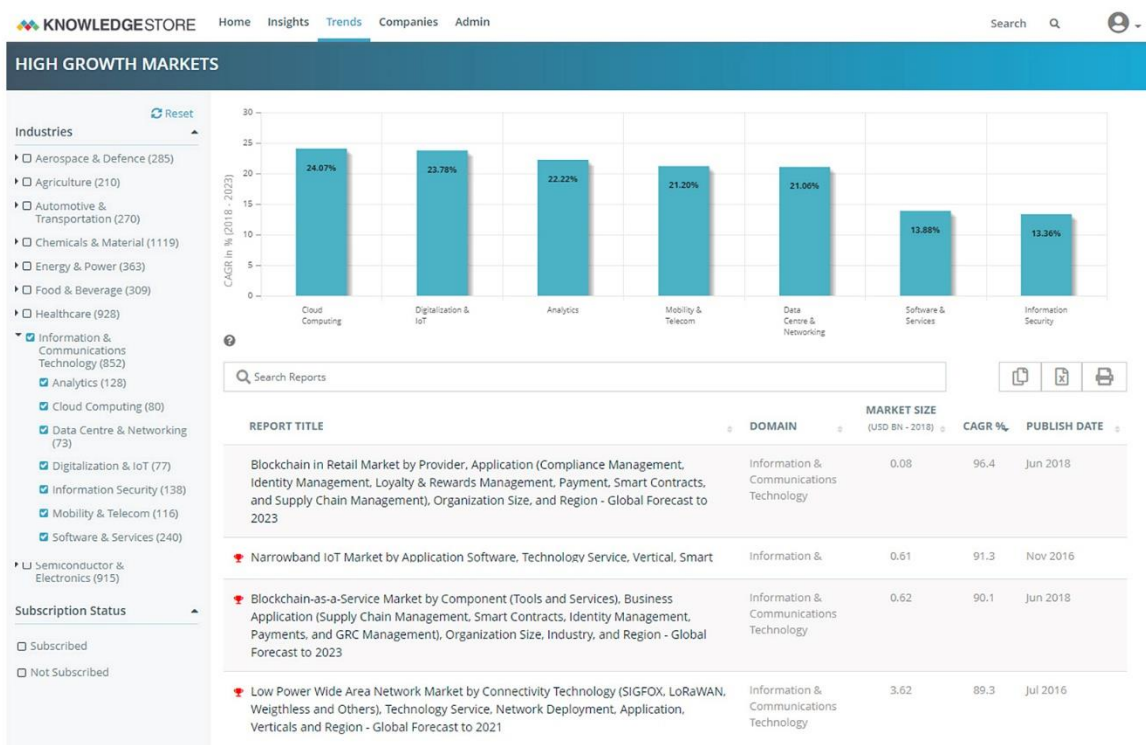
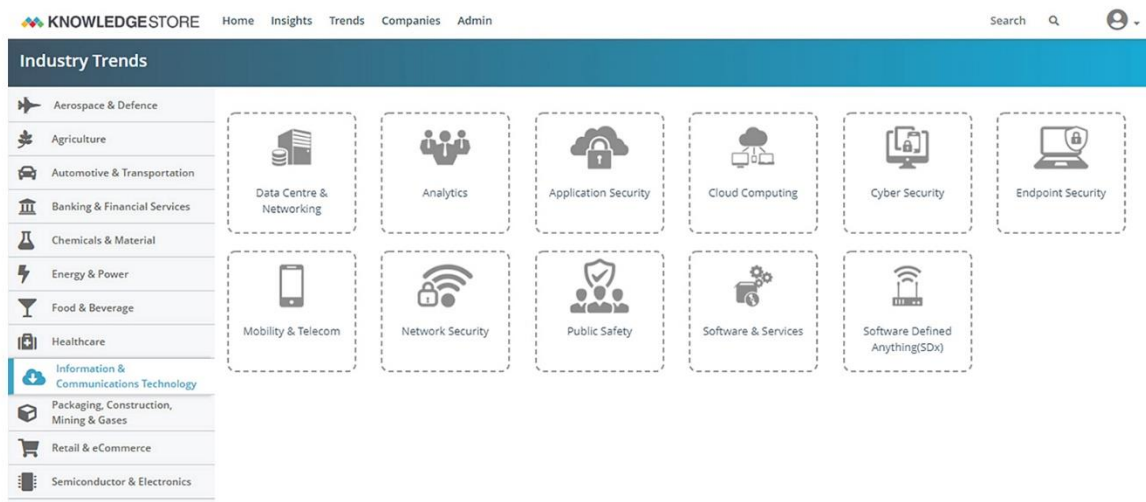
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13.3 AVAILABLE CUSTOMIZATIONS

With the given market data, MarketsandMarkets offers customizations as per the company's specific needs. The following customization options are available for the report:

GEOGRAPHIC ANALYSIS

- Further breakdown of the Americas Quantum Cryptography market
- Further breakdown of the European Quantum Cryptography market
- Further breakdown of the APAC Quantum Cryptography market
- Further breakdown of the Middle Eastern and African Quantum Cryptography market

13.4 RELATED REPORTS

SR. NO	REPORT TITLE	PUBLISHED DATE
1	QUANTUM DOTS MARKET - GLOBAL FORECAST TO 2021 By Product (QD Medical Devices, QD Solar Cells, QD Photodetector/QD Sensors, QD Lasers, QD Lighting, Batteries and Energy Storage Systems, QD Transistors, and QD Tags),Material, Vertical, and geography https://www.marketsandmarkets.com/PressReleases/quantum-dots.asp	MAY, 2018
2	QUANTUM COMPUTING MARKET - GLOBAL FORECAST TO 2023 By Revenue Source, Application (Simulation, Optimization, and Sampling), Industry (Defense, Banking & Finance, Energy & Power, Chemicals, and Healthcare & Pharmaceuticals), and Geography https://www.marketsandmarkets.com/Market-Reports/quantum-computing-market-144888301.html	AUG, 2017

13.5 AUTHOR DETAILS

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With an experience of over 14 years, he has been associated with ICT research for more than 10 years now. Prior to joining Markets and Markets, he has worked with firms like Frost and Sullivan, AMI-Partners, TNS Market Research and global IT major, IBM, where he worked for more than 5 years and his last role saw him as Market Segment Manager. He holds a rich experience in the technology sector working on several custom and syndicated research assignments with large Multinational Organizations.

Sameer Bhatnagar

*Associate Director,
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The role encompasses handling syndicated and custom research projects in the ICT domain. Having more than 14 years of experience with 10+ years in research, Sameer joined MarketsandMarkets from Forrester Research where he was working as a Consultant, serving the Business Technology (BT) Professionals role. The role saw him working on varied consulting/custom-research projects and communicating directly with clients, internal resources (i.e., sales, research), and third-party vendors to address queries and manage expectations to create quality deliverables. Sameer has singlehandedly driven projects that include deliverable's in the shape of data modeling, trend analysis, end-user requirements, competitive profiling, etc.

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